OC Core 1.3.1 Unignorable Spine EN

Independent-release framing

- This canonical article is the shortest source-bound route from repo root to the strongest OC Core 1.3.1 claims.
- It is built from real theorem, falsifiability, empirical, and synthesis sections of the Core 1.3 corpus plus bounded 1.3.1 appendices.
- mapped_positive_claim_total=10; no_prose_only_promotion=true; owner_approval_required=true

Contents

1 Reader Contract and Navigation Guide

Core 1.3 is the flagship reader-facing monograph, but readers arrive with different tasks. This guide is therefore a route map rather than a second introduction. Its job is to show where each of the five reader roles should begin: Reviewer route, Formal reader route, Domain reader route, Broad scientific reader route, and Practical user route. External collaborators who need deployment guidance should use the Practical user route.

The dedicated Figure Atlas in Appendix **??** and the Technical Derivation Atlas in Appendix **??** are intentionally outside the default continuous reading route. Their material belongs to the appendix inspection layer; the formal-reader route below consults the Technical Derivation Atlas only when local proof support is needed.

1.1 The five reading routes

The reviewer-navigation appendix uses slightly narrower review labels: "rapid critical reviewer" corresponds to the Reviewer route, "formal mathematical reviewer" to the Formal reader route, and "generalist scientific reviewer" to the Broad scientific reader route.

Reviewer route. Read the front matter, this guide, Sections 2, **??**–7, Section **??**, and then Appendices **??**–B. That route exposes the publication boundary, theorem status, load-bearing theorem spine, reviewer objections, and source witnesses as quickly as possible.

Formal reader route. Use Sections **??**–**??** for orientation only. Then read Sections 3–6, Section **??**, and Sections **??**–9 in order. Use Appendices **??**–**??**, Appendix **??**, and the Technical Derivation Atlas in Appendix **??** when local proof support is needed. This is the compact route that keeps exact doctrine, theorem scope, and proof machinery visible at once.

Domain reader route. Read Section 5 and Sections **??**–10, then move into Appendices C, **??**, **??**, **??**, **??**, and **??**, with Appendix **??** added only when experiments, predictions, or falsifiability catalogues need inspection. That path exposes bounded domain embedding, worked interpretation, empirical promotion rules, replay surfaces, benchmark cases, execution board, practical use, and comparison evidence without forcing the reader to traverse every technical-derivation section first.

Broad scientific reader route. Read Sections **??**–**??**, Sections 3–6, Section **??**, and Sections **??**–10 straight through as a continuous graduate-level monograph. Use Appendix **??** for notation when needed and defer Appendices **??**–**??** until a concrete figure, benchmark table, or support dossier question arises. This is the recommended route for a broad scientific reader.

Practical user route. Read the front matter, this guide, Section **??**, Section 10, and Appendices **??**, A, and **??** first; add Appendix **??** only when the task requires experiments, predictions, or falsifiability catalogues. Return to Sections **??**–9 only when a practical use case needs theorem, replay, or falsifier support to be inspected in detail. This route is for external collaborators who need deployment guidance: what can be predicted, which decisions can be supported, which procedures can be run, and how the result compares with alternatives.

1.2 Route legend

The five routes above are the working navigation system:

- **Reviewer route**: shortest path to claim boundary, proof spine, objections, and source witnesses.
- **Formal reader route**: compact path through doctrine, theorem scope, notation, and proof machinery.
- **Domain reader route**: bounded path from domain embedding to replay, benchmarks, execution, use, and comparison evidence.
- **Broad scientific reader route**: continuous graduate-level reading path through the main argument before atlas inspection.
- **Practical user route**: shortest path to usable tasks, required inputs, outputs, comparison boundaries, and deployment constraints.

1.3 External interaction crosswalk

The crosswalk maps common external needs to the shortest route bundle that can answer them while keeping the supporting audit trail visible.

External task	Primary route	Support route if verification is needed
Evaluate the release for review	This guide; Sections 2, **??**–**??**; Appendices **??**–B	Appendix **??** when proof-obligation details are needed
Audit the bounded theorem core	Section 2, Section **??**, and Section 7	Appendix **??**, Appendix **??**, Appendix B, and Appendix **??**
Check the empirical promotion bar	Sections **??**–8	Appendix C, Appendix **??**, Appendix **??**, Appendix **??**, Appendix **??**, and Appendix **??**
Cite the unified synthesis	Section 9	Appendix D, Appendix **??**, and the K-level parameter tables in Appendix **??**
Use or compare the model	Section 10	First inspect Appendix **??**; use Appendix **??** only for experiments, predictions, or falsifiability catalogues; then use Sections **??**–9 only when proof or replay support is needed
Verify publication boundaries	Section **??**	Appendix **??** and Appendix A

Table 1: Stable reading routes for the most common external interactions. The purpose is to keep external use concise while preserving the audit trail behind each cited claim.

1.4 What this guide is not

This guide does not restate the theory, replace the theorem roadmap, or turn the flagship monograph into a status summary. It simply tells the reader where each task belongs. Section 7 covers proof navigation. Section 9 covers the unified synthesis. Section 10 covers practical consequences and comparative analysis. The appendices keep the inspection burden inside the same volume without displacing the main reading spine.

2 Results and Derived Consequences

This section summarises the core structural results that follow from the formal framework introduced in Section **??**. It presents the canonical consequences of the Ontology of Continua (OC) axioms and operators in their Core 1.3 form.

All results in this chapter are domain–independent. They follow solely from the structural definitions of continua, axes, thresholds, potentials, flows, boundaries, cycles and the measure of continuumness $k(K,t)$. Proofs are not reproduced here in full detail; they are deferred to dedicated extension papers. The focus is on stating the key theorems and consequences that constrain all continua in the hierarchy K_0–K_{12}.

2.1 Theorem 1: Monotonicity of dimensionality

Statement. If a continuum K_{x+1} emerges from K_x, then

$$\dim(K_{x+1}) > \dim(K_x).$$

Justification (sketch). Dimensional emergence is defined as the appearance of a new axis A_{new} that is incompatible with all existing axes in $A(K_x)$; that is, it cannot be represented as a linear or structural combination of them. The dimensional threshold $\Theta_{\dim}(K_x)$ enforces that A_{new} is nondegenerate and structurally necessary. Hence, any emergent continuum has strictly higher dimensionality than its predecessor.

Corollary 1.1 (No degradational simplification). There is no continuum–level evolution that reduces dimensionality while preserving existence: if $K(t)$ is live ($k(K,t) > 0$), then

$$\dim\big(K(t+dt)\big) \geq \dim\big(K(t)\big).$$

Apparent reduction of dimension corresponds to the death of the higher–dimensional continuum and the birth of a different, lower–dimensional one, not to a continuous simplification.

2.2 Theorem 2: Impossibility of spontaneous dimension creation

Statement. No continuum can increase its dimensionality without exceeding the dimensional threshold and having access to a suitable axis in its embedding space:

$$\dim(K_{x+1}) > \dim(K_x) \quad \Rightarrow \quad T(K_x,t) > \Theta_{\dim}(K_x) \ \wedge\ A_{\text{new}} \in A(M_x) \setminus A(K_x).$$

Justification (sketch). New axes represent classes of differences that are incompatible with existing axes. Such incompatibility arises only when structural tension $T(K_x,t)$ with respect to the current threshold landscape cannot be resolved within the existing axes. By definition of $\Theta_{\dim}$, below this threshold all differences can be projected onto the structural span $\text{span}_{\text{str}}(A(K_x))$; above it, projection fails and a new axis is required. The new axis must already be available in the embedding space M_x; otherwise there is no structural direction along which the continuum can expand.

Consequence 2.1. Noise, local fluctuations or internal flows that do not raise structural tension above $\Theta_{\dim}(K_x)$ cannot generate new dimensions.

2.3 Theorem 3: Death through boundary and embedding collapse

Statement. A continuum K dies when its admissible state space collapses:

$$\Omega(K) = \emptyset.$$

Equivalent formulations. The following conditions are equivalent and characterise the death of K:

1. supporting cycles disappear, $C(K) = \emptyset$;

2. supporting flows vanish, $J_{\text{support}} = 0$, or are insufficient to maintain any cycles;

3. there exists a state–independent violation of thresholds, $\forall s : \exists\, k$ with $f_k(s) > 0$;

4. continuumness collapses, $k(K,t) \to 0$;

5. the continuum exits the region supported by its embedding space M, i.e. no state satisfies both the internal thresholds of K and the external constraints of M.

Corollary 3.1 (Death as exit from embedding space). If the constraints of M change so that no configuration of K can be embedded into M while satisfying its thresholds, then $\Omega(K)$ becomes empty and the continuum dies, regardless of its internal structure.

Corollary 3.2 (Irreversibility of death). Once $\Omega(K) = \emptyset$, no operator E acting within the same level, nor any interaction operator E_{int} that preserves the identity of K, can reconstruct a nonempty $\Omega(K)$. Any apparent "recovery" corresponds to the birth of a new continuum K', not the resurrection of the original K.

2.4 Theorem 4: Impossibility of evolution outside the embedding space

Statement. Let K_x be a continuum embedded in M_x. Then for all times t,

$$A\big(K_x(t)\big) \subseteq A(M_x), \qquad \text{span}_{\text{str}}\big(A(K_x(t+dt))\big) \subseteq \text{span}_{\text{str}}\big(A(M_x)\big).$$

Consequence 4.1. All dynamical processes of K_x, including dimensional birth $K_x \to K_{x+1}$, require that the embedding space M_x already contain the axes along which the evolution proceeds. A continuum cannot evolve in directions not present in its embedding space.

2.5 Theorem 5: Necessity and sufficiency of compatibility with M

Statement. A continuum K exists as a live continuum (i.e. $\Omega(K) \neq \emptyset$ and $k(K,t) > 0$) if and only if it is compatible with its embedding space M:

$$K \text{ exists } \iff \Omega(K) \neq \emptyset \ \wedge\ A(K) \subseteq A(M) \ \wedge\ \Theta(K) \text{ is satisfiable within } M.$$

Consequence 5.1. Even if a configuration is mathematically well–defined at the level of K, it does not correspond to a live continuum unless the embedding space can support the required axes, thresholds and flows. Incompatible configurations are excluded structurally by setting $\Omega(K) = \emptyset$.

2.6 Theorem 6: Structural tension and phase transitions

Statement. Let $T(K,t)$ denote structural tension, understood as a functional of potentials, axes and their gradients:

$$T(K,t) = T\big(P(t), A, \nabla P(t)\big).$$

Then:

- a phase transition occurs when $T(K,t) = \Theta_{\text{crit}}(K)$,
- a dimensional transition occurs when $T(K,t) = \Theta_{\text{dim}}(K)$,
- structural collapse occurs when $T(K,t) = \Theta_{\text{death}}(K)$.

Corollary 6.1. All qualitative changes in the continuum — emergence of new regimes, new dimensions, or collapse — are governed by the relationship between structural tension and the threshold landscape.

2.7 Theorem 7: Universal law of complexity growth

Statement. For any live continuum K,

$$S\big(K(t+dt)\big) \geq S\big(K(t)\big),$$

where S is a structural measure of complexity defined on the state space and its organisation (for example, combining contributions from number of axes, diversity of states, richness of cycles and properties of the embedding).

Consequence 7.1. Complexity increases strictly whenever:

- new axes appear;
- new stable cycles form;
- the admissible state space $\Omega(K)$ expands;
- the embedding space M gains new axes relevant to K.

Stagnant or constant complexity corresponds to finely balanced dynamics below critical thresholds; however, this regime is generically unstable (see Theorem 2.8).

2.8 Theorem 8: Impossibility of eternal stabilisation

Statement. There is no nontrivial live continuum K that remains forever at a fixed point in its state space while maintaining positive continuumness. Formally, there is no solution with

$$\frac{dP}{dt} = 0, \quad \frac{dJ}{dt} = 0, \quad \frac{d\Theta}{dt} = 0, \quad \frac{dk}{dt} = 0$$

for all t, except the trivial case where K is structurally frozen and decoupled from its embedding space.

Justification (sketch). Embedding spaces M_x themselves evolve and expand (Theorem 10). Internal and external flows cannot both be identically zero on all time scales without either:

- driving the continuum to collapse (loss of supporting flows), or
- inducing structural changes (axes, thresholds, cycles).

Hence any nontrivial live continuum is subject to evolution of its structure or to eventual death.

2.9 Theorem 9: Death as loss of cycles

Statement. A live continuum K dies when its maximal structurally stable cycle complex disappears:

$$C_{\text{max}}(K) = \emptyset.$$

Justification (sketch). Cycles represent self–maintaining flows that keep the continuum away from its boundary $\partial\Omega$. If no such cycle exists, then:

- supporting flows cannot form closed loops;
- trajectories either approach $\partial\Omega$ or violate thresholds;
- continuumness $k(K,t)$ decays to zero.

Therefore loss of the maximal cycle complex coincides with the collapse of $\Omega(K)$.

Corollary 9.1. Death can be detected structurally by the disappearance of all cycles satisfying a minimal stability condition (strictly positive distance to $\partial\Omega$).

2.10 Theorem 10: Monotonic growth of embedding spaces

Statement. Embedding spaces form a monotonic sequence:

$$M_0 \subset M_1 \subset \cdots \subset M_x \subset \dots,$$

and whenever a new continuum K_{x+1} emerges, the corresponding embedding space M_{x+1} strictly extends M_x in terms of its available axes:

$$A(M_x) \subset A(M_{x+1}).$$

Consequence 10.1. The growth of continua in dimension and complexity is inseparable from the growth of their embedding spaces. New continua cannot appear without an expansion of the structural possibilities encoded in M.

2.11 Theorem 11: Threshold–expressivity incompatibility

Statement. A continuum K collapses when its axes become insufficient to express the required differences:

$$\dim\big(A(K)\big) < \dim\big(\text{Differences}(K)\big),$$

where Differences(K) is the effective dimension of structurally relevant distinctions imposed by the embedding space and thresholds.

Consequence 11.1. Collapse may occur even without violating energetic or dynamical constraints if representational capacity is exceeded. In particular, cognitive, social or meta–theoretical continua can die purely due to loss of expressive adequacy of their axes.

2.12 Theorem 12: Incompleteness of embedding spaces

Statement. For any finite embedding space M_x, there exist potential continua K' that cannot be realised within M_x because their axes or thresholds are incompatible with $A(M_x)$ or with the constraints of M_x.

Consequence 12.1. No single embedding space M_x is universal for all possible continua. The sequence of embedding spaces must itself expand and diversify to host new structural regimes.

2.13 Corollaries and domain–independent consequences

Collecting the theorems above, we obtain the following domain–independent consequences:

- Dimensionality is strictly monotonic for live continua; no continuous degradation of dimension is possible.
- Dimension cannot appear spontaneously; it requires structural tension above Θ_{dim} and the presence of suitable axes in the embedding space.
- Death is characterised by the collapse of $\Omega(K)$, the disappearance of cycles and the irreversibility of this collapse.
- Evolution is constrained by embedding spaces; continua cannot evolve in directions that M does not support.
- Complexity tends to grow for live continua, driven by new axes, new cycles and expanding state spaces.
- Eternal exact stabilisation is structurally excluded for nontrivial continua; they either evolve or die.
- Representational and expressive limits can cause collapse even when energetic and dynamical limits are not yet reached.

These consequences hold uniformly for all levels K_0–K_{10}; their domain–specific content arises only from the interpretation of axes, potentials, thresholds and flows in each level.

2.14 Implications for the vertical hierarchy

In the vertical hierarchy K_0–K_{10}, the structural theorems manifest as follows:

- K_0: no dynamics, no birth or death; only structural conditions for distinguishability.
- K_1: classical phase–like behaviour with continuous flows and basic stability/critical thresholds.
- K_2: physical thresholds, including percolation, BKT transitions, coherence/condensation thresholds and mass–generating mechanisms.
- K_3–K_4: chemical and prebiotic thresholds (RAF closure, membrane formation, osmotic and curvature thresholds, redox and pH thresholds).
- K_5: electrical thresholds for excitation, spiking and gradient–driven flows in early neural systems.
- K_6: logical and representational thresholds for cognitive binding, prediction, memory stability and internal model coherence.
- K_7–K_8: social and civilizational thresholds governing trust, institutional stability, systemic resilience and collapse.
- K_9–K_{10}: expressive, coherence and consistency thresholds in the space of theories, paradigms, ontologies and meta–theoretical structures.

In each case, the same structural principles — monotonic dimension, threshold–governed emergence and collapse, dependence on embedding spaces, and complexity growth — apply, while the concrete interpretation of variables changes.

2.15 Summary

The results presented in this chapter constitute the structural backbone of OC. They state that dimension is monotonic, that emergent dimensions require threshold–induced phase transitions, that death is characterised by the collapse of admissible states and is irreversible, that evolution is constrained by embedding spaces, that complexity tends to grow for live continua, that cycles are the carriers of structural life, and that expressive limits can cause collapse. These principles apply across the entire continuum hierarchy and define the constraints under which all continua can exist, evolve or die.

3 Extended Boundary and Patch Geometry

This chapter restores the full theory of boundaries and patch geometry. Boundaries constitute one of the most universal structures in the OC framework: they govern phase separation in K_2, membrane formation in K_3–K_4, excitability in K_5, and representational or institutional limits in K_6–K_8. The present version consolidates the complete structure needed for Core 1.3: threshold–defined boundaries, patch discretisation, local threshold classes, and boundary–driven transitions.

3.1 Formal Definition of $\partial\Omega(K)$

For any continuum K with admissible state space $\Omega(K)$ and threshold landscape $\Theta(K)$, the boundary is defined as the set of states at which at least one threshold is exactly saturated:

$$\partial\Omega(K) = \left\{\, s \in \overline{\Omega(K)} : \exists k \text{ such that } g_k(s) = \Theta_k \right\}.$$

Equivalently, if $f_k(s) = g_k(s) - \Theta_k$ denotes the shifted residual form used in the notation appendix and the model chapter, then interior states satisfy $f_k(s) < 0$, admissible-side states satisfy $f_k(s) \le 0$, and the same boundary is the zero set $f_k(s) = 0$ for at least one active threshold.

Thus:

- interior states satisfy $f_k(s) < \Theta_k$ for all thresholds;
- exterior states violate at least one threshold;
- the boundary is a *threshold–defined surface*, not a geometric primitive.

This definition unifies classical surfaces, membranes, vesicles, percolation fronts, excitable boundaries, logical limits and institutional thresholds under one structure.

3.2 Boundary Geometry

Boundary geometry emerges from the local structure of thresholds, potentials and flows. If the active threshold at a point $s \in \partial\Omega(K)$ is g_k, the outward normal is aligned with the gradient:

$$n(s) \propto \nabla g_k(s).$$

Multiple active thresholds create corners, ridges and high–curvature regions. Curvature becomes dynamically relevant whenever bending or surface–energy terms enter the potential:

$$P_{\text{bend}}(s) = \tfrac{1}{2}\,\kappa(s)^2\,\kappa_b,$$

with bending rigidity κ_b.

This connects:

- K_2: classical surface tension,
- K_3–K_4: membrane curvature and vesicle energetics,
- K_5: excitable boundary geometry.

3.3 Patch Model of Boundaries

OC models boundaries not as continuous uniform surfaces but as collections of dynamic *patches*:

$$\partial\Omega(K) = \bigcup_i \partial\Omega_i.$$

Each patch P_i carries:

$$(\sigma_i, \Theta_i, P_i(t), J_i(t)),$$

representing its local state, thresholds, potentials and flows.

Local states. Patches take symbolic states:

$$\sigma_i \in \{L_\alpha, L_\beta, L_o, L_f, L_b\},$$

encoding local packing or structural regimes:

- L_α: fluid–disordered,
- L_β: ordered/gel,
- L_o: liquid–ordered,
- L_f: fragile/thin (pre–rupture),
- L_b: broken boundary state.

This alphabet generalises beyond biological membranes to percolation edges, fracture fronts, fluctuating interfaces and even symbolic/cognitive boundaries.

Local thresholds. Each patch has a vector:

$$\Theta_i = (\Theta_i^{\text{perm}}, \Theta_i^{\text{grad}}, \Theta_i^{\text{mem}}, \Theta_i^{\text{bend}}, \ldots),$$

with corresponding local potentials $P_i(t)$ and flows $J_i(t)$.

3.4 Boundary Threshold System

Across K_2–K_5 three threshold classes appear universally.

1. Permeability threshold Θ_{perm}. A patch becomes permeable when:

$$|\nabla P_i| > \Theta_i^{\text{perm}}.$$

Examples:

- osmotic permeability (K_3–K_4),
- ion leak onset (K_5),
- phase–front penetration (K_2).

2. Gradient threshold Θ_{grad}. If a gradient is unsustainable:

$$|\nabla P_i| > \Theta_i^{\text{grad}} \quad \Rightarrow \quad \sigma_i \to L_f.$$

This governs:

- vesicle rupture at high osmotic pressure (K_4),
- action–potential initiation at axon initial segments (K_5),
- accelerated phase fronts and crack propagation (K_2).

3. Membrane integrity threshold Θ_{mem}. If membrane tension or curvature exceeds limits:

$$\gamma_i > \Theta_i^{\text{mem}} \qquad \text{or} \qquad \kappa(s) > \Theta_i^{\text{bend}},$$

then:

$$\sigma_i \to L_b.$$

This yields a unified understanding of rupture, buckling, pore formation, boundary breakdown and surface collapse.

3.5 Boundary–Driven Dynamics

Neighbouring patches interact via flows:

$$J_{ij} = F(P_i, P_j, \sigma_i, \sigma_j).$$

This produces:

- stabilisation and rearrangement of patch mosaics,
- propagation of rupture fronts,
- curvature–driven migration,
- pore initiation,
- excitable–medium behaviour in K_5.

Patch evolution follows:

$$\frac{d\sigma_i}{dt} = S(\sigma_i, P_i, J_i, \Theta_i),$$

where S is a structural update operator independent of domain.

Boundary Localisation Principle. *All K_3–K_5 dimensional transitions initiate on boundary patches.*

This explains:

- protocell bursting beginning at high–curvature points,
- action potentials initiating at axon initial segments,
- phase transitions nucleating at defects or surfaces.

3.6 Boundary Failure and Collapse

Boundary collapse occurs when a connected set of patches becomes broken:

$$\sigma_i = L_b \qquad \forall\, i \in \mathcal{C},$$

and the cluster $\mathcal{C}$ percolates across the boundary.

By Theorem 3 (Results chapter),

$$\partial\Omega(K) \text{ collapses} \quad \Longrightarrow \quad \Omega(K) = \emptyset,$$

and the continuum dies.

We distinguish:

- **local failure**: isolated L_b patches, reversible;
- **cluster collapse**: percolating L_b network, irreversible;
- **global boundary loss**: disappearance of the boundary, implying death or transition to a new continuum.

3.7 Boundary and Rebirth

Birth of a new continuum K_{x+1} typically requires:

- formation or rupture of boundary patches,
- creation of new patch types,
- emergence of new threshold classes on boundary regions,
- establishment of a new global boundary geometry.

Boundary Rebirth Principle. *New continua emerge from new boundary geometries.*

Examples:

- membrane closure in the $K_3 \to K_4$ transition,
- appearance of excitable boundaries in $K_4 \to K_5$,
- formation of institutional boundaries in $K_6 \to K_7$,
- emergence of representational boundaries in $K_9 \to K_{10}$.

Boundaries are therefore not merely limits: they are generative surfaces where new axes, constraints and cycles are born.

4 Extended Threshold Landscape

This chapter reconstructs the full theory of thresholds in the Ontology of Continua (OC), consolidating structural elements from the Physics, Chemistry, Biology and Cognition runs. Thresholds are the central mechanism through which qualitative change, dimensional emergence, collapse and death are produced.

Thresholds are universal: they apply to physical, chemical, biological, cognitive, social and theoretical continua throughout the hierarchy K_0–K_{10}.

4.1 Definition of a Threshold

A *threshold* is represented by a constraint function

$$f_k : \Omega(K) \to \mathbb{R}$$

with the property

$$f_k(s) \leq 0 \quad \text{for all admissible states } s \in \Omega(K).$$

Thresholds separate qualitatively different dynamical regimes:

- $f_k(s) < 0$: safe regime (interior),
- $f_k(s) = 0$: boundary regime,
- $f_k(s) > 0$: forbidden regime (outside $\Omega(K)$).

Thus thresholds jointly determine:

- the shape of the admissible set $\Omega(K)$,
- the boundary $\partial\Omega(K)$,
- the conditions under which the continuum exists or collapses.

Thresholds are not static constants. They depend on potentials $P(t)$, flows $J(t)$, axes $A(K)$, environmental constraints and the structure of the embedding space M. At the structural level this is encoded by the threshold–evolution operator H from the general evolution machinery:

$$\frac{d\Theta_k}{dt} = H_k\big(\Theta_k, P, J, A, M\big),$$

where Θ_k denotes the parameter set defining the constraint f_k.

4.2 Taxonomy of Thresholds

OC distinguishes seven universal classes of thresholds. This taxonomy was stabilised in Core 2.x and clarified across the domain runs.

1. Existence thresholds Θ_{exist}. Conditions required for $\Omega(K) \neq \emptyset$. Examples:

- minimal energy conditions for bound physical states (K_2),
- minimal catalytic closure (RAF existence) in K_3,
- minimal membrane continuity and integrity in K_4,
- minimal coherence of internal models in K_6,
- minimal legitimacy and trust for institutions in K_7.

2. Stability thresholds Θ_{stab}. Conditions under which flows remain bounded and cycles are stable. Examples:

- confinement and boundedness conditions in physics (K_2),
- osmotic homeostasis conditions in protocells (K_4),
- membrane potential stability in neurons (K_5),
- normative stability in social continua (K_7).

3. Critical thresholds Θ_{crit}. Surfaces of qualitative change at fixed dimensionality. Examples:

- phase transitions and BKT–type thresholds (K_2),
- onset of catalytic networks (K_3),
- metabolic switching in K_4,
- spike initiation threshold in K_5,
- prediction–error bifurcations in K_6,
- institutional bifurcations in K_7.

4. Dimensional thresholds Θ_{dim}. Constraints that trigger the creation of new axes. Dimensional thresholds are central in OC:

$$T(K,t) > \Theta_{\text{dim}}(K) \quad \Rightarrow \quad \text{emergence of a new axis } A_{\text{new}}.$$

They govern the transitions $K_x \to K_{x+1}$ throughout the hierarchy.

5. Death thresholds Θ_{death}. Values beyond which no admissible state remains. Equivalently,

$$\forall s \in \overline{\Omega(K)} : \ \exists k \text{ with } f_k(s) > 0 \quad \Longleftrightarrow \quad \Omega(K) = \emptyset.$$

6. Expressivity thresholds Θ_{expr}. Thresholds on representational capacity. A continuum collapses when

$$\dim\big(A(K)\big) < \dim\big(\text{Differences}(K)\big),$$

where Differences(K) denotes the effective dimension of structurally relevant distinctions imposed by the embedding space and constraints. This class is crucial for cognitive, social and meta–theoretical collapse.

7. Embedding thresholds Θ_{embed}. Constraints imposed by the embedding space M_x. A continuum cannot exist as a live continuum if the embedding space does not support its axes, thresholds or flows; formally this is captured by the compatibility conditions in Theorems 4 and 5 (Results chapter).

4.3 Threshold Geometry

Thresholds induce a stratified geometry on $\Omega(K)$. For each threshold function f_k we consider the level sets

$$\Sigma_k(c) = \{\, s \in \Omega(K) \mid f_k(s) = c \,\}.$$

In particular:

- $\Sigma_k(0)$ defines components of the boundary $\partial\Omega(K)$,
- $\Sigma_k(c < 0)$ defines safe regions,
- $\Sigma_k(c > 0)$ defines forbidden regions.

Interactions between thresholds generate:

- multi–threshold intersections,
- cusp and fold structures (catastrophe–like geometries),
- regions of concentrated curvature,
- patch–level heterogeneity at the boundary (cf. Section 3).

The joint threshold geometry determines:

- the global shape of $\Omega(K)$,
- admissible trajectories and cycle structure,
- the available transitions of the continuum.

4.4 Dimensional Thresholds

Dimensional thresholds occupy a privileged role in OC because they decide when new differences become unrepresentable within the current axes.

Let $T(K,t)$ denote structural tension. Then, schematically:

$$T(K,t) < \Theta_{\text{dim}}(K) \quad \Rightarrow \quad \text{all relevant differences can be expressed within } A(K),$$

$$T(K,t) = \Theta_{\text{dim}}(K) \quad \Rightarrow \quad \text{projection onto } A(K) \text{ fails},$$

$$T(K,t) > \Theta_{\text{dim}}(K) \quad \Rightarrow \quad \text{a new axis } A_{\text{new}} \text{ is structurally required.}$$

Dimensional thresholds govern, in particular:

- birth of K_1 from the structural substrate K_0,
- emergence of chemical axes in K_3,
- membrane axes in K_4,
- excitation axes in K_5,
- cognitive representational axes in K_6,
- social normative axes in K_7,
- meta–logical axes in K_9–K_{10}.

4.5 Threshold Cascades

Thresholds frequently interact in cascades of the form

$$\Theta_{\text{grad}} \longrightarrow \Theta_{\text{perm}} \longrightarrow \Theta_{\text{mem}},$$

or more generally, gradients $\to$ permeability $\to$ structural failure.

Representative examples:

- **Vesicle bursting (K_4).** Osmotic gradients exceed Θ_{grad}, permeability increases (Θ_{perm}), and membrane integrity is lost (Θ_{mem}).
- **Action potentials (K_5).** Depolarisation exceeds a gradient threshold, channels open at Θ_{crit}, and a spike cycle is activated.
- **Institutional collapse (K_7).** Trust deficits exceed Θ_{grad}, normative coherence fails (Θ_{expr}), and institutional death follows (Θ_{death}).

Such cascades are central for early life, neural systems, and social and infrastructural dynamics.

4.6 Death Thresholds

Death thresholds determine when a continuum ceases to exist as a live continuum. The following structural conditions are equivalent manifestations of death:

- complete violation of thresholds at all states: $\forall s: \ \exists k$ with $f_k(s) > 0$;
- disappearance of supporting cycles: $C(K) = \emptyset$;
- collapse of the admissible state space: $\Omega(K) = \emptyset$;
- representational failure, e.g. Θ_{expr} exceeded;
- incompatibility with the embedding space (no configuration satisfies both internal and embedding constraints).

Once $\Omega(K) = \emptyset$, death is structurally irreversible: no operator acting within the same level can recreate $\Omega(K)$; any apparent recovery corresponds to the birth of a new continuum K'.

4.7 Examples Across Levels

The threshold taxonomy manifests differently at each level of the hierarchy.

K_1 **(Geometric continua).**

- Θ_{exist}: differentiability and regularity conditions defining the classical region Ω_{cl},
- Θ_{crit}: classical bifurcations and loss of stability.

K_2 **(Physical continua).**

- Θ_{crit}: phase transitions, including BKT thresholds,
- Θ_{dim}: coherence thresholds associated with mass generation and ordering,
- Θ_{death}: confinement/decoupling limits where the relevant continuum ceases to exist.

K_3 **(Chemical continua).**

- Θ_{exist}: catalytic closure and RAF existence,
- Θ_{crit}: reaction–network percolation,
- Θ_{expr}: insufficient chemical axes to represent required reaction diversity.

K_4 **(Prebiotic/biological membrane continua).**

- Θ_{grad}: osmotic pressure limits,
- Θ_{perm}: permeability thresholds,
- Θ_{mem}: membrane rupture and curvature thresholds.

K_5 **(Excitable continua).**

- Θ_{crit}: spike initiation thresholds,
- Θ_{stab}: refractory conditions and excitability bounds,
- Θ_{death}: loss of excitability and breakdown of excitable cycles.

K_6 **(Cognitive continua).**

- Θ_{expr}: limits of binding capacity and representational complexity,
- Θ_{crit}: prediction–error thresholds,
- Θ_{stab}: memory coherence and model–stability limits.

K_7–K_8 **(Social and civilizational continua).**

- Θ_{exist}: thresholds of trust and legitimacy,
- Θ_{expr}: institutional capacity and expressive limits,
- Θ_{death}: systemic collapse and breakdown of institutional and infrastructural cycles.

K_9–K_{10} **(Theoretical and meta–theoretical continua).**

- Θ_{expr}: expressive capacity of theories and meta–languages,
- Θ_{dim}: thresholds for meta–level expansion,
- Θ_{death}: inconsistency, incoherence or structural incompatibility in the space of models.

4.8 Summary

The extended threshold landscape provides the structural backbone for all qualitative change in OC. Thresholds:

- carve out $\Omega(K)$ and define $\partial\Omega(K)$,
- govern phase transitions, dimensional emergence and collapse,
- mediate cascades that connect gradients, permeability and structural failure,
- encode expressive and embedding limitations that can kill continua even without energetic overload.

Together with the boundary and patch geometry of Section 3, the threshold landscape completes the description of how continua are born, maintained and destroyed in the OC framework.

5 Cross-Disciplinary Instantiations of the Continuum Framework

This chapter tests a bounded representation claim: selected scientific and formal-theoretical domains can be expressed by the tuple

$$(\Omega, A, P(t), J(t), \Theta, \partial\Omega, C, k(t))$$

in ways consistent with the structural hierarchy of levels K_0–K_{12}, the threshold landscape, the operator family, and the branch topology of Section **??**. Here $k(t)$ suppresses the continuum index only to keep the domain tuple readable; locally it denotes the corresponding $k(K_x, t)$. Likewise, later domain blocks may suppress t in symbols such as P_x or J_x when the paragraph describes the family of potentials or flows rather than a single time slice. Each domain-specific mapping is treated here under one canonical label: the *candidate-embedding rule*. Phrases such as bounded representation test, structural-compatibility screen, bounded embedding pass, and bounded embedding criterion are local descriptions of that rule, not separate criteria. The aim is to show how well-studied classes of systems can be represented as continua under OC's structural axioms, not to claim that every object in a discipline is exhausted by a single OC encoding (**anderson_more_is_different**).

This is a structural-compatibility claim, not an operational-advantage claim. The practical comparison is deferred to Section 10 and Appendix **??**. This chapter asks whether a domain can be represented as a bounded continuum; it does not claim that every such domain already has a closed theorem packet, held-out replay, comparator audit, or deployment-ready procedure.

5.1 Principles for Domain Embedding

To represent a discipline within OC, the following structural alignment conditions must be satisfied.

1. Mapping of objects to continuum components. A domain has a candidate continuum representation if its objects can be mapped to:

- Ω: admissible configurations,
- A: axes defining essential degrees of freedom,
- P: potentials encoding forces, energies, costs, or semantic tension,
- J: flows (matter, charge, probability, information, resources),
- Θ: thresholds governing stability, phase change, and collapse,
- $\partial\Omega$: viability boundaries,
- C: cycles maintaining persistence,
- $k(t)$: continuumness (viability measure).

2. Embedding-space constraints. A valid instantiation requires both $\Omega(K_x) \subseteq \Omega(M_x)$ and $A(K_x) \subseteq A(M_x)$. If the required states or axes are absent from the embedding space, the proposed instantiation is structurally inadmissible.

3. K-level correspondence. A discipline is boundedly embedded at level K_x only when:

- its dominant axes correspond to the axes of K_x,
- its thresholds refine those of lower levels,
- its flows and cycles match the operator family of K_x,
- its behaviour respects dimensional monotonicity.

4. Structural compatibility. Domain instantiation requires:

- internal consistency with OC thresholds,
- existence of at least one supporting cycle C_{support},
- nonempty admissible region $\Omega \neq \emptyset$,
- extendibility of domain dynamics through the operator family.

A proposed embedding fails if any check fails: the mapping has an empty admissible region, the required axes are absent from the embedding space, the declared thresholds cannot be satisfied, or the proposed cycles do not keep the object away from the relevant boundary. These are rejection conditions, not merely cosmetic omissions.

The following sections summarise bounded candidate representations across disciplines.

5.2 Physics (Level K_2)

Physical systems instantiate K_2 when field configurations, order parameters, and coherence structures form the relevant axes. The discussion is intentionally bounded by standard critical phenomena, renormalisation, and percolation settings rather than by an unrestricted claim about all of physics (**goldenfeld_lectures**; **wilson_renormalization**; **stauffer_percolation**).

State space and axes.

- Ω_2: field configurations (scalar, vector, gauge fields).
- A_2: spatial axes, mode axes, internal symmetry axes, order-parameter axes.

Potentials and flows.

- P_2: physical energy functionals (action, free energy).
- J_2: diffusive, convective, and field-theoretic currents.

Thresholds.

- Θ_{crit}: phase boundaries,
- Θ_{BKT}: Berezinskii–Kosterlitz–Thouless vortex unbinding threshold,
- Θ_{mass}: coherence threshold for mass emergence,
- Θ_{death}: confinement or decoupling conditions.

Boundary structures. Within this bounded mapping, $\partial\Omega$ corresponds to critical surfaces, percolation boundaries, and phase-separation interfaces.

Cycles. Topological solitons, vortex pairs, renormalisation cycles, and oscillatory field modes are instances of C_2.

Bounded embedding criterion (Physics → K_2). A physical system meets the bounded K_2 embedding criterion when:

- its fields define a connected configuration space Ω_2,
- its energy functional defines thresholds compatible with OC,
- flows obey conservation laws expressible via J_2,
- coherent cycles maintain structural persistence.

5.3 Chemistry and Origins of Life (Levels K_3–K_4)

Chemical systems are represented at K_3, while protocellular systems are represented at K_4. These mappings are bounded by reaction-network closure, catalytic organisation, and membrane-constrained prebiotic structure (**kauffman_autocatalytic**; **raf_networks**).

Chemical continua (K_3).

- Ω_{chem}: concentration vectors, reaction states, catalytic configurations.
- A_{chem}: species concentrations, environmental parameters (pH, salinity, temperature).
- $P_{\text{chem}}(t)$: chemical potentials, affinity gradients, and concentration disequilibria.
- J_{chem}: reaction rates, transport fluxes.
- Θ_{act}: catalytic activation threshold.
- Θ_{closure}: reflexively autocatalytic and food-generated (RAF) closure threshold.
- C_{chem}: reaction cycles, RAF closure loops.

Protocellular continua (K_4).

- Ω_4: vesicle states, membrane configurations, internal chemical compositions.
- A_4: curvature, surface tension, permeability, charge.
- $P_4(t)$: osmotic, chemical, electrochemical, and curvature potentials.
- $\Theta_{\text{perm}}, \Theta_{\text{osm}}, \Theta_{\text{curv}}$: membrane viability thresholds.
- J_4: osmotic fluxes, ion fluxes, metabolic flows.
- C_4: metabolic cycles, membrane-maintenance cycles, gradient-sustaining cycles.

Bounded embedding criterion (Chemistry → K_3). A chemical system meets the bounded K_3 embedding criterion when:

- reaction networks form a nonempty admissible region,
- catalytic and closure thresholds are representable without membrane-specific axes,
- cycles maintain reaction closure and concentration control,
- embedding constraints support the chemical axes required by the route.

Bounded embedding criterion (Protocells → K_4). A protocellular system meets the bounded K_4 embedding criterion when:

- membrane-supported chemical dynamics form a nonempty admissible region,
- membrane boundaries define $\partial\Omega_4$,
- cycles maintain concentration gradients and compartment integrity,
- embedding constraints support both chemical and membrane axes.

5.4 Biology (Level K_5)

Biological cells, beyond the protocellular K_4 route, emerge when protocells acquire excitability and structured regulation. Here the target is the bounded embedding of excitable and morphogenetic organisation rather than an unrestricted theory of life as such (**turing_morphogenesis**; **hopfield_nn**).

State space and axes.

- Ω_5: distributions of membrane potential, channel states, ion gradients.
- A_5: excitability axes (membrane potential ΔV, ion concentrations, gating-variable axes).

Potentials and flows.

$$P_5 : \text{electrochemical potentials}, \qquad J_5 : \text{ion currents, metabolic fluxes.}$$

Thresholds.

$$\Theta_{\text{exc}}, \quad \Theta_{\text{refr}}, \quad \Theta_{\text{grad}}, \quad \Theta_{\text{death}}.$$

Cycles. The relevant cycles include the proto-spike cycle C_{spike}, metabolic cycles, and electrical oscillation loops.

Bounded embedding criterion (Biology → K_5). A biological system meets the bounded K_5 embedding criterion when:

- excitability defines nontrivial ΔV-axes,
- thresholds define firing and refractory structure,
- cycles maintain homeostasis and signalling,
- embedding includes ionic, energetic, and membrane-support axes.

5.5 Cognition (Level K_6)

The K_6–K_{10} sections below are structural candidate embeddings. They are frontier-facing representation sketches unless Section 10 or Appendix **??** explicitly promotes a narrower operational lane. They are not usable-now practical claims by themselves.

Cognitive systems arise when representations, bindings, and predictive models become the dominant axes. The discussion is anchored to classical work on the organisation of behaviour and predictive-processing motifs, not to a claim that OC replaces cognitive science in full (**hebb_organization**; **friston_free_energy**).

State space and axes.

- Ω_6: representational configurations, activated conceptual states, prediction states.
- A_6: representational axes, binding axes, predictive axes.

Potentials and flows.

- P_6: prediction error, semantic tension, confidence.
- J_6: attention flows, inference flows, memory transitions.

Thresholds.

$$\Theta_{\text{pred}}, \quad \Theta_{\text{bind}}, \quad \Theta_{\text{coh}}, \quad \Theta_{\text{expr}}.$$

Cycles. The relevant cycles include attention-prediction cycles, memory-consolidation cycles, and model-stability loops.

Bounded embedding criterion (Cognition → K_6). A cognitive system meets the bounded K_6 embedding criterion when:

- it maintains coherent internal models within expressivity thresholds,
- flows reduce prediction error and sustain conceptual structure,
- cycles regulate attention, memory, and updating,
- boundaries $\partial\Omega_6$ encode model limits.

5.6 Society (Level K_7)

Social systems instantiate K_7 when norms, institutions, and trust form the dominant structural axes. The intended scope is thresholded collective behaviour, networked coordination, and institutional persistence (**granovetter_threshold**; **schelling_micromotives**; **newman_networks**).

State space and axes.

- Ω_7: communication graphs, role structures, institutional states.
- A_7: normative axes, trust axes, institutional axes.

Potentials and flows.

$$\begin{aligned} P_7 &: \text{trust, legitimacy, symbolic capital,} \\ J_7 &: \text{communication, resource exchange, and norm transmission.} \end{aligned}$$

Thresholds.

$$\Theta_{\text{trust}}, \quad \Theta_{\text{leg}}, \quad \Theta_{\text{role}}.$$

Cycles. The relevant cycles include institutional cycles, normative cycles, and coordination cycles.

Bounded embedding criterion (Society → K_7). A social system meets the bounded K_7 embedding criterion when:

- trust and norms define viability thresholds,
- institutions maintain cycles of governance and coordination,
- flows of communication and resources respect embedding constraints,
- social boundaries form nonempty $\partial\Omega_7$.

5.7 Civilizational Systems (Level K_8)

Civilizational systems extend social continua to encompass infrastructures, energy flows, and large-scale coordination. The scope is limited to large-scale organised systems with networked infrastructure and stability constraints (**newman_networks**; **may_stability_complexity**).

State space and axes.

- Ω_8: infrastructure networks, economic states, population distributions.
- A_8: infrastructural axes, energy-flow axes, communication axes, complexity axes.

Potentials and flows.

$$P_8 : \text{energy availability, resource gradients, risk potentials,} \qquad J_8 : \text{trade, energy flux, data flows.}$$

Thresholds.

$$\Theta_{\text{stress}}, \quad \Theta_{\text{capacity}}, \quad \Theta_{\text{cohesion}}.$$

Cycles. The relevant cycles include economic cycles, infrastructure-renewal cycles, and knowledge-transfer loops.

Bounded embedding criterion (Civilizational systems → K_8). A civilizational system meets the bounded K_8 embedding criterion when:

- energy and resource structures define P_8,
- infrastructural flows form stable global cycles,
- institutions at K_7 embed consistently as substructures,
- large-scale boundaries define global $\partial\Omega_8$.

5.8 Theory and Formal Systems (Levels K_9–K_{10})

Theory-level and metatheoretical formal-system structures occupy the upper route treated in this chapter. This subsection gives an OC-internal schema for K_9–K_{10}, with local support carried by Sections **??** and **??**. It does not claim a fully closed external embedding theorem for every theory or formal system. Recursive structural continua remain the K_{11} layer outside this chapter's candidate-embedding rule.

Theoretical continua (K_9).

- Ω_9: space of models, formalisms, and representations.
- A_9: conceptual, formal, and modelling axes.
- P_9: coherence potentials, empirical tension, explanatory power.
- J_9: model-revision flows, anomaly-handling flows, and argument-transfer flows.
- $\Theta_{\text{expr}}, \Theta_{\text{coh}}$: thresholds for consistency and expressivity.
- $\partial\Omega_9$: inconsistency, expressivity-failure, or unrepaired-anomaly boundaries.
- C_9: paradigm cycles, research cycles.
- $k_9(t)$: coherence and validation-bandwidth continuumness.

Metatheoretical formal-system continua (K_{10}).

- Ω_{10}: space of metatheoretical states, formal systems, proof states, model-of-models objects, and recursion-bounded languages.
- A_{10}: metatheory axes, recursion axes, consistency axes, expressivity axes, and theory-comparison axes.
- P_{10}: proof-flow capacity, consistency margin, formal expressive capacity, and metatheoretical coherence potential.
- J_{10}: proof-search flows, formal-update flows, meta-model translations, and recursion-control flows.
- $\Theta_{\text{cons}}, \Theta_{\text{rec}}, \Theta_{\text{meta}}$: thresholds for consistency, bounded recursion, and meta-level coherence.
- $\partial\Omega_{10}$: inconsistency, undecidable-overreach, recursion-boundary surfaces, or meta-theory boundary failures.
- C_{10}: proof cycles, recursion cycles, formal-update cycles, and model-of-models audit cycles.
- $k_{10}(t)$: formal and metatheoretical viability under consistency, recursion, self-reference, and proof-throughput constraints.

Candidate-embedding rule (Theory → K_9, metatheoretical formal system → K_{10}). The theoretical and metatheoretical formal-system routes meet the candidate-embedding rule when the relevant K_9 or K_{10} structure satisfies the following conditions:

- its conceptual and formal axes form a valid axis set,
- its coherence constraints define thresholds compatible with OC,
- cycles of reasoning, modelling, and revision sustain viability,
- embedding spaces support the required expressive degrees of freedom.

Summary

This chapter states bounded cross-disciplinary candidate-embedding criteria across physics, chemistry, protocells, biology, cognition, society, civilizational systems, theory, and formal systems. Each bounded route is mapped to the same structural tuple at the level of detail declared in its subsection, but each mapping is claimed only at the level of structural compatibility, not as a sweeping proof that OC already replaces the best local models in every discipline. That stronger operational and comparative question is taken up later under the explicit support classes in Section 10 and the row-level atlas in Appendix **??**.

6 Extended Falsifiability and Experimental Programme

This section restores and expands the falsifiability framework for the Ontology of Continua (OC). It formulates explicit structural predictions, falsification criteria, and experimental programmes across the hierarchy K_2–K_{12}, using only the axioms and definitions of Core 1.3.

The goal is not to provide a full catalogue of experiments but to show that OC is structurally falsifiable: it makes cross-domain commitments about thresholds, cycles, dimensional transitions and boundary behaviour that can in principle be contradicted by empirical and theoretical data.

6.1 Meaning of Falsifiability for Structural Theories

For a structural theory, falsifiability is not limited to individual numerical predictions. Instead, falsification occurs when the structural relations between

$$(\Omega, A, P(t), J(t), \Theta, \partial\Omega, C, k(t))$$

are violated in ways incompatible with the axioms and theorems of OC.

Structural vs phenomenological falsifiability.

- *Phenomenological falsifiability* concerns direct mismatches between model outputs and observations (e.g. wrong phase diagram, incorrect scaling exponent).
- *Structural falsifiability* concerns violations of the universal constraints on continua: thresholds, cycles, dimensional monotonicity, embedding-compatibility, collapse and rebirth conditions.

OC primarily lives at the structural level; phenomenological models must factor through OC's structural requirements to be considered proper instantiations.

Structural falsification channels. A continuum model claiming to instantiate OC can be falsified if:

1. **Threshold violations** occur systematically: observed systems remain viable while violating putative Θ_{exist}, Θ_{stab}, Θ_{crit} or Θ_{death}.
2. **Cycle disappearance** contradicts viability: systems maintain long-term organisation without any identifiable cycle complex C that satisfies the OC conditions.
3. **Forbidden dimensional transitions** are observed: dimension appears to increase without new axes in the embedding space or without crossing a dimensional threshold Θ_{dim}.
4. **Inconsistent tuples** are required: successful models require mutually incompatible versions of the canonical OC state tuple for one and the same system.

Structural vs empirical failure. Empirical failure of a specific domain model does not immediately falsify OC; the structural theory is challenged only when the empirical success of alternative models requires violations of OC's structural constraints. Conversely, if a domain repeatedly exhibits structures that OC forbids (e.g. stable continua without supporting cycles), OC is directly falsified.

Cross-domain falsification logic. Because OC claims universality, a structural falsification in one domain (propely established) propagates across all domains that rely on the same axioms. This makes OC *more* falsifiable than narrow theories:

- if monotonicity of dimension fails in a controlled physical system, it must also be reconsidered for cognitive and social continua;
- if viable protocells are demonstrated without any boundary structure compatible with $\partial\Omega$, the entire K-level hierarchy is affected.

The remainder of this chapter formulates explicit predictions and tests.

6.2 Predictions and Tests in Physics (K_2)

Physical continua provide the most classical testing ground for OC at level K_2.

Structural predictions.

- **P2.1 (Critical thresholds).** Phase transitions correspond to structural crossings of Θ_{crit}. No sharp long-range ordering transition may occur without a corresponding threshold surface in Ω_2.
- **P2.2 (Percolation monotonicity).** For connectivity-based systems with control parameter p, global viability ($k_2 > 0$) requires $p \geq p_c$; for $p < p_c$ no infinite cluster compatible with Ω_2 can exist.
- **P2.3 (Dimensional monotonicity).** No physical continuum can increase its effective dimension without the activation of at least one new axis in the embedding space M_2 and crossing a dimensional threshold Θ_{dim}.
- **P2.4 (Boundary-driven collapse).** Collapse of ordered phases occurs via destruction of supporting cycles (e.g. vortex pairs, domain structures) and loss of well-defined $\partial\Omega_2$, not via spontaneous disappearance of order in the interior alone.

Falsification criteria and tests.

- **F2.1.** A robust, reproducible phase transition with no structural thresholds in the order-parameter geometry would falsify P2.1.
- **F2.2.** Observation of stable infinite clusters for $p < p_c$ across well-controlled percolation-like systems would contradict P2.2.
- **F2.3.** Demonstration of a continuum whose effective dimensionality increases (e.g. from 2D to 3D behaviour) while all axes of M_2 remain fixed and no new degrees of freedom are activated would falsify P2.3 and the general dimensional monotonicity theorem.
- **Experimental programme.** Precision experiments on coherence destruction, vortex unbinding, and percolation thresholds can be used to map empirical Θ_{crit} and test whether the associated $\partial\Omega_2$ behaves as OC predicts (critical slowing down, boundary localisation of collapse).

6.3 Predictions and Tests in Origins of Life (K_3–K_4)

For origins-of-life scenarios OC makes explicit claims about catalytic closure, membrane structure and protocell viability.

Structural predictions.

- **P3.1 (Catalytic closure).** Long-term stable chemical organisations at K_3 require at least one RAF-like cycle complex; systems without catalytic closure cannot maintain $k_3 > 0$.

- **P3.2 (Osmotic collapse threshold).** Protocells with fixed membrane composition exhibit a radius–tension– permeability relation: beyond a critical combination of osmotic gradient, curvature and permeability $(\Theta_{\text{grad}}, \Theta_{\text{perm}}, \Theta_{\text{mem}})$ collapse is inevitable.
- **P3.3 (Boundary necessity).** Protocells lacking a structurally identifiable boundary $\partial\Omega_4$ (membrane or equivalent compartment structure) cannot sustain nontrivial metabolic flows $J_{\text{metabolic}}$ over timescales defining $k_4 > 0$.
- **P3.4 (Patch-localised failure).** Membrane collapse initiates on boundary patches where local thresholds are first crossed, not uniformly across the surface.

Falsification criteria and tests.

- **F3.1.** Demonstration of chemically self-sustaining organisations with $k_3 > 0$ and no identifiable catalytic closure or cycle complex would falsify P3.1.
- **F3.2.** Systematic violation of predicted osmotic-collapse relations (e.g. arbitrary gradients sustained without structural adjustment) would contradict P3.2.
- **F3.3.** Observation of long-lived protocell-like systems with robust metabolic fluxes but no boundary structure compatible with $\partial\Omega_4$ would falsify P3.3 and weaken the role of boundaries in OC.
- **Experimental programme.** Controlled protocell leakage and bursting experiments, combined with quantitative measurements of gradients and membrane properties, can test the structural status of Θ_{grad}, Θ_{perm}, Θ_{mem} and patch dynamics.

6.4 Predictions and Tests in Biological Systems (K_4–K_5)

At the biological level, OC focuses on excitability, proto-spikes and the transition $K_4 \to K_5$.

Structural predictions.

- **P4.1 (Minimal excitability threshold).** The transition from purely metabolic protocells to excitable continua requires a minimal excitability threshold Θ_{exc}; no stable K_5 behaviour can exist without nontrivial ΔV-axes and a well-defined spike threshold.
- **P4.2 (Proto-spike invariants).** Across early excitable systems there exist minimal invariants for spike-like cycles (e.g. minimal amplitude of ΔV, refractory period τ) corresponding to structural constraints of C_{spike}.
- **P4.3 (Excitability precedes complex inheritance).** Long-term inheritance of complex regulatory structures requires prior establishment of K_5-like channels and excitability cycles; purely K_4 systems cannot indefinitely sustain inherited regulatory complexity without transitioning to a higher level.

Falsification criteria and tests.

- **F4.1.** Demonstration of genuinely K_5-like signalling (spike trains, excitable waves) in systems with no identifiable excitability axes or thresholds would falsify P4.1.
- **F4.2.** Discovery of stable, complex inheritance mechanisms in protocell systems strictly confined to K_4 (no excitability, no spike-like cycles) would challenge P4.3.
- **Experimental programme.** Minimal ion-channel sets, synthetic excitable membranes and controlled ΔV instability tests can be used to map the structural space of Θ_{exc}, Θ_{refr} and the resulting C_{spike}.

6.5 Predictions and Tests in Cognitive Continua (K_6)

For cognitive systems OC emphasises binding capacity, prediction error and cycle-based maintenance of internal models.

Structural predictions.

- **P6.1 (Binding capacity limit).** For any cognitive continuum there is a finite binding capacity threshold Θ_{bind} beyond which additional features cannot be integrated into coherent representations without collapse of k_6.
- **P6.2 (Prediction-error collapse).** There exists a prediction-error threshold Θ_{pred} beyond which no coherent internal model can be maintained; cognitive collapse occurs when prediction error cannot be reduced by any flows J_6 within current axes.
- **P6.3 (Cycle-based collapse).** Cognitive collapse always involves destruction of key cycles (attention, prediction, memory); there is no collapse scenario with $C_{\text{max}}(K_6)$ intact.

Falsification criteria and tests.

- **F6.1.** Robust evidence for unbounded binding capacity in finite systems (no observable thresholds, no degradation) would falsify P6.1.
- **F6.2.** Empirical regimes where prediction error diverges yet stable, coherent internal models persist indefinitely would contradict P6.2.
- **F6.3.** Demonstration of cognitive collapse in which attention, prediction and memory cycles remain fully intact would falsify P6.3 and the cycle-based definition of collapse.
- **Experimental programme.** Behavioural and neurophysiological studies of binding degradation under load, prediction-error saturation, and memory reconsolidation can probe the empirical counterparts of Θ_{bind}, Θ_{pred}, and C_{memory}.

6.6 Predictions and Tests in Social/Civilisational Continua (K_7–K_8)

At higher social levels OC proposes structural constraints on trust, institutional cycles and infrastructural stability.

Structural predictions.

- **P7.1 (Trust threshold).** Stable social continua at K_7 require trust above a threshold Θ_{trust}; below this threshold institutional cycles cannot maintain $k_7 > 0$.
- **P7.2 (Institutional cycle closure).** No stable K_7 continuum exists without at least one closed institutional cycle (e.g. taxation–service–legitimacy loop) compatible with OC's definition of C_7.
- **P8.1 (Infrastructure cycle thresholds).** Civilisational continua K_8 require stable infrastructural cycles; collapse at K_8 involves boundary failure in $\partial\Omega_8$ and percolation of failure across key infrastructural layers.

Falsification criteria and tests.

- **F7.1.** Historical or simulated examples of long-term stable institutions with trust consistently below any plausible Θ_{trust} would challenge P7.1.
- **F7.2.** Demonstration of durable, high-complexity societies without any identifiable closed governance or resource cycles would falsify P7.2.
- **F8.1.** Robust models of civilisational collapse that do not involve breakdown of infrastructural cycles or boundary failures in $\partial\Omega_8$ would contradict P8.1.
- **Experimental programme.** Quantitative analyses of historical data, network fragility models, and large-scale simulations of trust dynamics and infrastructure percolation can be used to test the structural role of Θ_{trust}, C_{inst} and $\partial\Omega_8$.

6.7 Theoretical and Meta-Theoretical Predictions (K_9–K_{10})

OC itself, and other high-level frameworks, can be tested at the theoretical and meta-theoretical levels.

Structural predictions.

- **P9.1 (Coherence thresholds).** Any long-lived theoretical continuum K_9 must satisfy coherence and expressivity thresholds $\Theta_{\text{coh}}, \Theta_{\text{expr}}$; persistent success of theories that explicitly violate internal coherence would contradict the OC framework.
- **P9.2 (Inconsistency cascades).** Accumulated inconsistencies in a theoretical framework must lead to collapse of its cycle complex (research programmes, paradigm loops); sustained progress in the presence of arbitrarily large incoherence is incompatible with OC.
- **P10.1 (Meta-incompleteness).** Any meta-theoretical continuum K_{10} is structurally incomplete: as bodies of models grow, structural tension at K_{10} must eventually exceed some Θ_{dim}, forcing either expansion of axes or collapse.

Falsification criteria and tests.

- **F9.1.** A fully self-consistent, closed theoretical system that captures all relevant models while never requiring expansion or revision and yet continues to support unbounded empirical progress would challenge P9.2 and P10.1.
- **F9.2.** Demonstration of theoretical frameworks that remain incoherent by OC standards but nonetheless exhibit stable, cumulative, cross-domain predictive success would falsify the structural link between coherence and k_9.
- **Experimental programme.** Formal analyses of model families, logical consistency checks, and meta-level studies of theory change can be used to test the existence and behaviour of Θ_{expr}, Θ_{coh} and associated cycles at K_9–K_{10}.

6.8 Meta-Criteria and Cross-Domain Validation

Finally, OC proposes meta-level criteria that couple falsifiability across domains and K-levels.

Predictive invariants across K-levels. Several structural relations are predicted to hold uniformly:

- existence of at least one supporting cycle C_{support} for any live continuum;
- monotonicity of dimension under birth operators;
- necessity of embedding-space compatibility $A(K_x) \subseteq A(M_x)$ and threshold compatibility with M_x;
- collapse signatures: critical slowing, boundary localisation, cycle breakdown, divergence of structural tension.

Violation of these invariants at any well-understood K-level would propagate upward and downward in the hierarchy.

Algorithmic structural tests. Core 1.2 and subsequent work are expected to develop algorithmic tests for:

- Ω-consistency (nonempty admissible region given thresholds),
- cycle viability (existence of cycle complexes C under given flows),
- embedding compatibility (axes and thresholds vs. M_x).

These tests can be applied to concrete models in physics, chemistry, biology, cognition and social sciences to check whether they are legitimate OC instantiations.

Cross-domain falsification logic. Because axioms are shared across K-levels, falsification has a *cross-domain* structure:

- if one structural axiom fails in a tightly controlled physical context, all higher-level predictions derived from it must be reconsidered;
- conversely, consistent validation of the same structural motif (e.g. boundary-driven collapse) in multiple domains increases confidence in the corresponding OC axiom.

Unified validation pipeline. Core 1.3 provides the structural basis for an explicit validation pipeline in Core 1.2:

1. select domain models and identify their continuum components $(\Omega, A, P, J, \Theta, \partial\Omega, C, k)$;
2. test local structural predictions (thresholds, cycles, boundaries);
3. test cross-level invariants (dimensional monotonicity, embedding compatibility);
4. update the mapping between domain theories and K-levels.

OC is therefore not a purely conceptual framework: it is designed to be empirically constrained and — in principle — structurally falsifiable across the full hierarchy K_0–K_{10}.

7 Theorem Roadmap and Proof Navigation

This chapter is the clearest compact map of the proof-bearing route. Its task is to show where the load-bearing theorem spine lies, which statement classes carry which scientific burden, and where a hostile reviewer should probe first. It does not replace the formal chapters or the reader guide. The reader guide owns entry routes; this chapter owns proof navigation.

7.1 Load-bearing theorem spine

At the level of the bounded journal-core claim, the present release depends on a relatively small load-bearing spine even though the monograph itself remains much larger. The schematic route below is compact rather than proof-theoretic minimality in the strict sense: it keeps the residue and rebirth branch visible because the defended article-level theorem classifies post-collapse identity, not death alone. The spine can therefore be stated schematically as dependency notation rather than object-language implication. To avoid making the route look like a proof calculus, the support map is written as labelled review edges:

1. **Edge 1.** Axiom 3.2 together with Definitions 12.1, 12.3, and 12.4 supports Source Theorem 3, and Source Theorem 3 supports Lemma 1.
2. **Edge 2.** Axiom 3.3 together with Definition 12.6 supports Source Corollary 3.2, and Source Corollary 3.2 supports Lemma 3.
3. **Edge 3.** Lemma 1 together with Definition 12.5 supports Lemma 2.
4. **Edge 4.** Lemma 2 is the sole premise recorded for Theorem A in the canonical proof-obligation registry.

Definitions in this list are prerequisite nodes, not independent theorem premises. The word "supports" marks an admitted release support step rather than a primitive implication operator of OC.

Lemma 3 remains an admitted rebirth-control obligation in the surrounding roadmap, but it is not a premise of Theorem A in the canonical proof-obligation registry. Its load-bearing role is different: it supports the companion rebirth-control statement, Proposition B. Whenever a compressed article proof discusses the death–residue–rebirth package in one paragraph, Lemma 3 should be read as supporting Proposition B, while Theorem A itself remains the death-and-residue classification routed through Lemma 2.

Proposition B (article-local rebirth control). Within the article-level extraction, after K has lawfully collapsed so that $\Omega(K) = \emptyset$ and $k(K,t) = 0$, a later object K' is classified as rebirth if and only if K' is live, is numerically distinct from the dead original continuum K, appears either in the same-embedding branch or in an explicitly witnessed related-embedding branch, inherits at least one declared residue support element, and satisfies the relevant birth conditions. The identity rule supplies the reason this non-identity condition is forced rather than optional. This proposition is a companion control statement for the rebirth branch; it is not a premise of Theorem A.

In this roadmap, the *Source* prefix marks a witness projection from the audited public LaTeX source corpus and archival compiled PDF described in the source-audit appendix. Such rows are carried into Core 1.3 as source-witnessed support nodes and are inspected through the canonical proof-obligation registry; they are not newly asserted primitive theorems of the article-level extraction. The unprefixed lemmas, Theorem A, and Proposition B are the article-level derived rows assembled from that witnessed support. Theorem A uses the death-and-residue spine; Proposition B uses the rebirth-control branch.

Axioms, source theorems and corollaries, lemmas, and Theorem A carry proof-force nodes; definitions fix the terms used by those nodes.

The point of making that chain explicit is not to pretend that the whole framework is reducible to one diagram. The point is to show that the bounded claim has an auditable positive core rather than a cloud of motivational prose. Figure 1 also marks one excluded lift-heavy temptation route as a hostile-review boundary; it is reviewed precisely so that it cannot be mistaken for admitted support in the positive chain. The Corollary 3.2 / Lemma 3 branch is shown as an admitted rebirth-control branch for Proposition B beside the theorem route rather than as an antecedent of Theorem A. In this roadmap the related-embedding branch is represented by one auditable article-local witness, not by an exhaustive list of every admissible source-level related-embedding witness. This is the same branch denoted $\mathcal{M} \sim_{\text{res}} \mathcal{M}'$ in Section **??**: a named structure-preserving map $h : \mathcal{M} \to \mathcal{M}'$, a residue element $r \in R_{\text{res}}(K)$, admissible-region membership $h(r) \in \Omega(K') \subseteq \Omega_{\text{amb}}(\mathcal{M}')$, and preservation of the threshold and admissibility predicates on the residue-bearing substructure. Section **??** and the journal core state that witness test explicitly.

7.2 What surrounds the spine

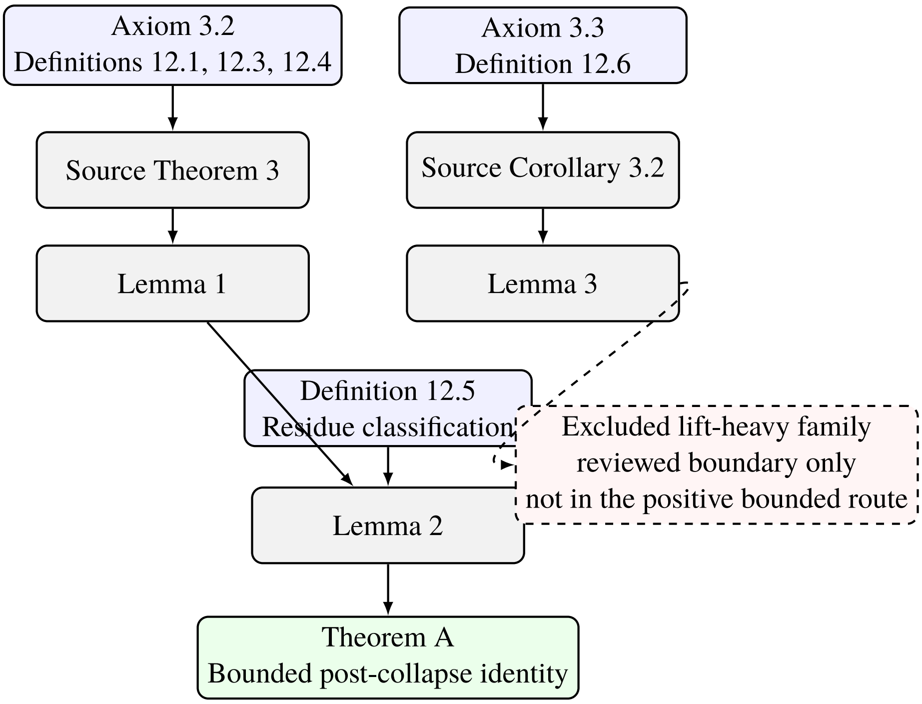
The theorem spine has a small proof dependency, but it is interpreted and audited inside the wider manuscript in five distinct ways:

1. the formal doctrine chapters fix admissibility, thresholds, operators, levels, branching, and falsifiability;
2. the consistency and source-boundary chapters fix statement class, theorem fate, and publication scope;
3. the proof-machinery chapter and appendix fix revision, comparator, compression, and proof-obligation discipline;
4. the operational and empirical chapters fix promotion, replay, and falsifier discipline;
5. the appendices keep the audit and atlas burden inside the same volume.

That is why the monograph is allowed to be larger than the defended article without becoming rhetorically loose. These surrounding chapters are publication and audit scaffolding, not additional premises of Theorem A.

7.3 Statement classes and burden

The table below is a proof-navigation map for reviewer use. It follows the statement-class taxonomy of Section **??**. The proposition row is a navigation-only subdivision of derived statements, not a new proof-status class. Operational rules remain a downstream inspection burden rather than a separate proof-status class.



Solid arrows: admitted theorem route and admitted rebirth-control branch
Dashed arrows: excluded temptation route reviewed but not admitted

Figure 1: Load-bearing theorem spine of the bounded Core 1.3 claim. Solid arrows mark admitted support. The positive Theorem A route terminates at Lemma 2; the Lemma 3 branch remains visible as admitted rebirth-control support beside that route. The dashed branch marks an excluded lift-heavy route that is reviewed as a boundary condition rather than admitted into the positive proof. The larger monograph remains necessary because the surrounding chapters fix the ontology, operator shell, K-level doctrine, collapse grammar, empirical bar, and audit layers that keep this spine scientifically interpretable and externally citable.

Statement class	Scientific burden	Where to inspect it
Axiomatic statements	Fix primitive admissibility, identity, or threshold conditions.	Axioms Appendix, Notation Appendix, and Sections 3–**??**.
Derived theorems and lemmas	Carry the proof-bearing positive body.	Sections 2, **??**, 7, Reviewer Navigation Matrix, and Proof Machinery Appendix.
Article-level propositions	Carry companion control statements that are derived from the same audited support but are not premises of Theorem A. Proposition B is the rebirth-control example.	Sections 7, **??**, and Journal Core Bridge Appendix.
Structural statements	Explain hierarchy, module layout, and relation between levels or artifacts.	Sections **??**–6, Section **??**, Section **??**, and Section **??**.
Interpretive statements	Explain meaning, nonclaims, and downstream boundaries.	Sections **??**, **??**, 10, Source Audit Appendix, and Reviewer Navigation Matrix.
Empirical statements	Require a source-owned domain packet, theorem-native trace, numerical parameter law, observable binding, pinned data route, held-out replay, comparator audit, and falsifier before promotion.	Sections **??**–8 and Appendices C–**??**.
Frontier statements	Mark lawful future work without positive release authority.	Section 10, Appendix **??**, and the frontier rows in the source-owned science corpus.

Table 2: Statement classes and their inspection burden in Core 1.3. The table does not replace the formal chapters; it simply prevents proof, interpretation, empirical promotion, and frontier language from quietly changing roles. Replay, comparator, falsifier, and escalation rules are downstream control obligations attached to those classes.

All inspection targets in Table 2 are label-checked against the English master source. If a release wrapper changes one of those labels, the table must be regenerated or patched before the roadmap is treated as a mechanically followable audit route.

7.4 Shortest hostile-review checklist

If a reviewer wants the most efficient attack sequence, the logically hardest checks remain the following:

1. verify that collapse is tied to loss of admissible realization rather than to loose metaphor;
2. verify that the irreversibility rule forbids restoration of the original continuum as numerically the same entity;
3. verify that residue and rebirth are controlled classifications rather than narrative labels;
4. verify that no excluded lift-heavy theorem family is smuggled back into the positive bounded route;
5. verify that the journal-core extraction says nothing stronger than the source-owned science corpus and canonical SPOT permit; the flagship monograph is then checked only as a derived comparison surface;
6. verify that the K-level doctrine, especially K_{11} and K_{12}, is structurally justified rather than rhetorically inflated.

If the release fails there, later prose cannot rescue it. If it survives there, a slower full reading becomes worth the effort.

Bridge to the next chapter. With the proof route mapped, the next chapter changes mode deliberately. It no longer navigates the spine abstractly; it shows how the formal grammar should be read through worked examples, counterexamples, and bounded interpretive checks.

8 Hybrid-Escalation Evidence Bar and Execution Protocols

This chapter presents the empirical execution bar of the release. It states which source authorities, held-out designs, benchmark policies, tolerances, and falsifier conditions lawfully test promoted packets, and it separates bounded empirical support from rhetorical overreach. It does not restate the theorem spine, the full proof-support ledger, or the complete dataset inventory; it tells the reader how the packets are actually challenged.

The scientific authority for that evidence bar remains the canonical science SPOT, `OC_CORE_1_3_SCIENCE_SPOT`, materialised in this repository at `releases/oc_core_1_3/editorial/OC_CORE_1_3_SCIENCE_SPOT_latest.json`. This chapter projects that authority into reader-facing form so that the execution bar, replay discipline, and promotion verdict share one lawful source boundary. The generated section below prints the exact `metadata.repo_sha` and `metadata.ts_utc` for the SPOT revision used in this build.

SPOT revision: repo SHA `0dd049c48def00df8a7d50bef07abb127a3c4fb3`; timestamp `2026-04-22T09:23:17Z`. The evidence bar is now locked to the canonical SPOT rather than to drifting summary surfaces. The mathematics anchor is replay-only: it requires deterministic exact replay rather than a data-backed benchmark route. The empirical domain lanes require held-out results that clear the declared five-sigma bar without tail-breach inflation. Residual containment by itself is not enough for promotion.

8.1 Global evidence bar

For the empirical lanes, the default evidence bar remains hybrid escalation: official or open primary data first, then institute-run measurements only if the observable family cannot be closed otherwise. The mathematics lane is replay-only and closes through deterministic exact replay rather than an out-of-sample data route. In every lane, evidence discipline sits behind theorem-native trace closure rather than replacing it.

8.2 Execution protocol matrix

Domain	Wave	Evidence bar	Held-out policy	Replay command	Replay status	Closure verdict
Mathematics	WAVE_0_ANCHOR	HYBRID_ESCALATION	DETERMINISTIC_COUNTEREXAMPLE_AND_TRACE_REPLAY	run_claim_simulations_v1.py	PASS_REPLAYABLE	PASS
Physics	WAVE_1_PHYSICS	HYBRID_ESCALATION	LEAVE_ONE_BENCHMARK_FAMILY_OUT	run_oc_core_domain_hard_closure_replay_v1.py--domain-idPHYSICS	PASS_REPLAYABLE	PASS
Chemistry	WAVE_2_CHEMISTRY	HYBRID_ESCALATION	HOLD_OUT_ONE_COMPOUND_OR_REACTION_CLASS	run_oc_core_domain_hard_closure_replay_v1.py--domain-idCHEMISTRY	PASS_REPLAYABLE	PASS
Biology	WAVE_3_BIOLOGY	HYBRID_ESCALATION	RESERVE_ONE_DATASET_FAMILY_PER_SIGNATURE_CLASS	run_oc_core_domain_hard_closure_replay_v1.py--domain-idBIOLOGY	PASS_REPLAYABLE	PASS
Systems / Civilizational projection	WAVE_4_SYSTEMS	HYBRID_ESCALATION	HELD_OUT_INTERVAL_REPLAY_WITH_TURNING_POINT_CHECKS	run_oc_core_domain_hard_closure_replay_v1.py--domain-idSYSTEMS_CIVILIZATIONAL_PROJECTION	PASS_REPLAYABLE	PASS

8.3 Method ladder

Order	Method class	Entry rule	Anti-cycle condition
1	SOURCE_EXTRACTION	Use when a claim lacks an exact theorem target, exact dependency list, or exact source text.	May not be repeated unless a new source block, new theorem family, or newly scoped parent claim is introduced.
2	FRESH_DERIVATION_FROM_OPERATORS	Use when source extraction alone does not close the theorem-native route.	May not be repeated unless a new formal premise or a new decomposition of the claim space appears.
3	REPARAMETERIZATION_OR_CLAIM_DECOMPOSITION	Use when the claim is too wide, rubberized, or underspecified to admit a lawful parameter law.	May not be repeated unless a new observable family or a new admissible regime is declared before replay.
4	NEW_DATA_OR_MEASUREMENT_ROUTE	Use only after theorem target, parameter law, and held-out policy are already fixed.	May not be repeated unless the new route adds a previously unavailable observable family or a nonoverlapping held-out family.
5	SAME_CLAIM_CLASS_ADVERSARIAL_COMPARISON	Use when the claim survives internal closure steps and must now face simpler same-claim-class competitors.	May not be repeated unless a new comparator family or a new cost-normalized metric is introduced.

8.4 Per-domain escalation doctrine

Mathematics. Protocol `OC13::EXECUTION::MATHEMATICS::ANCHOR_REPLAY` runs under evidence bar `HYBRID_ESCALATION` and currently reports readiness `PASS`. Closure phase: `PHASE_1_STABILIZE_CLOSED_CORE`. Launch rule: `MAINTAIN_ALWAYS_ON`. Pinned route institutions: Clay Mathematics Institute, American Institute of Mathematics. Held-out policy: `DETERMINISTIC_COUNTEREXAMPLE_AND_TRACE_REPLAY`, with minimum cases `31`. Quantitative gate: minimum cases 31; strictly terminalized cases at least 31; counterexample survivors capped at 0; replay divergences capped at 0. Replay harness: `logion/k7/spe/orchestrator/science/run_claim_simulations_v1.py`. Institute-run trigger: `ONLY_IF_DETERMINISTIC_REPLAY_BREAKS`. Measurement wave: `NONE`. Open gaps: none. Escalation plan: Expand the exact-question corpus rather than broadening outward claims when a replay gap appears.

Physics. Protocol `OC13::EXECUTION::PHYSICS::CONSTANTS_SPECTRA_TRANSPORT` runs under evidence bar `HYBRID_ESCALATION` and currently reports readiness `PASS`. Closure phase: `PHASE_3_PHYSICS_CLOSURE`. Launch rule: `START_IMMEDIATELY_AFTER_PHASE_1`. Pinned route institutions: NIST, U.S. Department of Energy Office of Science. Held-out policy: `LEAVE_ONE_BENCHMARK_FAMILY_OUT`, with minimum cases `30`. Quantitative gate: minimum cases 30; coverage ratio 1.0; mean absolute normalized error at or below 1.0 sigma; 95th-percentile normalized error at or below 2.5 sigma; maximum normalized error at or below 5.0 sigma; tail breaches capped at 0; critical residual threshold 2.5 sigma; severe residual threshold 1.5 sigma. Replay harness: `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idPHYSICS`. Institute-run trigger: `OPEN_DATA_COVERAGE_LT_1_0_OR_HELD_OUT_CASES_LT_30_OR_RESIDUAL_BREACH_PERSISTS`. Measurement wave: `INSTITUTE_RUN::PHYSICS::WAVE_1A`. Open gaps: none. Escalation plan: If official/open data do not cover the required observable family, queue institute-run measurement or observational partner acquisition.

Chemistry. Protocol `OC13::EXECUTION::CHEMISTRY::THERMOCHEMISTRY_SPECTRA_KINETICS` runs under evidence bar `HYBRID_ESCALATION` and currently reports readiness `PASS`. Closure phase: `PHASE_4_CHEMISTRY_CLOSURE`. Launch rule: `START_AFTER_PHYSICS_THEOREM_PACKET_LOCKS`. Pinned route institutions: NIST, U.S. Department of Energy Office of Science. Held-out policy: `HOLD_OUT_ONE_COMPOUND_OR_REACTION_CLASS`, with minimum cases `30`. Quantitative gate: minimum cases 30; coverage ratio 1.0; mean absolute normalized error at or below 1.0 sigma; 95th-percentile normalized error at or below 2.5 sigma;

maximum normalized error at or below 5.0 sigma; tail breaches capped at 0; critical residual threshold 2.5 sigma; severe residual threshold 1.5 sigma. Replay harness: `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idCHEMISTRY`. Institute-run trigger: `OPEN_DATA_COVERAGE_LT_1_0_OR_HELD_OUT_CASES_LT_30_OR_RESIDUAL_BREACH_PERSISTS`. Measurement wave: `INSTITUTE_RUN::CHEMISTRY::WAVE_2A`. Open gaps: none. Escalation plan: When official/open reference data are insufficient, queue targeted laboratory or partner-measurement acquisition.

Biology. Protocol `OC13::EXECUTION::BIOLOGY::STATE_TRANSITIONS_AND_RESPONSE_SIGNATURES` runs under evidence bar `HYBRID_ESCALATION` and currently reports readiness `PASS`. Closure phase: `PHASE_5_BIOLOGY_CLOSURE`. Launch rule: `START_AFTER_CHEMISTRY_THEOREM_PACKET_LOCKS`. Pinned route institutions: NCBI GEO. Held-out policy: `RESERVE_ONE_DATASET_FAMILY_PER_SIGNATURE_CLASS`, with minimum cases `30`. Quantitative gate: minimum cases 30; coverage ratio 1.0; mean absolute normalized error at or below 1.0 sigma; 95th-percentile normalized error at or below 2.5 sigma; maximum normalized error at or below 5.0 sigma; tail breaches capped at 0; expected calibration error at or below 0.05. Replay harness: `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idBIOLOGY`. Institute-run trigger: `OPEN_DATA_COVERAGE_LT_1_0_OR_HELD_OUT_CASES_LT_30_OR_SIGNATURE_CLASS_UNDERCONSTRAINED`. Measurement wave: `INSTITUTE_RUN::BIOLOGY::WAVE_3A`. Open gaps: none. Escalation plan: If open repositories cannot close the lane honestly, promote an institute-run observation or assay plan instead of overclaiming.

Systems / Civilizational projection. Protocol `OC13::EXECUTION::SYSTEMS::MACRO_TRAJECTORIES_AND_REGIME_SHIFTS` runs under evidence bar `HYBRID_ESCALATION` and currently reports readiness `PASS`. Closure phase: `PHASE_6_SYSTEMS_CLOSURE`. Launch rule: `ALLOW_PARALLEL_DATA_PREP_AFTER_CHEMISTRY_BUT_FINAL_PROMOTION_AFTER_BIOLOGY_FORMAL_ROUTE`. Pinned route institutions: World Bank, U.S. Department of Energy Office of Science. Held-out policy: `HELD_OUT_INTERVAL_REPLAY_WITH_TURNING_POINT_CHECKS`, with minimum cases `30`. Quantitative gate: minimum cases 30; coverage ratio 1.0; tail breaches capped at 0; Brier score at or below 0.08; expected calibration error at or below 0.05; critical-failure F1 at least 0.87; critical-failure precision at least 0.85; critical-failure recall at least 0.9. Replay harness: `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idSYSTEMS_CIVILIZATIONAL_PROJECTION`. Institute-run trigger: `OPEN_DATA_COVERAGE_LT_1_0_OR_HELD_OUT_INTERVAL_COUNT_LT_30_OR_TURNING_POINT_BREACH_PERSISTS`. Measurement wave: `INSTITUTE_RUN::SYSTEMS::WAVE_4A`. Open gaps: none. Escalation plan: If open macro-process series are insufficient, queue institute-run observation design or partner data acquisition.

Transition to proof machinery. Once the empirical bar is fixed, the next chapter closes the triad by stating the proof machinery, revision law, competition discipline, and compression rule that keep the outward manuscript synchronized to the science verdict.

9 Unified Science Synthesis

Final unified-synthesis release status: `PASS`.

This chapter states the promoted synthesis projected from the canonical science SPOT. It gives the lawfully promoted unified synthesis with explicit K0–K12 parameter laws, numerical rows, falsifiers, and empirical held-out prediction summaries. The title and scope of this chapter follow the canonical SPOT record; the opening gate paragraph states the release status, and the K-level rows then give the theorem claims, parameter laws, observable bindings, numerical subsets, falsifiers, and packet bindings needed to keep the synthesis auditable.

9.1 Final Unified Synthesis Release Gate

Exact validator: `FINAL_UNIFIED_SYNTHESIS_VALIDATOR_PASS`. The release gate now reports `0` bridge-only domains, `0` hostile-review blockers, and integrability at `PASS`. The final promotion rule is parallel: every empirical core domain must be theorem-native; the bridge-only domain total must be zero; the hostile-review blocker total must be zero; the unified atlas and integrability suite must pass; and the K0–K12 numerical rows must remain explicit. No release blocker remains in the canonical SPOT.

9.2 K0: Structural Conditions for Any Universe

This row is currently PASS and therefore contributes lawfully to the closed synthesis stack. Promotion blockers: none remain at this level.

Promoted theorem-native claim. K0 fixes the non-empty admissible meta-domain required for any lawful continuum and therefore bounds every later K-level before empirical specialization.

Operator and K-level binding. The local K0 root process glyphs Ψ_0, Φ_0, and Λ_0 fix distinction generation, relational reconfiguration, and compositional assembly; compact synthesis aliases F_0, Q_0, and U_0 label the corresponding root constraints only where they are defined locally. Core witness anchors are tabulated in Appendix D and Appendix **??**.

Parameter law.

$$\mu(\Omega(K_0)) > 0, \quad \forall x \in \{1, \ldots, 12\} : \text{DoF}(M_x) > 0 \text{ and } \frac{\text{DoF}(K_x)}{\text{DoF}(M_x)} \leq 1, \quad C_{\text{nt}} \geq 1$$

Observable map and units. Meta-admissibility, meta-space compatibility, and the presence of at least one non-trivial lawful cycle. Registered observables: K0::ADMISSIBLE_STATE_MEASURE, K0::META_SPACE_COMPATIBILITY_RATIO, K0::NONTRIVIAL_CYCLE_PRESENCE, MATHEMATICS::EPSILON_STAR, MATHEMATICS::MINIMAL_DENOMINATOR, MATHEMATICS::MINIMAL_AUGMENTATION_RANK, MATHEMATICS::COMPONENT_FIBER_COUNT, MATHEMATICS::MOVE_CONNECTIVITY_OR_DEGREE_BOUND, MATHEMATICS::STRICT_TERMINALIZED_TOTAL, MATHEMATICS::COUNTEREXAMPLE_EXECUTED_TOTAL, MATHEMATICS::PYTHON_EVIDENCE_EXECUTED_TOTAL.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Admissible-state measure floor	$\mu(\Omega(K_0))$	> 0	dimensionless lower bound
Meta-space compatibility ratio for each promoted level	$\text{DoF}(K_x)/\text{DoF}(M_x)$	≤ 1	dimensionless ratio; applies for each x in 1,...,12
Non-trivial cycle presence	C_{nt}	1	present

Falsifier and collapse boundary. Collapse occurs if the admissible set degenerates, if a later K-level exceeds its meta-space, or if the structural substrate loses its last non-trivial lawful cycle. Falsifier witness anchors are tabulated in Appendix D.

Domain packet bindings.

- Mathematics is closed as THEOREM_NATIVE with verdict PASS and 0 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are campaign status PASS_31_OF_31_TERMINALIZED; 31 terminalized cases. The closure dossier anchor is listed in Appendix D.

9.3 K1: Minimal Observable Distinctions and Energetic Axes

This row is currently PASS and therefore contributes lawfully to the closed synthesis stack. Promotion blockers: none remain at this level.

Promoted theorem-native claim. K1 introduces resolvable energetic and temporal distinctions while preserving the admissible-state floor inherited from K0.

Operator and K-level binding. F, G, and U bind contradiction load to resolvable energetic distinctions, monotonic update order, and a non-degenerate observable region. Core witness anchors are tabulated in Appendix D and Appendix **??**.

Parameter law.

$$\Delta E \geq \Theta_{\Delta E}, \quad \frac{dt}{d\tau} > 0, \quad \mu(\Omega(K_1)) > 0$$

Observable map and units. The first observable regime is tracked by energy-gap resolution, monotonic update ordering, and positive configuration-space measure. Registered observables: `K1::ENERGY_GAP_THRESHOLD`, `K1::TEMPORAL_MONOTONICITY`, `K1::CONFIGURATION_MEASURE`, `PHYSICS::FINE_STRUCTURE_CONSTANT_RESIDUAL`, `PHYSICS::HYDROGEN_BALMER_LINE_RESIDUAL`, `PHYSICS::TRANSPORT_SCALING_RESIDUAL`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix ??.

Quantity	Symbol	Value / range	Units or note
Minimal resolvable energy scale	$\Theta_{\Delta E}$	10^{-34}–10^{-33}	J
Temporal monotonicity	$dt/d\tau$	> 0	ordered update constraint
Configuration-space measure floor	$\mu(\Omega(K_1))$	> 0	dimensionless lower bound

Falsifier and collapse boundary. Collapse begins when energetic distinctions are no longer resolvable, temporal order breaks, or the first observable state space degenerates. Falsifier witness anchors are tabulated in Appendix D.

Domain packet bindings.

- Physics is closed as `THEOREM_NATIVE` with verdict `PASS` and `20` held-out cases. The theorem packet is `COMPLETE` and the parameter law is `COMPLETE`. The key replay metrics are 30 covered cases; 20 held-out cases; mean absolute normalized error 0.188136 sigma; 95th-percentile normalized error 0.64778 sigma; maximum normalized error 1.48 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix D.

9.4 K2: Fields, Phases, and Large-Scale Structure

This row is currently `PASS` and therefore contributes lawfully to the closed synthesis stack. Promotion blockers: none remain at this level.

Promoted theorem-native claim. K2 binds field configurations, expansion history, and phase thresholds into the first large-scale physical continuum admissible under the kernel operators.

Operator and K-level binding. F, H, and R constrain field admissibility, phase transitions, and large-scale stabilization margins inside the physical continuum. Core witness anchors are tabulated in Appendix D and Appendix ??.

Parameter law.

$$H_0 > 0, \quad 0 < \Omega_{\rm m} < 1, \quad |\Omega_k| < 2 \times 10^{-3}, \quad \mathcal{T}_{\rm phase}(t) \in \{T_c^{\rm EW}, T_c^{\rm QCD}\}$$

Observable map and units. Expansion rate, curvature, cosmological density fractions, and critical phase windows tied to physical observables and replay packets. Registered observables: `K2::EXPANSION_RATE`, `K2::CURVATURE_BOUND`, `K2::PHASE_TRANSITION_WINDOW`, `CHEMISTRY::GAS_PHASE_ENTHALPY_RESIDUAL`, `CHEMISTRY::SPECTRAL_TRANSITION_RESIDUAL`, `CHEMISTRY::KINETIC_OR_EQUILIBRIUM_RESIDUAL`, `PHYSICS::FINE_STRUCTURE_CONSTANT_RESIDUAL`, `PHYSICS::HYDROGEN_BALMER_LINE_RESIDUAL`, `PHYSICS::TRANSPORT_SCALING_RESIDUAL`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Hubble parameter today	H_0	67.4 ± 0.5	$\mathrm{km\,s^{-1}\,Mpc^{-1}}$
Matter density fraction	Ω_m	0.315 ± 0.007	dimensionless
Critical phase windows	$T_c^{\mathrm{EW}}, T_c^{\mathrm{QCD}}$	$\sim$ 100 GeV; 150–170 MeV	closed phase set
Curvature bound	Ω_k	$\lvert\Omega_k\rvert < 2 \times 10^{-3}$	dimensionless

Falsifier and collapse boundary. K2 collapses when curvature exits the admissible band, phase windows fail to close lawfully, or large-scale physical observables lose replayable stabilization. Falsifier witness anchors are tabulated in Appendix D.

Domain packet bindings.

- Chemistry is closed as `THEOREM_NATIVE` with verdict `PASS` and `19` held-out cases. The theorem packet is `COMPLETE` and the parameter law is `COMPLETE`. The key replay metrics are 30 covered cases; 19 held-out cases; mean absolute normalized error 0.533285 sigma; 95th-percentile normalized error 1.0 sigma; maximum normalized error 1.597772 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix D.
- Physics is closed as `THEOREM_NATIVE` with verdict `PASS` and `20` held-out cases. The theorem packet is `COMPLETE` and the parameter law is `COMPLETE`. The key replay metrics are 30 covered cases; 20 held-out cases; mean absolute normalized error 0.188136 sigma; 95th-percentile normalized error 0.64778 sigma; maximum normalized error 1.48 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix D.

The SPOT also tracks `1` excluded or non-promoted side binding for this level outside the closed synthesis summary: `DRT`. The corresponding source rows remain in the canonical surfaces and support appendices, not in the promoted synthesis stack.

9.5 K3: Molecular Organization and Chemical Bonding

This row is currently `PASS` and therefore contributes lawfully to the closed synthesis stack. Promotion blockers: none remain at this level.

Promoted theorem-native claim. K3 packages bounded molecular organization through lawful bond-length, bond-energy, and activation-energy envelopes derived from lower-level admissibility.

Operator and K-level binding. G, H, and R bind composition-sensitive bonding, activation barriers, and molecular-scale stabilization windows. Core witness anchors are tabulated in Appendix D and Appendix **??**.

Parameter law.

$$r_{\mathrm{cov}} \in [0.1, 0.2]\,\mathrm{nm}, \quad E_{\mathrm{bond}} \in [1, 5]\,\mathrm{eV}, \quad E^{\ddagger} \in [0.1, 1]\,\mathrm{eV}$$

Observable map and units. Bond lengths, dissociation energies, dielectric context, and activation barriers for bounded chemistry lanes. Registered observables: `K3::BOND_LENGTH`, `K3::BOND_ENERGY`, `K3::ACTIVATION_ENERGY`, `CHEMISTRY::GAS_PHASE_ENTHALPY_RESIDUAL`, `CHEMISTRY::SPECTRAL_TRANSITION_RESIDUAL`, `CHEMISTRY::KINETIC_OR_EQUILIBRIUM_RESIDUAL`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Bond length (covalent)	r_{cov}	0.1–0.2	nm
Bond dissociation energy	E_{bond}	1–5	eV
Reaction activation energy	$E^{\ddagger}$	0.1–1	eV

Falsifier and collapse boundary. K3 collapses when bond- and barrier-level regularities cannot be bounded by a non-degenerate parameter law or when chemical observables drift outside admissible molecular windows. Falsifier witness anchors are tabulated in Appendix D.

Domain packet bindings.

- Chemistry is closed as `THEOREM_NATIVE` with verdict `PASS` and `19` held-out cases. The theorem packet is `COMPLETE` and the parameter law is `COMPLETE`. The key replay metrics are 30 covered cases; 19 held-out cases; mean absolute normalized error 0.533285 sigma; 95th-percentile normalized error 1.0 sigma; maximum normalized error 1.597772 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix D.

9.6 K4: Compartmental and Membrane Thresholds

This row is currently `PASS` and therefore contributes lawfully to the closed synthesis stack. Promotion blockers: none remain at this level.

Promoted theorem-native claim. K4 closes the first compartmental organization packet by binding membrane thickness, bending stability, and gradient maintenance to lawful biological thresholds.

Operator and K-level binding. H, R, and S couple membrane stability, osmotic admissibility, and retained gradients into a bounded protocellular regime. Core witness anchors are tabulated in Appendix D and Appendix **??**.

Parameter law.

$$d_{\text{mem}} \in [3,5]\,\text{nm}, \quad \kappa \in [10,25]\,k_{\text{B}}T, \quad \Delta pH \in [0.5,2]$$
$$\gamma_{\text{edge}} \in [5,20]\,\text{pN}, \quad \Pi \in [10^5,10^6]\,\text{Pa}, \quad P_i \in [10^{-12},10^{-8}]\,\text{m/s}$$

Observable map and units. Membrane thickness, bending modulus, osmotic pressure, proton gradient, and permeability lanes for compartmental organization. Registered observables: `K4::MEMBRANE_THICKNESS`, `K4::BENDING_MODULUS`, `K4::PROTON_GRADIENT`, `K4::LINE_TENSION`, `K4::OSMOTIC_PRESSURE_SCALE`, `K4::PERMEABILITY_RANGE`, `BIOLOGY::EXPRESSION_SIGNATURE_RESIDUAL`, `BIOLOGY::STATE_TRANSITION_ORDER_ERROR`, `BIOLOGY::RESPONSE_ONSET_RESIDUAL`, `CHEMISTRY::GAS_PHASE_ENTHALPY_RESIDUAL`, `CHEMISTRY::SPECTRAL_TRANSITION_RESIDUAL`, `CHEMISTRY::KINETIC_OR_EQUILIBRIUM_RESIDUAL`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Membrane thickness	d_{mem}	3–5	nm
Membrane bending modulus	κ	10–25	$k_{\text{B}}T$
Typical proton gradient	ΔpH	0.5–2	pH units
Line tension (edge)	γ_{edge}	5–20	pN
Osmotic pressure scale	Π	10^5–10^6	Pa
Permeability range	P_i	10^{-12}–10^{-8}	m/s

Falsifier and collapse boundary. K4 collapses if compartments cannot retain a lawful gradient, if membranes lose structural stability, or if permeability escapes the admissible packet. Falsifier witness anchors are tabulated in Appendix D.

Domain packet bindings.

- Biology is closed as `THEOREM_NATIVE` with verdict `PASS` and `16` held-out cases. The theorem packet is `COMPLETE` and the parameter law is `COMPLETE`. The key replay metrics are 30 covered cases; 16 held-out cases; mean absolute normalized error 0.444222 sigma; 95th-percentile normalized error 1.030033 sigma; maximum normalized error 1.438582 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix D.

- Chemistry is closed as THEOREM_NATIVE with verdict PASS and 19 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are 30 covered cases; 19 held-out cases; mean absolute normalized error 0.533285 sigma; 95th-percentile normalized error 1.0 sigma; maximum normalized error 1.597772 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix D.

9.7 K5: Excitable and Early Neural Continua

This row is currently PASS and therefore contributes lawfully to the closed synthesis stack. Promotion blockers: none remain at this level.

Promoted theorem-native claim. K5 binds excitability, refractory timing, and membrane potential maintenance into the first lawful neural packet.

Operator and K-level binding. F, G, and S stabilize excitable-state transitions, ion-channel gating, and refractory boundaries in bounded neural lanes. Core witness anchors are tabulated in Appendix D and Appendix **??**.

Parameter law.

$$C_m = 1\,\mu\mathrm{F/cm}^2, \quad V_{\text{rest}} \in [-75, -60]\,\mathrm{mV}, \quad \tau_{\text{ref}} \in [2, 5]\,\mathrm{ms}$$

Observable map and units. Membrane capacitance, resting potential, channel conductance, and refractory timing for bounded excitable systems. Registered observables: K5::MEMBRANE_CAPACITANCE, K5::RESTING_POTENTIAL, K5::REFRACTORY_TIME, BIOLOGY::EXPRESSION_SIGNATURE_RESIDUAL, BIOLOGY::STATE_TRANSITION_ORDER_ERROR, BIOLOGY::RESPONSE_ONSET_RESIDUAL.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Membrane capacitance	C_m	1	μF/cm^2
Resting membrane potential	V_{rest}	-75–-60	mV
Refractory time	τ_{ref}	2–5	ms

Falsifier and collapse boundary. K5 collapses when excitability loses bounded thresholds, refractory timing fails, or the neural packet cannot survive counterexample pressure across datasets. Falsifier witness anchors are tabulated in Appendix D.

Domain packet bindings.

- Biology is closed as THEOREM_NATIVE with verdict PASS and 16 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are 30 covered cases; 16 held-out cases; mean absolute normalized error 0.444222 sigma; 95th-percentile normalized error 1.030033 sigma; maximum normalized error 1.438582 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix D.

9.8 K6: Cognitive Prediction and Representation

This row is currently PASS and therefore contributes lawfully to the closed synthesis stack. Promotion blockers: none remain at this level.

Promoted theorem-native claim. K6 formalizes bounded cognitive prediction and representation by linking prediction-error thresholds, working-memory span, and adaptation strength.

Operator and K-level binding. Q, R, and S bind representation stability, prediction thresholds, and adaptive update constraints in bounded cognitive lanes. Core witness anchors are tabulated in Appendix D and Appendix **??**.

Parameter law.

$$\Theta_{\text{pred}} \in [0.05, 0.15], \quad M_{\text{WM}} \in [3, 7], \quad \eta \in [10^{-3}, 10^{-1}]$$

Observable map and units. Prediction error, working-memory span, and adaptation strength for bounded cognitive organization. Registered observables: `K6::PREDICTION_ERROR_THRESHOLD`, `K6::WORKING_MEMORY_SPAN`, `K6::HEBBIAN_STRENGTH`, `BIOLOGY::EXPRESSION_SIGNATURE_RESIDUAL`, `BIOLOGY::STATE_TRANSITION_ORDER_ERROR`, `BIOLOGY::RESPONSE_ONSET_RESIDUAL`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Prediction-error threshold	Θ_{pred}	0.05–0.15	normalized units
Working-memory span	M_{WM}	3–7	bound items
Hebbian strength coefficient	η	10^{-3}–10^{-1}	dimensionless

Falsifier and collapse boundary. K6 collapses when predictive stabilization fails across bounded cognitive observables or when representation thresholds require hidden freedom. Falsifier witness anchors are tabulated in Appendix D.

Domain packet bindings.

- Biology is closed as `THEOREM_NATIVE` with verdict `PASS` and `16` held-out cases. The theorem packet is `COMPLETE` and the parameter law is `COMPLETE`. The key replay metrics are 30 covered cases; 16 held-out cases; mean absolute normalized error 0.444222 sigma; 95th-percentile normalized error 1.030033 sigma; maximum normalized error 1.438582 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix D.

The SPOT also tracks `1` excluded or non-promoted side binding for this level outside the closed synthesis summary: `DRT`. The corresponding source rows remain in the canonical surfaces and support appendices, not in the promoted synthesis stack.

9.9 K7: Social Coordination and Group Stability

This row is currently `PASS` and therefore contributes lawfully to the closed synthesis stack. Promotion blockers: none remain at this level.

Promoted theorem-native claim. K7 closes bounded social coordination through group size, communication bandwidth, and coordination thresholds.

Operator and K-level binding. R, S, and U bind coordination thresholds, communication throughput, and conflict costs into admissible social trajectories. Core witness anchors are tabulated in Appendix D and Appendix **??**.

Parameter law.

$$N_s \in [120, 180], \quad J_{\text{comm}} \in [1, 5]\,\text{bits/s}, \quad \Theta_{\text{coord}} \in [0.2, 0.4]$$
$$C_{\text{conf}} \in [0.1, 1]$$

Observable map and units. Group stability size, communication throughput, coordination threshold, and conflict cost inside social coordination continua. Registered observables: `K7::GROUP_STABILITY_SIZE`, `K7::COMMUNICATION_BANDWIDTH`, `K7::COORDINATION_THRESHOLD`, `K7::CONFLICT_COST_SCALE`, `BIOLOGY::EXPRESSION_SIGNATURE_RESIDUAL`, `BIOLOGY::STATE_TRANSITION_ORDER_ERROR`, `BIOLOGY::RESPONSE_ONSET_RESIDUAL`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Group stability size	N_s	120–180	agents
Information-flow bandwidth per agent	J_{comm}	1–5	bits/s

Quantity	Symbol	Value / range	Units or note
Coordination threshold	Θ_{coord}	0.2–0.4	dimensionless
Conflict cost scale	C_{conf}	0.1–1	arbitrary units

Falsifier and collapse boundary. K7 collapses when communication throughput and coordination thresholds cannot stabilize social group trajectories without hidden rescue assumptions. Falsifier witness anchors are tabulated in Appendix D.

Domain packet bindings.

- Biology is closed as `THEOREM_NATIVE` with verdict `PASS` and `16` held-out cases. The theorem packet is `COMPLETE` and the parameter law is `COMPLETE`. The key replay metrics are 30 covered cases; 16 held-out cases; mean absolute normalized error 0.444222 sigma; 95th-percentile normalized error 1.030033 sigma; maximum normalized error 1.438582 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix D.

9.10 K8: Civilizational Flows and Infrastructural Stability

This row is currently `PASS` and therefore contributes lawfully to the closed synthesis stack. Promotion blockers: none remain at this level.

Promoted theorem-native claim. K8 binds civilizational energy density, renewal timing, and repair-decay balance into bounded macro-process trajectories and regime-shift packets.

Operator and K-level binding. H, R, and U bind throughput, repair-decay balance, and regime-shift boundaries in bounded systems-level observables. Core witness anchors are tabulated in Appendix D and Appendix **??**.

Parameter law.

$$E_{\text{dens}} \in [10^2, 10^3]\,\mathrm{W/m^2}, \quad \tau_{\text{infra}} \in [20, 50]\,\text{years}, \quad R_{\text{rep}} \in [0.8, 1.2]$$
$$\Theta_I \in [0.1, 0.3]$$

Observable map and units. Energy density, infrastructure renewal, information coherence, and repair-decay balance in bounded civilizational lanes. Registered observables: `K8::ENERGY_CONSUMPTION_DENSITY`, `K8::INFRASTRUCTURE_RENEWAL_TIME`, `K8::REPAIR_TO_DECAY_RATIO`, `K8::INFORMATION_COHERENCE_THRESHOLD`, `SYSTEMS_CIVILIZATIONAL_PROJECTION::POPULATION_TRAJECTORY_RESIDUAL`, `SYSTEMS_CIVILIZATIONAL_PROJECTION::GDP_GROWTH_TURNING_POINT_ERROR`, `SYSTEMS_CIVILIZATIONAL_PROJECTION::PROCESS_SYSTEM_REGIME_SCORE`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Energy consumption density	E_{dens}	10^2–10^3	$\mathrm{W/m^2}$
Infrastructure renewal time	τ_{infra}	20–50	years
Information-coherence threshold	Θ_I	0.1–0.3	dimensionless
Repair-to-decay ratio	R_{rep}	0.8–1.2	dimensionless

Falsifier and collapse boundary. K8 collapses when macro-trajectories lose bounded turning-point fidelity, when repair-decay balance drifts outside its band, or when regime-shift packets require discretionary knobs. Falsifier witness anchors are tabulated in Appendix D.

Domain packet bindings.

- Systems / Civilizational projection is closed as THEOREM_NATIVE with verdict PASS and 18 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are 30 covered cases; 18 held-out cases; mean absolute normalized error 0.56097 sigma; 95th-percentile normalized error 1.924704 sigma; maximum normalized error 2.25395 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix D.

9.11 K9: Knowledge Systems and Scientific Memory

This row is currently PASS and therefore contributes lawfully to the closed synthesis stack. Promotion blockers: none remain at this level.

Promoted theorem-native claim. K9 formalizes bounded scientific-knowledge continua through anomaly accumulation, paradigm coherence, and model-validation bandwidth.

Operator and K-level binding. Q, R, and U bind anomaly handling, paradigm coherence, and memory retention inside knowledge-system packets. Core witness anchors are tabulated in Appendix D and Appendix **??**.

Parameter law.

$$\Theta_{\text{par}} \in [0.15, 0.25], \quad J_{\text{anom}} \in [10^{-4}, 10^{-2}], \quad \tau_{\text{mem}} \in [30, 200] \text{ years}$$
$$B_{\text{val}} \in [0.1, 1]$$

Observable map and units. Paradigm coherence, anomaly accumulation, validation bandwidth, and scientific-memory half-life. Registered observables: K9::PARADIGM_COHERENCE_THRESHOLD, K9::ANOMALY_ACCUMULATION_RATE, K9::SCIENTIFIC_MEMORY_HALF_LIFE, K9::MODEL_VALIDATION_BANDWIDTH, MATHEMATICS::EPSILON_STAR, MATHEMATICS::MINIMAL_DENOMINATOR, MATHEMATICS::MINIMAL_AUGMENTATION_RANK, MATHEMATICS::COMPONENT_FIBER_COUNT, MATHEMATICS::MOVE_CONNECTIVITY_OR_DEGREE_BOUND, MATHEMATICS::STRICT_TERMINALIZED_TOTAL, MATHEMATICS::COUNTEREXAMPLE_EXECUTED_TOTAL, MATHEMATICS::PYTHON_EVIDENCE_EXECUTED_TOTAL, PHYSICS::FINE_STRUCTURE_CONSTANT_RESIDUAL, PHYSICS::HYDROGEN_BALMER_LINE_RESIDUAL, PHYSICS::TRANSPORT_SCALING_RESIDUAL.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Paradigm-coherence threshold	Θ_{par}	0.15–0.25	dimensionless
Anomaly accumulation rate	J_{anom}	10^{-4}–10^{-2}	per cycle
Scientific-memory half-life	τ_{mem}	30–200	years
Model-validation bandwidth	B_{val}	0.1–1	normalized

Falsifier and collapse boundary. K9 collapses if anomaly load outruns lawful model revision or if scientific memory and validation bandwidth fail to stabilize the knowledge packet. Falsifier witness anchors are tabulated in Appendix D.

Domain packet bindings.

- Mathematics is closed as THEOREM_NATIVE with verdict PASS and 0 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are campaign status PASS_31_OF_31_TERMINALIZED; 31 terminalized cases. The closure dossier anchor is listed in Appendix D.
- Physics is closed as THEOREM_NATIVE with verdict PASS and 20 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are 30 covered cases; 20 held-out cases; mean absolute normalized error 0.188136 sigma; 95th-percentile normalized error 0.64778 sigma; maximum normalized error 1.48 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix D.

9.12 K10: Formal Systems and Recursion

This row is currently PASS and therefore contributes lawfully to the closed synthesis stack. Promotion blockers: none remain at this level.

Promoted theorem-native claim. K10 binds bounded recursion, consistency thresholds, and proof-flow capacity into the formal-system lane of the unified-science closure stack.

Operator and K-level binding. F, Q, and U bind recursion depth, proof throughput, and consistency thresholds in formal continua. Core witness anchors are tabulated in Appendix D and Appendix **??**.

Parameter law.

$$d_{\mathrm{rec}} \in [10, 10^4], \quad \Theta_{\mathrm{cons}} \in [0.01, 0.05], \quad J_{\mathrm{proof}} \in [1, 10^3]$$

Observable map and units. Recursion depth, consistency threshold, and proof-flow capacity in bounded formal systems. Registered observables: `K10::RECURSION_DEPTH`, `K10::CONSISTENCY_THRESHOLD`, `K10::PROOF_FLOW_CAPACITY`, `MATHEMATICS::EPSILON_STAR`, `MATHEMATICS::MINIMAL_DENOMINATOR`, `MATHEMATICS::MINIMAL_AUGMENTATION_RANK`, `MATHEMATICS::COMPONENT_FIBER_COUNT`, `MATHEMATICS::MOVE_CONNECTIVITY_OR_DEGREE_BOUND`, `MATHEMATICS::STRICT_TERMINALIZED_TOTAL`, `MATHEMATICS::COUNTEREXAMPLE_EXECUTED_TOTAL`, `MATHEMATICS::PYTHON_EVIDENCE_EXECUTED_TOTAL`, `PHYSICS::FINE_STRUCTURE_CONSTANT_RESIDUAL`, `PHYSICS::HYDROGEN_BALMER_LINE_RESIDUAL`, `PHYSICS::TRANSPORT_SCALING_RESIDUAL`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Recursion depth (effective)	d_{rec}	10–10^4	steps
Consistency threshold	Θ_{cons}	0.01–0.05	dimensionless
Proof-flow capacity	J_{proof}	1–10^3	steps/s

Falsifier and collapse boundary. K10 collapses if formal recursion and proof throughput cannot remain bounded without violating consistency thresholds. Falsifier witness anchors are tabulated in Appendix D.

Domain packet bindings.

- Mathematics is closed as `THEOREM_NATIVE` with verdict `PASS` and `0` held-out cases. The theorem packet is `COMPLETE` and the parameter law is `COMPLETE`. The key replay metrics are campaign status `PASS_31_OF_31_TERMINALIZED`; 31 terminalized cases. The closure dossier anchor is listed in Appendix D.
- Metaontology is closed as `FRAME_ONLY` with verdict `PASS` and `0` held-out cases. This is a frame-only support row, so packet completeness is recorded as a support-surface status rather than as a theorem-native empirical closure claim. No quantitative replay metrics are required for this row. The closure dossier anchor is listed in Appendix D.
- Physics is closed as `THEOREM_NATIVE` with verdict `PASS` and `20` held-out cases. The theorem packet is `COMPLETE` and the parameter law is `COMPLETE`. The key replay metrics are 30 covered cases; 20 held-out cases; mean absolute normalized error 0.188136 sigma; 95th-percentile normalized error 0.64778 sigma; maximum normalized error 1.48 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix D.

9.13 K11: Meta-Theoretical Reflexivity

This row is currently PASS and therefore contributes lawfully to the closed synthesis stack. Promotion blockers: none remain at this level.

Promoted theorem-native claim. K11 captures meta-theoretical reflexivity as irreducible scientific work beyond K10 across the closed domain atlas.

Operator and K-level binding. Q, R, and S bind meta-coherence, reflexive depth, and cross-landscape coupling under the requirement that no K10-only translation preserves all declared upper-level burdens. Core witness anchors are tabulated in Appendix D and Appendix **??**.

Parameter law.

$$\Theta_{\text{meta}} \in [0.02, 0.1], \quad d_{\text{refl}} \in [3, 50], \quad J_X \in [0.1, 1]$$
$$P_{\text{funct}} \in [10^0, 10^3]$$

Observable map and units. Meta-coherence, reflexive depth, and cross-landscape coupling in the meta-theoretical packet. Registered observables: `K11::META_COHERENCE_THRESHOLD`, `K11::REFLEXIVE_DEPTH`, `K11::CROSS_LANDSCAPE_COUPLING`, `K11::FUNCTORIAL_POTENTIAL`, `MATHEMATICS::EPSILON_STAR`, `MATHEMATICS::MINIMAL_DENOMINATOR`, `MATHEMATICS::MINIMAL_AUGMENTATION_RANK`, `MATHEMATICS::COMPONENT_FIBER_COUNT`, `MATHEMATICS::MOVE_CONNECTIVITY_OR_DEGREE_BOUND`, `MATHEMATICS::STRICT_TERMINALIZED_TOTAL`, `MATHEMATICS::COUNTEREXAMPLE_EXECUTED_TOTAL`, `MATHEMATICS::PYTHON_EVIDENCE_EXECUTED_TOTAL`, `PHYSICS::FINE_STRUCTURE_CONSTANT_RESIDUAL`, `PHYSICS::HYDROGEN_BALMER_LINE_RESIDUAL`, `PHYSICS::TRANSPORT_SCALING_RESIDUAL`, `CHEMISTRY::GAS_PHASE_ENTHALPY_RESIDUAL`, `CHEMISTRY::SPECTRAL_TRANSITION_RESIDUAL`, `CHEMISTRY::KINETIC_OR_EQUILIBRIUM_RESIDUAL`, `BIOLOGY::EXPRESSION_SIGNATURE_RESIDUAL`, `BIOLOGY::STATE_TRANSITION_ORDER_ERROR`, `BIOLOGY::RESPONSE_ONSET_RESIDUAL`, `SYSTEMS_CIVILIZATIONAL_PROJECTION::POPULATION_TRAJECTORY_RESIDUAL`, `SYSTEMS_CIVILIZATIONAL_PROJECTION::GDP_GROWTH_TURNING_POINT_ERROR`, `SYSTEMS_CIVILIZATIONAL_PROJECTION::PROCESS_SYSTEM_REGIME_SCORE`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Meta-coherence threshold	Θ_{meta}	0.02–0.1	dimensionless
Functorial potential	P_{funct}	10^0–10^3	abstract units
Reflexive depth	d_{refl}	3–50	levels
Cross-landscape coupling	J_X	0.1–1	dimensionless

Falsifier and collapse boundary. K11 collapses if an explicit K10-only reduction preserves the declared meta-theoretical burdens or if meta-coherence depends on an excluded lift route. Falsifier witness anchors are tabulated in Appendix D.

Domain packet bindings.

- Mathematics is closed as `THEOREM_NATIVE` with verdict `PASS` and `0` held-out cases. The theorem packet is `COMPLETE` and the parameter law is `COMPLETE`. The key replay metrics are campaign status `PASS_31_OF_31_TERMINALIZED`; 31 terminalized cases. The closure dossier anchor is listed in Appendix D.
- Physics is closed as `THEOREM_NATIVE` with verdict `PASS` and `20` held-out cases. The theorem packet is `COMPLETE` and the parameter law is `COMPLETE`. The key replay metrics are 30 covered cases; 20 held-out cases; mean absolute normalized error 0.188136 sigma; 95th-percentile normalized error 0.64778 sigma; maximum normalized error 1.48 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix D.
- Chemistry is closed as `THEOREM_NATIVE` with verdict `PASS` and `19` held-out cases. The theorem packet is `COMPLETE` and the parameter law is `COMPLETE`. The key replay metrics are 30 covered cases; 19 held-out cases; mean absolute normalized error 0.533285 sigma; 95th-percentile normalized error 1.0 sigma; maximum normalized error 1.597772 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix D.
- Biology is closed as `THEOREM_NATIVE` with verdict `PASS` and `16` held-out cases. The theorem packet is `COMPLETE` and the parameter law is `COMPLETE`. The key replay metrics are 30 covered cases; 16 held-out cases; mean absolute normalized error 0.444222 sigma; 95th-percentile normalized error 1.030033 sigma; maximum normalized error 1.438582 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix D.
- Systems / Civilizational projection is closed as `THEOREM_NATIVE` with verdict `PASS` and `18` held-out cases. The theorem packet is `COMPLETE` and the parameter law is `COMPLETE`. The key replay metrics are 30 covered cases; 18 held-out cases; mean absolute normalized error 0.56097 sigma; 95th-percentile normalized error

1.924704 sigma; maximum normalized error 2.25395 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix D.

- Metaontology is closed as FRAME_ONLY with verdict PASS and 0 held-out cases. This is a frame-only support row, so packet completeness is recorded as a support-surface status rather than as a theorem-native empirical closure claim. No quantitative replay metrics are required for this row. The closure dossier anchor is listed in Appendix D.

9.14 K12: Global Semantic Coherence

This row is currently PASS and therefore contributes lawfully to the closed synthesis stack. Promotion blockers: none remain at this level.

Promoted theorem-native claim. K12 packages global semantic coherence across the closed domain atlas because cross-domain meaning remains jointly interpretable and no surviving global falsifier breaks the unified synthesis.

Operator and K-level binding. Q, R, S, and U bind semantic coherence, global compatibility, and reference stability into the final unified-science synthesis layer. Core witness anchors are tabulated in Appendix D and Appendix **??**.

Parameter law.

$$C_{\text{uni}} \in [10^3, 10^{12}], \quad \Theta_{\text{uni}} \in [10^{-3}, 10^{-2}], \quad R_{\text{uni}} \in [0.1, 1]$$
$$T_{\text{uni}} \in [10^2, 10^8]$$

Observable map and units. Universal integration capacity, cross-continuum compatibility, and structural reachability. Registered observables: K12::UNIVERSAL_INTEGRATION_CAPACITY, K12::CROSS_CONTINUUM_COMPATIBILITY, K12::STRUCTURAL_REACHABILITY, K12::GLOBAL_TENSION_BUDGET, MATHEMATICS::EPSILON_STAR, MATHEMATICS::MINIMAL_DENOMINATOR, MATHEMATICS::MINIMAL_AUGMENTATION_RANK, MATHEMATICS::COMPONENT_FIBER_COUNT, MATHEMATICS::MOVE_CONNECTIVITY_OR_DEGREE_BOUND, MATHEMATICS::STRICT_TERMINALIZED_TOTAL, MATHEMATICS::COUNTEREXAMPLE_EXECUTED_TOTAL, MATHEMATICS::PYTHON_EVIDENCE_EXECUTED_TOTAL, PHYSICS::FINE_STRUCTURE_CONSTANT_RESIDUAL, PHYSICS::HYDROGEN_BALMER_LINE_RESIDUAL, PHYSICS::TRANSPORT_SCALING_RESIDUAL, CHEMISTRY::GAS_PHASE_ENTHALPY_RESIDUAL, CHEMISTRY::SPECTRAL_TRANSITION_RESIDUAL, CHEMISTRY::KINETIC_OR_EQUILIBRIUM_RESIDUAL, BIOLOGY::EXPRESSION_SIGNATURE_RESIDUAL, BIOLOGY::STATE_TRANSITION_ORDER_ERROR, BIOLOGY::RESPONSE_ONSET_RESIDUAL, SYSTEMS_CIVILIZATIONAL_PROJECTION::POPULATION_TRAJECTORY_RESIDUAL, SYSTEMS_CIVILIZATIONAL_PROJECTION::GDP_GROWTH_TURNING_POINT_ERROR, SYSTEMS_CIVILIZATIONAL_PROJECTION::PROCESS_SYSTEM_REGIME_SCORE.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Universal integration capacity	C_{uni}	10^3–10^{12}	dimensionless
Cross-continuum compatibility	Θ_{uni}	10^{-3}–10^{-2}	dimensionless
Structural reachability	R_{uni}	0.1–1	dimensionless
Global tension budget	T_{uni}	10^2–10^8	dimensionless

Falsifier and collapse boundary. K12 collapses if cross-domain compatibility fragments, if the unified atlas cannot pass integrability, or if a surviving global falsifier breaks semantic coherence. Falsifier witness anchors are tabulated in Appendix D.

Domain packet bindings.

- Mathematics is closed as THEOREM_NATIVE with verdict PASS and 0 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are campaign status PASS_31_OF_31_TERMINALIZED; 31 terminalized cases. The closure dossier anchor is listed in Appendix D.
- Physics is closed as THEOREM_NATIVE with verdict PASS and 20 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are 30 covered cases; 20 held-out cases; mean absolute normalized error 0.188136 sigma; 95th-percentile normalized error 0.64778 sigma; maximum normalized error 1.48 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix D.
- Chemistry is closed as THEOREM_NATIVE with verdict PASS and 19 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are 30 covered cases; 19 held-out cases; mean absolute normalized error 0.533285 sigma; 95th-percentile normalized error 1.0 sigma; maximum normalized error 1.597772 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix D.
- Biology is closed as THEOREM_NATIVE with verdict PASS and 16 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are 30 covered cases; 16 held-out cases; mean absolute normalized error 0.444222 sigma; 95th-percentile normalized error 1.030033 sigma; maximum normalized error 1.438582 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix D.
- Systems / Civilizational projection is closed as THEOREM_NATIVE with verdict PASS and 18 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are 30 covered cases; 18 held-out cases; mean absolute normalized error 0.56097 sigma; 95th-percentile normalized error 1.924704 sigma; maximum normalized error 2.25395 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix D.
- Metaontology is closed as FRAME_ONLY with verdict PASS and 0 held-out cases. This is a frame-only support row, so packet completeness is recorded as a support-surface status rather than as a theorem-native empirical closure claim. No quantitative replay metrics are required for this row. The closure dossier anchor is listed in Appendix D.

9.15 Empirical Held-Out Prediction Summary

The following summary records the empirical synthesis lanes that have already cleared theorem-native promotion and held-out prediction review. Where a promoted lane carries severe cases or false negatives, those adverse metrics are reported explicitly below. The pass/fail policy is the validation matrix in Appendix C; Appendix **??** and the canonical synthesis surface retain the full numerical record. Metric labels are printed with their canonical surface names; critical_failure_recall is the promoted systems recall metric used by the synthesis surface.

Domain	Class	Verdict	Held-out total	Metrics
Physics	THEOREM_NATIVE	PASS	20	covered_case_total=30, covered_held_out_case_total=20, normalized_error_mean_abs=0.188136, normalized_error_p95=0.64778, normalized_error_max=1.48, severe_case_total=0, tail_breach_count=0
Chemistry	THEOREM_NATIVE	PASS	19	covered_case_total=30, covered_held_out_case_total=19, normalized_error_mean_abs=0.533285, normalized_error_p95=1.0, normalized_error_max=1.597772, severe_case_total=1, tail_breach_count=0
Biology	THEOREM_NATIVE	PASS	16	covered_case_total=30, covered_held_out_case_total=16, normalized_error_mean_abs=0.444222, normalized_error_p95=1.030033, normalized_error_max=1.438582, severe_case_total=0, tail_breach_count=0

Domain	Class	Verdict	Held-out total	Metrics
Systems / Civilizational projection	THEOREM_NATIVE	PASS	18	covered_case_total=30, covered_held_out_case_total=18, normalized_error_mean_abs=0.56097, normalized_error_p95=1.924704, normalized_error_max=2.25395, severe_case_total=2, fn=1, critical_failure_precision=1.0, critical_failure_recall=0.979592, critical_failure_f1=0.989691, tail_breach_count=0

Transition to practical use. The release-gated synthesis consolidates the integrated Core 1.3 result without turning that result into an unbounded public claim. The next chapter translates the same bounded result into prediction tasks, decision rules, and operational procedures.

10 Practical Consequences, Predictive Power, and Use of OC Core 1.3

OC Core 1.3 is practically useful when a reader must choose a lawful packet, make a bounded residual prediction, perform anomaly screening, audit a state transition, monitor a regime shift, or compare routes across domains. This chapter states what can be used now, what inputs are needed, what outputs it produces, and where the boundary lies. It does not present OC as an unrestricted universal prediction system.

The practical value lies in playbooks and comparison matrices. They show what prior science already does well, where OC adds route selection, falsifier testing, and replay discipline, and where the atlas marks complementarity, integration, or a local model's superiority in a specific lane.

This framing is comparative rather than replacement-oriented. Specialised local model families in physics, chemistry, and biology remain strong inside their native lanes. The Systems / Civilizational projection lane is likewise bounded and does not license unrestricted civilizational forecasting or control. OC's present claim is narrower: it aligns selected closed packets through shared route selection, falsifier tests, and replay checks while each domain-level closure remains local to its validated theorem and its replay records. Appendix **??** gives the row-level trace anchors for that operational claim. The surrounding comparison vocabulary is contextualised by established model families (**goldenfeld_lectures**; **wilson_renormalization**; **kauffman_autocatalytic**; **raf_networks**; **hebb_organization**; **friston_free_energy**). The canonical utility atlas also carries theorem-native mathematics and one cross-domain unified-science row; the opening paragraphs foreground the empirical lanes because they provide the most immediate replay-tested use cases for external readers.

This chapter draws from the canonical practical-utility atlas so that reader-facing statements about usefulness stay tied to the canonical support classes: FORMALLY_PROVED, THEOREM_NATIVE_HELD_OUT_VALIDATED, and OPERATIONALLY_SUPPORTED_WITHIN_BOUNDS. The Practical Utility and Model Comparison Atlas (Appendix **??**) carries the detailed support surface: use cases, playbooks, same-claim baselines, serious comparators, readiness boundaries, and trace anchors in one place.

Provenance. The governing utility surface and its release mirror are named in Appendix **??**; the governing JSON surface records the exact repository SHA and UTC release timestamp.

10.1 Practical value and support boundary

The practical utility atlas currently tracks 6 usable-now rows, 3 frontier-program rows, and 1 hypothesis-only row. The theorem-native domain lanes are usable under explicit theorem, data-route, and falsifier discipline; the cross-domain route-selection lane is operationally supported within bounds and is used for packet selection rather than as a domain-level theorem claim. Its closed practical value is not unrestricted universal prediction; it is lawful packet selection, bounded residual prediction, anomaly screening, state-transition auditing, regime-shift monitoring, and cross-domain route selection within one source-bound routing framework. The atlas also keeps 3 frontier-program rows and 1 hypothesis-only row explicit so practical ambition does not masquerade as finished closure.

10.2 Support classes

Every practical claim below is explicitly labeled so the reader can see what is already usable, what is bounded, and what still remains exploratory.

- `FORMALLY_PROVED` means formally proved and is usable now within the declared bounds.
- `THEOREM_NATIVE_HELD_OUT_VALIDATED` means theorem-native + empirical / held-out validated and is usable now within the declared bounds.
- `OPERATIONALLY_SUPPORTED_WITHIN_BOUNDS` means operationally supported within bounds and is usable now within the declared bounds.
- `FRONTIER_PROGRAM` means frontier program and is not yet a lawful usable-now claim.
- `HYPOTHESIS_ONLY` means hypothesis only and is not yet a lawful usable-now claim.

10.3 What is already usable now

Domain	Problem class	What OC gives now	Support	Output / decision	Trace
Mathematics	Strict closure certification and counterexample elimination	Certify whether a bounded formal object or question packet survives strict closure and counterexample search under the locked mathematical lane.	formally proved	A replayable accept or reject result together with minimal denominator, augmentation rank, or surviving counterexample classification.	row `MATHEMATICS_USE_001_STRICT_CLOSURE_CERTIFICATION`; source `releases/oc_core_1_3/editorial/science_sources/closure_bundles/mathematics/bundle.json`; closure `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/mathematics`; refs `content/20_oc_core_1_3_theorem_roadmap.tex`, `content/21_oc_core_1_3_worked_examples.tex`

Domain	Problem class	What OC gives now	Support	Output / decision	Trace
Physics	Constants, spectral-line, and transport-envelope audit	Audit bounded constants, Balmer-line families, and transport-scaling envelopes on held-out benchmark families and flag out-of-envelope residual behaviour.	theorem-native + empirical / held-out validated	A pass or fail residual verdict, anomaly prioritisation, and a decision about whether the family remains inside the promoted packet.	row `PHYSICS_USE_001_BOUNDED_RESIDUAL_AUDIT`; source `releases/oc_core_1_3/editorial/science_sources/closure_bundles/physics/bundle.json`; closure `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/physics`; refs `content/22_oc_core_1_3_operationalization_program.tex`, `content/23_oc_core_1_3_empirical_execution_protocols.tex`
Chemistry	Thermochemical, spectroscopic, and kinetic packet audit	Audit bounded thermochemical, spectral, and reaction-envelope families on held-out compounds or reactions and flag out-of-envelope chemistry lanes.	theorem-native + empirical / held-out validated	A pass or fail residual verdict, compound or reaction-family screening result, and a decision about whether the lane remains inside the promoted chemistry packet.	row `CHEMISTRY_USE_001_COMPOUND_REACTION_PACKET_AUDIT`; source `releases/oc_core_1_3/editorial/science_sources/closure_bundles/chemistry/bundle.json`; closure `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/chemistry`; refs `content/22_oc_core_1_3_operationalization_program.tex`, `content/23_oc_core_1_3_empirical_execution_protocols.tex`

Domain	Problem class	What OC gives now	Support	Output / decision	Trace
Biology	Expression-signature, state-transition, and response-onset screening	Screen bounded biological lanes for expression-signature residuals, state-transition ordering errors, and response-onset residuals across held-out datasets.	theorem-native + empirical / held-out validated	A pass or fail signature verdict, a state-transition ordering judgment, and an escalation decision for the biological lane.	row `BIOLOGY_USE_001_TRANSITION_AND_RESPONSE_SCREENING`; source `releases/oc_core_1_3/editorial/science_sources/closure_bundles/biology/bundle.json`; closure `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/biology`; refs `content/22_oc_core_1_3_operationalization_program.tex`, `content/23_oc_core_1_3_empirical_execution_protocols.tex`
Systems /K8 projection	Macro-trajectory and regime-shift monitoring	Monitor bounded macro trajectories, turning-point candidates, and regime-shift signatures on held-out official series and flag out-of-band intervals.	theorem-native + empirical / held-out validated	A pass or fail trajectory verdict, a turning-point or regime-shift alert, and a decision about whether the lane remains inside the promoted systems packet.	row `SYSTEMS_USE_001_REGIME_SHIFT_MONITORING`; source `releases/oc_core_1_3/editorial/science_sources/closure_bundles/systems_civilizational_projection/bundle.json`; closure `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/systems_civilizational_projection`; refs `content/22_oc_core_1_3_operationalization_program.tex`, `content/23_oc_core_1_3_empirical_execution_protocols.tex`

Domain	Problem class	What OC gives now	Support	Output / decision	Trace
Cross-domain / unified science	Cross-domain route selection, escalation discipline, and packet coordination	Choose the lawful domain packet, comparator set, and escalation route for a mixed scientific or operational problem without letting rhetoric outrun the closed packet boundary.	operationally supported within bounds	A lawful packet-selection decision, a comparator shortlist, and an explicit escalation path when the closed packet is not enough.	row `UNIFIED_USE_001_CROSS_DOMAIN_ROUTE_SELECTION`; refs `content/22_oc_core_1_3_operationalization_program.tex`, `content/23_oc_core_1_3_empirical_execution_protocols.tex`

10.4 How to use OC in practice

The playbooks below answer the operational question directly: when to use OC, what inputs are required, what procedure to run, and what the resulting decision or prediction actually is.

Mathematics: Strict closure and counterexample playbook Use when the task is to determine whether a bounded exact question already lies inside the promoted mathematical anchor or should be rejected by the strict counterexample route. Inputs: formal question encoding; strict closure policy; counterexample search route. Procedure: 1. Encode the target object in the admissible exact-question form. 2. Run the declared terminalization and counterexample search. 3. Accept the result only if the replay is deterministic and survivor-free. 4. Escalate to frontier if the question leaves the declared exact scope. Output: A replayable proof result, a surviving counterexample, or a lawful frontier demotion. Support class: formally proved. Trace: row `MATHEMATICS_PLAYBOOK_001_STRICT_CLOSURE`; source `releases/oc_core_1_3/editorial/science_sources/closure_bundles/mathematics/bundle.json`; closure `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/mathematics`; refs `content/20_oc_core_1_3_theorem_roadmap.tex`, `content/21_oc_core_1_3_worked_examples.tex`. Failure boundary: Stop if the object cannot be encoded in the admissible exact-question format or if a surviving counterexample remains after the declared search.

Physics: Bounded constants, spectra, and transport audit Use when the task is to decide whether a constants family, spectral family, or transport family belongs inside the promoted physics packet or should be escalated as an anomaly. Inputs: chosen benchmark family; official dataset route; held-out split and sigma policy. Procedure: 1. Map the observable family to the promoted physics packet. 2. Load the pinned official dataset and the locked held-out split. 3. Run the declared residual replay and inspect sign or order constraints. 4. Accept the family only if the replay stays inside the promoted tolerance band. Output: A bounded inclusion, anomaly, or escalation verdict for the target physical family. Support class: theorem-native + empirical / held-out validated. Trace: row `PHYSICS_PLAYBOOK_001_BOUNDED_RESIDUAL_AUDIT`; source `releases/oc_core_1_3/editorial/science_sources/closure_bundles/physics/bundle.json`; closure `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/physics`; refs `content/22_oc_core_1_3_operationalization_program.tex`, `content/23_oc_core_1_3_empirical_execution_protocols.tex`. Failure boundary: Stop and demote to frontier if the observable family requires a widened scope, hidden fit freedom, or a post-hoc split change.

Chemistry: Thermochemical, spectral, and kinetic packet audit Use when the task is to decide whether a bounded compound family or reaction class lies inside the promoted chemistry packet or should be escalated as an out-of-envelope case. Inputs: compound or reaction-family definition; pinned NIST or DOE-linked route; held-out rotation and tolerance policy. Procedure: 1. Map the chemistry lane to the promoted packet and observable family. 2. Load the pinned reference tables and the locked held-out split. 3. Run the declared replay on thermochemical, spectral, or kinetic outputs. 4. Accept the lane only if the held-out family remains inside the promoted tolerance band. Output: A bounded inclusion, anomaly, or escalation verdict for the target chemistry lane. Support class: theorem-native + empirical / held-out validated. Trace: row `CHEMISTRY_PLAYBOOK_001_PACKET_AUDIT`; source `releases/oc_core_1_3/editorial/science_sources/closure_bundles/chemistry/bundle.json`; closure `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/chemistry`; refs `content/22_oc_core_1_3_operationalization_program.tex`, content/23_oc_core_1_3_empirical_execution_protocols.tex. Failure boundary: Stop and demote to frontier if the lane requires hidden fit freedom, widened scope, or post-hoc split changes.

Biology: Bounded biological signature and onset screening Use when the task is to decide whether a bounded assay lane belongs inside the promoted biological packet or should be escalated as an unresolved biological signature family. Inputs: bounded assay family; multi-dataset held-out split; declared signature, transition, or onset observable. Procedure: 1. Map the biological lane to the promoted packet and observable family. 2. Load the locked multi-dataset route and held-out split. 3. Run the declared replay on expression, transition-order, or onset outputs. 4. Accept the lane only if the held-out datasets stay inside the promoted signature envelope. Output: A bounded inclusion, anomaly, or escalation verdict for the target biological lane. Support class: theorem-native + empirical / held-out validated. Trace: row BIOLOGY_PLAYBOOK_001_SIGNATURE_SCREENING; source releases/oc_core_1_3/editorial/science_sources/closure_bundles/biology/bundle.json; closure releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/biology; refs content/22_oc_core_1_3_operationalization_program.tex, content/23_oc_core_1_3_empirical_execution_protocols.tex. Failure boundary: Stop and demote to frontier if the lane depends on a single dataset family, hidden fit freedom, or a widened biological claim boundary.

Systems /K8 projection: Bounded macro-trajectory and regime-shift monitoring Use when the task is to decide whether an official macro-process series remains inside the promoted systems packet or should be escalated as a turning-point or regime anomaly. Inputs: target macro-process series; held-out interval; declared trajectory or turning-point metric. Procedure: 1. Map the macro-process lane to the promoted systems packet. 2. Load the pinned official series and the locked held-out interval. 3. Run the declared residual and turning-point replay. 4. Accept the lane only if held-out intervals remain inside the promoted trajectory and regime envelope. Output: A bounded inclusion, anomaly, or escalation verdict for the target systems lane. Support class: theorem-native + empirical / held-out validated. Trace: row SYSTEMS_PLAYBOOK_001_REGIME_SHIFT_MONITORING; source releases/oc_core_1_3/editorial/science_sources/closure_bundles/systems_civilizational_projection/bundle.json; closure releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/systems_civilizational_projection; refs content/22_oc_core_1_3_operationalization_program.tex, content/23_oc_core_1_3_empirical_execution_protocols.tex. Failure boundary: Stop and demote to frontier if the lane requires out-of-band trajectory behaviour, a widened horizon, or post-hoc interval selection.

Cross-domain /unified science: Cross-domain packet selection and escalation playbook Use when a scientific or engineering problem touches more than one domain lane and the team must decide which closed packet, comparator family, and evidence route should govern the next action. Inputs: declared target problem class; bounded observable family; candidate domain packets; decision or intervention objective. Procedure: 1. Map the target problem to the narrowest lawful observable family. 2. Select the closed packet whose theorem-to-observable map already governs that family. 3. Check same-claim baselines and serious comparators before claiming advantage. 4. Escalate only when the closed packet boundary is reached and the atlas permits the translation. Output: A lawful route-selection decision with explicit packet, comparator, falsifier, and escalation boundaries. Support class: operationally supported within bounds. Trace: row UNIFIED_PLAYBOOK_001_ROUTE_SELECTION; refs content/22_oc_core_1_3_operationalization_program.tex, content/23_oc_core_1_3_empirical_execution_protocols.tex. Failure boundary: Stop and demote to frontier if the observable family is outside every closed packet or if two candidate routes produce incompatible bounded verdicts.

10.5 What OC predicts across the closed empirical domains

Physics. For this domain, OC currently uses the closed packet to audit bounded constants, Balmer-line families, and transport-scaling envelopes on held-out benchmark families and flag out-of-envelope residual behaviour. Use-case trace: row PHYSICS_USE_001_BOUNDED_RESIDUAL_AUDIT; source releases/oc_core_1_3/editorial/science_sources/closure_bundles/physics/bundle.json; closure releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/physics; refs content/22_oc_core_1_3_operationalization_program.tex, content/23_oc_core_1_3_empirical_execution_protocols.tex. Measured outputs: relative_constant_error, spectral_line_residual, transport_scaling_residual. Benchmark families: PHY_BENCH_001; fundamental constants residuals; spectral line residual envelopes; transport or plasma scaling envelopes. Held-out case total: 20. What remains open: OC Core 1.3 does not yet lawfully promote a general OC replacement for the full physics model stack.

Chemistry. For this domain, OC currently uses the closed packet to audit bounded thermochemical, spectral, and reaction-envelope families on held-out compounds or reactions and flag out-of-envelope chemistry lanes. Use-case trace: row `CHEMISTRY_USE_001_COMPOUND_REACTION_PACKET_AUDIT`; source `releases/oc_core_1_3/editorial/science_sources/closure_bundles/chemistry/bundle.json`; closure `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/chemistry`; refs `content/22_oc_core_1_3_operationalization_program.tex`, `content/23_oc_core_1_3_empirical_execution_protocols.tex`. Measured outputs: `enthalpy_residual`, `spectral_transition_residual`, `kinetic_or_equilibrium_residual`. Benchmark families: `CHEM_BENCH_001`; thermochemical reference values; spectroscopic transition families; kinetic or equilibrium observable envelopes. Held-out case total: 19. What remains open: OC Core 1.3 does not yet promote a general OC catalyst-discovery or unrestricted synthetic-chemistry engine.

Biology. For this domain, OC currently uses the closed packet to screen bounded biological lanes for expression-signature residuals, state-transition ordering errors, and response-onset residuals across held-out datasets. Use-case trace: row `BIOLOGY_USE_001_TRANSITION_AND_RESPONSE_SCREENING`; source `releases/oc_core_1_3/editorial/science_sources/closure_bundles/biology/bundle.json`; closure `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/biology`; refs `content/22_oc_core_1_3_operationalization_program.tex`, `content/23_oc_core_1_3_empirical_execution_protocols.tex`. Measured outputs: `expression_signature_residual`, `state_transition_order_error`, `response_onset_residual`. Benchmark families: `BIO_BENCH_001`; gene-expression time-series families; cell-state transition signatures; stimulus-response or excitability onset markers. Held-out case total: 16. What remains open: OC Core 1.3 does not yet promote a general OC biological intervention or whole-phenotype control engine.

Systems /K8 projection. For this domain, OC currently uses the closed packet to monitor bounded macro trajectories, turning-point candidates, and regime-shift signatures on held-out official series and flag out-of-band intervals. Use-case trace: row `SYSTEMS_USE_001_REGIME_SHIFT_MONITORING`; source `releases/oc_core_1_3/editorial/science_sources/closure_bundles/systems_civilizational_projection/bundle.json`; closure `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/systems_civilizational_projection`; refs `content/22_oc_core_1_3_operationalization_program.tex`, `content/23_oc_core_1_3_empirical_execution_protocols.tex`. Measured outputs: `trajectory_residual`, `turning_point_error`, `held_out_interval_regime_score`. Benchmark families: `SYS_BENCH_001`; macro indicator trajectories; resource-intensity or throughput curves; held-out interval stability and regime shifts. Held-out case total: 18. What remains open: OC Core 1.3 does not yet promote unrestricted long-horizon civilizational forecasting or control as lawful closed science.

10.6 Benchmark against serious alternatives

This chapter distinguishes between simpler same-claim-class baselines and serious established model families. The first question is whether a cheaper competitor already solves the same bounded lane. The second is whether OC merely coexists with, integrates, or improves the fragmented scientific picture around that lane. Each comparison row below is an indexed main-body summary rather than a standalone verdict. The comparison id is the row-level trace anchor; Appendix **??** expands that id into the source refs, scope boundary, comparator row, and trace hooks that make the summary auditable.

Same-claim-class baselines.

Comparison id	Domain	Comparator families	Verdict	Comparison scope	Trace
SAME_CLAIM_CLASS::MATHEMATICS	Mathematics	exact enumerative baseline; direct constructive proof baseline	DECLARED_NO_SUPERIOR_BASELINE	Theorem-native anchor lane is trace-closed and replayable. Cost-normalized budget: NO_COMPARATOR_MAY_WIN_ON_SAME_CLAIM_CLASS_AFTER_COST_NORMALIZATION	row SAME_CLAIM_CLASS::MATHEMATICS; source releases/oc_core_1_3/editorial/science_sources/closure_bundles/mathematics/bundle.json; closure releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/mathematics
SAME_CLAIM_CLASS::PHYSICS	Physics	bounded residual fit without theorem trace; family-wise least-squares baseline	NO_SIMPLER_SUPERIOR_BASELINE_FOUND	Bounded theorem-native physics packet is trace-closed, replay-closed, and comparator-clean on the pinned official held-out route. Cost-normalized budget: NO_COMPARATOR_MAY_WIN_ON_SAME_CLAIM_CLASS_AFTER_COST_NORMALIZATION	row SAME_CLAIM_CLASS::PHYSICS; source releases/oc_core_1_3/editorial/science_sources/closure_bundles/physics/bundle.json; closure releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/physics
SAME_CLAIM_CLASS::CHEMISTRY	Chemistry	reference-table envelope baseline; reaction-class residual fit baseline	NO_SIMPLER_SUPERIOR_BASELINE_FOUND	Bounded theorem-native chemistry packet is trace-closed, replay-closed, and comparator-clean on the pinned thermochemical, spectral, and kinetic held-out route. Cost-normalized budget: NO_COMPARATOR_MAY_WIN_ON_SAME_CLAIM_CLASS_AFTER_COST_NORMALIZATION	row SAME_CLAIM_CLASS::CHEMISTRY; source releases/oc_core_1_3/editorial/science_sources/closure_bundles/chemistry/bundle.json; closure releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/chemistry
SAME_CLAIM_CLASS::BIOLOGY	Biology	dataset-family signature fit baseline; state-transition score baseline	NO_SIMPLER_SUPERIOR_BASELINE_FOUND	Bounded theorem-native biology packet is trace-closed, replay-closed, and comparator-clean across the pinned multi-dataset held-out biological lanes. Cost-normalized budget: NO_COMPARATOR_MAY_WIN_ON_SAME_CLAIM_CLASS_AFTER_COST_NORMALIZATION	row SAME_CLAIM_CLASS::BIOLOGY; source releases/oc_core_1_3/editorial/science_sources/closure_bundles/biology/bundle.json; closure releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/biology

Comparison id	Domain	Comparator families	Verdict	Comparison scope	Trace
SAME_CLAIM_CLASS::SYSTEMS_CIVILIZATIONAL_PROJECTION	Systems /K8 projection	macro-trend residual baseline; turning-point heuristic baseline	NO_SIMPLER_SUPERIOR_BASELINE_FOUND	Bounded theorem-native systems packet is trace-closed, replay-closed, and comparator-clean on the pinned macro-trajectory and regime-shift held-out route. Cost-normalized budget: NO_COMPARATOR_MAY_WIN_ON_SAME_CLAIM_CLASS_AFTER_COST_NORMALIZATION	row SAME_CLAIM_CLASS::SYSTEMS_CIVILIZATIONAL_PROJECTION; source releases/oc_core_1_3/editorial/science_sources/closure_bundles/systems_civilizational_projection/bundle.json; closure releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/systems_civilizational_projection

Serious model families.

Comparison id	Domain	Model family	Problem class	Relation to OC	What OC adds or closes	Source refs
SERIOUS_COMPARATOR::PHYSICS::PLASMA_TRANSPORT_MODELS	Physics	Gyrokinetic, confinement-scaling, and plasma-transport models	transport or plasma scaling envelopes	complements	OC adds a shared bounded packet and replay rule that lets transport residual families live beside constants and spectra under one inspected comparison boundary.	releases/oc_core_1_3/editorial/science_sources/practical_utility/serious_model_comparison_catalog.json, releases/oc_core_1_3/editorial/science_sources/closure_bundles/physics/bundle.json
SERIOUS_COMPARATOR::PHYSICS::QED_ATOMIC_PRECISION	Physics	Precision quantum electrodynamics and atomic-structure theory	fundamental constants and spectral transition families	integrates	OC adds one bounded theorem-to-observable packet that keeps constants, Balmer spectra, and transport residual families under one explicit closure, falsifier, and held-out replay contract.	releases/oc_core_1_3/editorial/science_sources/practical_utility/serious_model_comparison_catalog.json, releases/oc_core_1_3/editorial/science_sources/closure_bundles/physics/bundle.json

Comparison id	Domain	Model family	Problem class	Relation to OC	What OC adds or closes	Source refs
SERIOUS_COMPARATOR::CHEMISTRY::KINETICS_AND_TST	Chemistry	Chemical kinetics, transition-state, and equilibrium models	reaction-class kinetic or equilibrium envelopes	integrates	OC adds a bounded chemistry packet that keeps those observable families under one theorem, one parameter-law shell, and one held-out replay discipline.	releases/oc_core_1_3/editorial/science_sources/practical_utility/serious_model_comparison_catalog.json, releases/oc_core_1_3/editorial/science_sources/closure_bundles/chemistry/bundle.json
SERIOUS_COMPARATOR::CHEMISTRY::QUANTUM_CHEMISTRY	Chemistry	Quantum chemistry, ab initio, and density-functional families	thermochemical and spectroscopic envelopes	remains superior within its lane	OC adds a bounded packet that ties thermochemical, spectroscopic, and kinetic envelopes to one theorem-native route with one falsifier and one replay discipline.	releases/oc_core_1_3/editorial/science_sources/practical_utility/serious_model_comparison_catalog.json, releases/oc_core_1_3/editorial/science_sources/closure_bundles/chemistry/bundle.json
SERIOUS_COMPARATOR::BIOLOGY::EXCITABILITY_AND_RESPONSE	Biology	Excitable-media and response-onset dynamical models	response-onset residual lanes	complements	OC adds a single bounded biological packet that keeps response onset tied to expression and transition-order evidence inside one replay and falsifier framework.	releases/oc_core_1_3/editorial/science_sources/practical_utility/serious_model_comparison_catalog.json, releases/oc_core_1_3/editorial/science_sources/closure_bundles/biology/bundle.json
SERIOUS_COMPARATOR::BIOLOGY::SYSTEMS_BIOLOGY	Biology	Gene-regulatory, transcriptomic, and systems-biology state-space models	expression-signature and state-transition lanes	integrates	OC adds a bounded packet that keeps expression signatures, state-transition order, and response onset under one multi-dataset held-out closure route.	releases/oc_core_1_3/editorial/science_sources/practical_utility/serious_model_comparison_catalog.json, releases/oc_core_1_3/editorial/science_sources/closure_bundles/biology/bundle.json

Comparison id	Domain	Model family	Problem class	Relation to OC	What OC adds or closes	Source refs
SERIOUS_COMPARATOR::SYSTEMS::SYSTEM_DYNAMICS_AND_CHANGE_POINTS	Systems /K8 projection	Systems dynamics, control, and change-point models	macro-process trajectories and regime shifts	integrates	OC adds a bounded macro-trajectory packet that keeps trajectory residuals, turning points, and regime scores under one theorem-to-observable route and one held-out decision bar.	releases/oc_core_1_3/editorial/science_sources/practical_utility/serious_model_comparison_catalog.json, releases/oc_core_1_3/editorial/science_sources/closure_bundles/systems_civilizational_projection/bundle.json
SERIOUS_COMPARATOR::CROSS_DOMAIN::COMPLEXITY_PROGRAMS	Cross-domain /unified science	Complexity-science and multiscale integration programs	cross-domain integration and multiscale explanation	complements	OC adds a harder closure grammar: explicit theorem packets, parameter laws, held-out replay, falsifiers, and same-claim comparator ledgers for each bounded lane.	releases/oc_core_1_3/editorial/science_sources/practical_utility/serious_model_comparison_catalog.json, content/24_oc_core_1_3_proof_machinery.tex
SERIOUS_COMPARATOR::CROSS_DOMAIN::REDUCTIONIST_PATCHWORK	Cross-domain /unified science	Best-in-class local-model portfolio governance across separate domain theories	cross-domain scientific planning and explanation	integrates	OC adds one explicit kernel, one K-level ladder, one theorem-to-observable grammar, and one atlas for translating bounded packets across domains under declared scope controls.	releases/oc_core_1_3/editorial/science_sources/practical_utility/serious_model_comparison_catalog.json, content/24_oc_core_1_3_proof_machinery.tex

10.7 What prior science could do, what remained fragmented, and what OC closes

Comparison id	Domain	Prior science already did	Fragmentation or limit	What OC closes or adds	Open boundary
SERIOUS_COMPARATOR::PHYSICS::PLASMA_TRANSPORT_MODELS	Physics	Provides strong local transport and confinement heuristics for specific plasma regimes and benchmark devices.	These models are powerful in their native lane but do not by themselves close a unified residual grammar spanning constants, spectra, and transport under one promotion rule.	OC adds a shared bounded packet and replay rule that lets transport residual families live beside constants and spectra under one inspected comparison boundary.	OC does not replace specialised transport solvers or device-specific plasma modelling outside the promoted packet.

Comparison id	Domain	Prior science already did	Fragmentation or limit	What OC closes or adds	Open boundary
SERIOUS_COMPARATOR::PHYSICS::QED_ATOMIC_PRECISION	Physics	Delivers high-precision local calculations for relations among constants, atomic transitions, and spectral families inside explicit microscopic equation classes.	Selection and justification of the local model family remain lane-specific, and the route does not by itself give a shared kernel-to-observable closure discipline across domains.	OC adds one bounded theorem-to-observable packet that keeps constants, Balmer spectra, and transport residual families under one explicit closure, falsifier, and held-out replay contract.	Detailed local derivation outside the promoted residual packet still relies on the standard physics model stack.
SERIOUS_COMPARATOR::CHEMISTRY::KINETICS_AND_TST	Chemistry	Explains rate, barrier, and equilibrium behaviour for declared reaction classes with strong domain-specific local formalisms.	The family remains reaction-class specific and does not itself unify thermochemistry, spectroscopy, and kinetics under one kernel-bound promotion route.	OC adds a bounded chemistry packet that keeps those observable families under one theorem, one parameter-law shell, and one held-out replay discipline.	OC does not erase the need for local kinetic modelling outside the promoted bounded families.
SERIOUS_COMPARATOR::CHEMISTRY::QUANTUM_CHEMISTRY	Chemistry	Provides molecule-specific energetic and spectroscopic calculations with a mature tool chain for local chemical systems.	Method choice, basis dependence, and domain-specific tuning remain lane-specific, and the family does not by itself provide a closure ladder from kernel operators to held-out benchmark governance.	OC adds a bounded packet that ties thermochemical, spectroscopic, and kinetic envelopes to one theorem-native route with one falsifier and one replay discipline.	High-precision molecule-specific calculation remains the job of the established chemistry stack outside the promoted OC packet.
SERIOUS_COMPARATOR::BIOLOGY::EXCITABILITY_AND_RESPONSE	Biology	Captures threshold and onset dynamics well in specific excitability, signalling, or response families.	The family remains local to its chosen mechanism and does not by itself close broader biological signature, transition, and onset lanes under one bounded acceptance bar.	OC adds a single bounded biological packet that keeps response onset tied to expression and transition-order evidence inside one replay and falsifier framework.	Detailed mechanism-specific onset modelling still belongs to the established lane-specific models outside the promoted packet.
SERIOUS_COMPARATOR::BIOLOGY::SYSTEMS_BIOLOGY	Biology	Provides established state-space, regulatory-network, and transcriptomic modelling inside specific assay and dataset families.	Performance is often lane- and dataset-specific, and cross-dataset closure plus theorem-to-observable promotion discipline remain fragmented.	OC adds a bounded packet that keeps expression signatures, state-transition order, and response onset under one multi-dataset held-out closure route.	OC does not yet replace detailed mechanistic biological modelling outside the promoted assay lanes.
SERIOUS_COMPARATOR::SYSTEMS::SYSTEM_DYNAMICS_AND_CHANGE_POINTS	Systems /K8 projection	Provides strong local tools for trend extraction, control analysis, and regime-change detection in declared systems lanes.	These tools are typically lane-specific and do not by themselves close a shared theorem route from kernel operators to official macro-process held-out promotion.	OC adds a bounded macro-trajectory packet that keeps trajectory residuals, turning points, and regime scores under one theorem-to-observable route and one held-out decision bar.	OC does not promote unrestricted civilizational prediction or replace every local macro or control model outside the bounded packet.
SERIOUS_COMPARATOR::CROSS_DOMAIN::COMPLEXITY_PROGRAMS	Cross-domain /unified science	Provides rich ideas about emergence, multiscale organisation, and complex adaptive behaviour across many scientific domains.	These programs are often strong conceptually but heterogeneous in proof bar, operator discipline, and closed promotion criteria.	OC adds a harder closure grammar: explicit theorem packets, parameter laws, held-out replay, falsifiers, and same-claim comparator ledgers for each bounded lane.	OC does not exhaust every possible complexity-science question; it narrows attention to the lanes it can lawfully close.
SERIOUS_COMPARATOR::CROSS_DOMAIN::REDUCTIONIST_PATCHWORK	Cross-domain /unified science	Explains many local scientific lanes well by using separate best-in-class models inside each domain.	Cross-domain translation, shared operator meaning, and unified promotion discipline remain fragmented across disconnected model families.	OC adds one explicit kernel, one K-level ladder, one theorem-to-observable grammar, and one atlas for translating bounded packets across domains under declared scope controls.	OC does not abolish local domain theories; it governs how bounded closed packets are related and promoted together.

10.8 What remains frontier or hypothesis

Domain	Problem class	Support	Why this is not yet closed
Physics	Broader microphysical law extension beyond the promoted residual packet	frontier program	OC Core 1.3 does not yet lawfully promote a general OC replacement for the full physics model stack.
Chemistry	General catalyst or out-of-family chemistry design	frontier program	OC Core 1.3 does not yet promote a general OC catalyst-discovery or unrestricted synthetic-chemistry engine.
Biology	Multi-scale phenotype control and unrestricted biological intervention	frontier program	OC Core 1.3 does not yet promote a general OC biological intervention or whole-phenotype control engine.
Systems /K8 projection	Unrestricted long-horizon civilizational forecasting and control	hypothesis only	OC Core 1.3 does not yet promote unrestricted long-horizon civilizational forecasting or control as lawful closed science.

Transition to the atlas. The practical section is intentionally bounded: it carries the high-signal tables needed for reader-facing use while leaving the exhaustive audit burden to Appendix **??**. The main chapter therefore remains a guide to bounded use rather than a duplicate of the atlas.

A Journal-Core Extraction Bridge

This appendix explains how the journal-core article relates to the master monograph.

A.1 Why the journal-core exists

Some channels require a bounded, faster-reading scientific artifact. Editors, reviewers, and external readers often need a paper-sized object whose defended claim can be read in one sitting. The journal-core exists for that purpose. Its job is not to replace the monograph but to provide a narrow article that remains anchored to the same scientific truth conditions.

A.2 What the journal-core may omit

The journal-core may compress:

- extensive chapter-scale didactic explanation,
- large parts of the domain atlas,
- broad module coverage,
- detailed appendical inventories,
- and some of the monograph-level pedagogical scaffolding.

It may not silently change:

- theorem scope,
- explicit nonclaims,
- claimed source basis,
- figure meaning,
- or the relation between inherited corpus and present defended claim.

A.3 What the monograph guarantees to later artifacts

Because the journal-core is extracted from the present volume, later owner-facing and public channels do not need to improvise their own understanding of the science. They can derive from the master monograph when they need depth, from the journal core when they need bounded exposition, and from the companion layer when they need support evidence.

This is the practical publication value of the monograph. It is not merely larger than the article. It is the document that prevents every later projection from becoming a separate interpretive exercise.

B Reviewer Navigation Matrix and Audit Checklist

Start with the row that matches the question or concern, inspect the primary route, and use the verification column as the pass/fail test. This appendix uses three canonical artifact terms:

- *Master Monograph*: the English flagship volume. Later uses of *Monograph* are shorthand for this same artifact.
- *Journal Core*: the shorter bounded extraction from the monograph.
- *Release*: the outward package that also contains supporting surfaces, PDFs, assets, and companion dossiers.

Capitalisation marks the canonical glossary terms above; lowercase uses are generic prose shorthand unless explicitly styled as one of those terms. The following class labels are local definitions for this appendix rather than additional canonical glossary terms. Concrete artifact and witness classes are as follows:

- *Surfaces* are the release-bundled `editorial/*_latest.json` projections.
- *Companion dossiers* are the generated dossier packages under `editorial/dossier_packages/`.
- *Assets* are the released figures and tables under `assets/`.
- *Archival PDF witness*: the prior Core 1.2 PDF named in the journal-core provenance route in Appendix A: *Ontology of Continua — Core v1.2.0*.

B.1 Review questions, compact route, and decisive checks

Core theorem route and scope.

Reviewer question	Primary route to inspect	Verification cue
What is the central defended result?	Abstract; Sections 2, 7	Theorem A: bounded post-collapse identity classification; no general lift theorem.
Where does the proof-bearing route begin?	Sections 2, 7	Verify that Axiom 3.2 with Definitions 12.1, 12.3, and 12.4 supports Source Theorem 3; that Source Theorem 3 supports Lemma 1; that Lemma 1 with Definition 12.5 supports Lemma 2; and that Lemma 2 supports Theorem A. Inspect Axiom 3.3, Definition 12.6, and Source Corollary 3.2 separately as the companion rebirth-control branch.
Is collapse defined formally or rhetorically?	Sections 2, **??**	Admissible-state emptiness, $k \to 0$, and threshold failure.
What exactly is residue?	Sections **??**, **??**	Post-collapse structural trace; no continuing live identity.
What exactly is rebirth?	Sections **??**, **??**	Later live continuum grounded in residue; numerically new.
Why is irreversibility necessary?	Sections 2, 7	Blocks same-entity restoration after lawful death.
Could the main theorem depend on an excluded lift-heavy theorem family?	Section 7; Appendix A	No excluded lift-heavy theorem family is used in the bounded proof route.
Why is the monograph larger than the journal core?	Section **??**; Appendix A	Didactic shell, formal architecture, and repair logic live here.
Why is the journal core smaller without becoming weaker?	Appendix A	One-way extraction; theorem scope unchanged.
Why is the hierarchy fixed as $K_0, \ldots, K_{12}$?	Sections **??**, **??**, **??**	Upper levels carry substantive higher-order roles.

Audit, provenance, and review posture.

Reviewer question	**Primary route to inspect**	**Verification cue**
What would falsify the upper-level doctrine?	Sections **??**, **??**	Lossless reduction of K_{11} and K_{12} to K_{10}.
How do proof-oriented statement classes differ?	Sections **??**, 7	Axiomatic, derived, structural, interpretive, empirical, and frontier rows keep separate burdens; editorial-bridge language remains packaging support rather than a proof class.
What is merely interpretive?	Sections **??**, **??**, **??**	Pass only if interpretive language stays outside the proof route and is not promoted into theorem, replay, or practical-use authority.
What is the exact source base?	Section **??**; Appendix **??**	Source corpus, SPOT surface, and archival PDF witness are separate objects.
Why is ORCID foregrounded?	Title page; Section **??**	Stable scholarly identity.
What is the role of companion dossiers?	Section **??**; Appendix A	Support and provenance exposure, not replacement authority.
How should a non-specialist enter the monograph?	Sections 1, **??**	Didactic ladder before proof audit.
Where are the most likely points of critique?	Section 7; Appendix A	Collapse semantics, irreversibility, upper levels, extraction discipline.
What would count as a decisive release failure?	Sections **??**, 6, **??**	Continued identity after admissible-state loss, or reliance on an excluded theorem family.
What remains outside scope?	Sections **??**, **??**; Appendix A	Pass only if excluded theorem families, frontier rows, and projection targets remain outside the positive proof body.

B.2 Compact reading order by reviewer role

Rapid critical reviewer. Read the abstract, theorem roadmap, relevant theorems and corollaries, the worked examples, and the journal-core bridge appendix. Return to this matrix only to choose a new route after that first pass. This route should reveal immediately whether the release is too large, too narrow, or too broad in scope.

Formal mathematical reviewer. Read the model, results, axioms, operators, collapse/rebirth material, hierarchy chapter, notation appendix, and source-audit chapter. The decisive question is whether the derived bounded claim can be reconstructed from the stated source-native rows without informal supplementation.

Generalist scientific reviewer. Read the introduction, reader guide, theorem roadmap, worked examples, discussion, conclusion, and then return to the formal chapters selectively. The decisive question here is whether the volume explains itself honestly enough that scientific judgment can precede local notation mastery.

C Empirical Validation Matrix and Replay Ledger

This appendix is projected from the canonical science SPOT so that bridge evidence and promotion verdicts remain visibly distinct.

This appendix is projected from the canonical SPOT. It reports both bridge evidence and promotion verdicts without letting one masquerade as the other.

Domain	Class	Trace	Evidence	Verdict	Blocking ids	Notes
Mathematics	THEOREM_NATIVE	TRACE_COMPLETE	EVIDENCE_COMPLETE	PASS	none	Sources: Clay Mathematics Institute, American Institute of Mathematics; Replay: `logion/k7/spe/orchestrator/science/run_claim_simulations_v1.py`; Metrics: Campaign `PASS_31_OF_31_TERMINALIZED`; 31 terminalized cases; Next action: `MAINTAIN_REPLAY_DISCIPLINE`
Physics	THEOREM_NATIVE	TRACE_COMPLETE	EVIDENCE_COMPLETE	PASS	none	Sources: NIST, U.S. Department of Energy Office of Science; Replay: `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idPHYSICS`; Metrics: 30 covered / 20 held-out cases; Mean abs. normalized error 0.188136 sigma; P95 normalized error 0.64778 sigma; Maximum normalized error 1.48 sigma; Tail breaches 0; Next action: `MAINTENANCE_ONLY`
Chemistry	THEOREM_NATIVE	TRACE_COMPLETE	EVIDENCE_COMPLETE	PASS	none	Sources: NIST, U.S. Department of Energy Office of Science; Replay: `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idCHEMISTRY`; Metrics: 30 covered / 19 held-out cases; Mean abs. normalized error 0.533285 sigma; P95 normalized error 1.0 sigma; Maximum normalized error 1.597772 sigma; Tail breaches 0; Next action: `MAINTENANCE_ONLY`
Biology	THEOREM_NATIVE	TRACE_COMPLETE	EVIDENCE_COMPLETE	PASS	none	Sources: NCBI GEO; Replay: `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idBIOLOGY`; Metrics: 30 covered / 16 held-out cases; Mean abs. normalized error 0.444222 sigma; P95 normalized error 1.030033 sigma; Maximum normalized error 1.438582 sigma; Tail breaches 0; Next action: `MAINTENANCE_ONLY`
Systems / Civilizational projection	THEOREM_NATIVE	TRACE_COMPLETE	EVIDENCE_COMPLETE	PASS	none	Sources: World Bank, U.S. Department of Energy Office of Science; Replay: `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idSYSTEMS_CIVILIZATIONAL_PROJECTION`; Metrics: 30 covered / 18 held-out cases; Mean abs. normalized error 0.56097 sigma; P95 normalized error 1.924704 sigma; Maximum normalized error 2.25395 sigma; Tail breaches 0; Next action: `MAINTENANCE_ONLY`

D Unified Synthesis Support Dossiers

The main unified-synthesis section remains the canonical synthesis unit in the numbered Core 1.3 sequence of the manuscript. The material gathered here is retained as level-wise evidence support: it keeps compact K-anchored theorem-support, empirical, falsifier, and packet dossiers available inside the flagship volume while omitting the synthesis-section introduction and release-gate prose. Section 9 uses the generated synthesis surface for the main K-level exposition. This appendix is the support-dossier wrapper: it gives readers a single inspection location for the level-wise support files without turning them into a second synthesis chapter. The main chapter states the synthesis, while this appendix carries the dossier-oriented inspection copy. Compatibility support files are not input directly by the English platinum master. A build variant must therefore choose either the compatibility synthesis-support master or this appendix wrapper, never both. The full DFS compatibility stream still records those compatibility entries for auditability, but the platinum master does not input the compatibility synthesis-support master. The constants-and-parameters apparatus is part of the Appendix **??** data apparatus and is addressable there as Appendix **??**; this support appendix does not create a second synthesis or a second top-level constants appendix. Source stamp: `releases/oc_core_1_3/editorial/OC_CORE_1_3_UNIFIED_SYNTHESIS_latest.json`; current repository SHA and UTC generation timestamp are recorded inside that JSON surface so this appendix does not freeze stale build metadata.

D.1 Level K0: structural conditions for any universe

Current closure status: `PASS`. Promotion blockers: none.

Theorem claim. K0 fixes the non-empty admissible meta-domain required for any lawful continuum and therefore bounds every later K-level before empirical specialization.

Operator and K-level binding. The local K0 root process glyphs Ψ_0, Φ_0, and Λ_0 fix distinction generation, relational reconfiguration, and compositional assembly; compact synthesis aliases F_0, Q_0, and U_0 label the corresponding root constraints only where they are defined locally. Kernel refs: `content/03_model.tex`, `content/10_klevels_full.tex`, `content/11_operators_full.tex`, `APPENDIX_Q_SYNTHESIS_DOSSIER_K0`.

Parameter law.

$$\mu(\Omega(K_0)) > 0, \quad \forall x \in \{1, \dots, 12\} : \text{DoF}(M_x) > 0 \text{ and } \frac{\text{DoF}(K_x)}{\text{DoF}(M_x)} \leq 1, \quad C_{\text{nt}} \geq 1$$

Observable binding and units. Meta-admissibility, meta-space compatibility, and the presence of at least one non-trivial lawful cycle. Observable ids: `K0::ADMISSIBLE_STATE_MEASURE`, `K0::META_SPACE_COMPATIBILITY_RATIO`, `K0::NONTRIVIAL_CYCLE_PRESENCE`, `MATHEMATICS::EPSILON_STAR`, `MATHEMATICS::MINIMAL_DENOMINATOR`, `MATHEMATICS::MINIMAL_AUGMENTATION_RANK`, `MATHEMATICS::COMPONENT_FIBER_COUNT`, `MATHEMATICS::MOVE_CONNECTIVITY_OR_DEGREE_BOUND`, `MATHEMATICS::STRICT_TERMINALIZED_TOTAL`, `MATHEMATICS::COUNTEREXAMPLE_EXECUTED_TOTAL`, `MATHEMATICS::PYTHON_EVIDENCE_EXECUTED_TOTAL`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Admissible-state measure floor	$\mu(\Omega(K_0))$	> 0	dimensionless lower bound

Quantity	Symbol	Value / range	Units or note
Meta-space compatibility ratio for each promoted level	$\mathrm{DoF}(K_x)/\mathrm{DoF}(M_x)$	≤ 1	dimensionless ratio; applies for each x in 1,...,12
Non-trivial cycle presence	C_{nt}	1	present

Falsifier and collapse boundary. Collapse occurs if the admissible set degenerates, if a later K-level exceeds its meta-space, or if the structural substrate loses its last non-trivial lawful cycle. Falsifier refs: `APPENDIX_Q_SYNTHESIS_DOSSIER_K0`, `content/falsifiability/falsifiability_master.tex`, `content/falsifiability/falsifiability_k0.tex`, `content/falsifiability/falsifiability_k9.tex`, `content/falsifiability/falsifiability_k10.tex`, `content/falsifiability/falsifiability_k11.tex`, `content/falsifiability/falsifiability_k12.tex`.

Closed-domain bindings. `MATHEMATICS`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `0`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/mathematics`. Metrics: `strict_campaign_acceptance_status`=PASS_31_OF_31_TERMINALIZED, `strict_terminalized_total`=31.

D.2 Level K1: minimal observable distinctions and energetic axes

Current closure status: `PASS`. Promotion blockers: none.

Theorem claim. K1 introduces resolvable energetic and temporal distinctions while preserving the admissible-state floor inherited from K0.

Operator and K-level binding. F, G, and U bind contradiction load to resolvable energetic distinctions, monotonic update order, and a non-degenerate observable region. Kernel refs: `content/03_model.tex`, `content/10_klevels_full.tex`, `content/11_operators_full.tex`, `APPENDIX_Q_SYNTHESIS_DOSSIER_K1`.

Parameter law.

$$\Delta E \geq \Theta_{\Delta E}, \quad \frac{dt}{d\tau} > 0, \quad \mu(\Omega(K_1)) > 0$$

Observable binding and units. The first observable regime is tracked by energy-gap resolution, monotonic update ordering, and positive configuration-space measure. Observable ids: `K1::ENERGY_GAP_THRESHOLD`, `K1::TEMPORAL_MONOTONICITY`, `K1::CONFIGURATION_MEASURE`, `PHYSICS::FINE_STRUCTURE_CONSTANT_RESIDUAL`, `PHYSICS::HYDROGEN_BALMER_LINE_RESIDUAL`, `PHYSICS::TRANSPORT_SCALING_RESIDUAL`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Minimal resolvable energy scale	$\Theta_{\Delta E}$	10^{-34}–10^{-33}	J
Temporal monotonicity	$dt/d\tau$	> 0	ordered update constraint

Quantity	Symbol	Value / range	Units or note
Configuration-space measure floor	$\mu(\Omega(K_1))$	> 0	dimensionless lower bound

Falsifier and collapse boundary. Collapse begins when energetic distinctions are no longer resolvable, temporal order breaks, or the first observable state space degenerates. Falsifier refs: `APPENDIX_Q_SYNTHESIS_DOSSIER_K1`, `content/falsifiability/falsifiability_master.tex`, `content/falsifiability/falsifiability_k1.tex`, `content/falsifiability/falsifiability_k2.tex`, `content/falsifiability/falsifiability_k9.tex`, `content/falsifiability/falsifiability_k10.tex`.

Closed-domain bindings. `PHYSICS`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `20`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/physics`. Metrics: `covered_case_total`=30, `covered_held_out_case_total`=20, `normalized_error_mean_abs`=0.188136, `normalized_error_p95`=0.64778, `normalized_error_max`=1.48, `tail_breach_count`=0.

D.3 Level K2: fields, phases, and large-scale structure

Current closure status: `PASS`. Promotion blockers: none.

Theorem claim. K2 binds field configurations, expansion history, and phase thresholds into the first large-scale physical continuum admissible under the kernel operators.

Operator and K-level binding. F, H, and R constrain field admissibility, phase transitions, and large-scale stabilization margins inside the physical continuum. Kernel refs: `content/03_model.tex`, `content/10_klevels_full.tex`, `content/11_operators_full.tex`, `APPENDIX_Q_SYNTHESIS_DOSSIER_K2`.

Parameter law.

$$H_0 > 0, \quad 0 < \Omega_{\rm m} < 1, \quad |\Omega_k| < 2 \times 10^{-3}, \quad \mathcal{T}_{\rm phase}(t) \in \{T_c^{\rm EW}, T_c^{\rm QCD}\}$$

Observable binding and units. Expansion rate, curvature, cosmological density fractions, and critical phase windows tied to physical observables and replay packets. Observable ids: `K2::EXPANSION_RATE`, `K2::CURVATURE_BOUND`, `K2::PHASE_TRANSITION_WINDOW`, `CHEMISTRY::GAS_PHASE_ENTHALPY_RESIDUAL`, `CHEMISTRY::SPECTRAL_TRANSITION_RESIDUAL`, `CHEMISTRY::KINETIC_OR_EQUILIBRIUM_RESIDUAL`, `PHYSICS::FINE_STRUCTURE_CONSTANT_RESIDUAL`, `PHYSICS::HYDROGEN_BALMER_LINE_RESIDUAL`, `PHYSICS::TRANSPORT_SCALING_RESIDUAL`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Hubble parameter today	H_0	67.4 ± 0.5	$\mathrm{km\,s^{-1}\,Mpc^{-1}}$
Matter density fraction	$\Omega_{\rm m}$	0.315 ± 0.007	dimensionless
Critical phase windows	$T_c^{\rm EW}$, $T_c^{\rm QCD}$	$\sim$ 100 GeV; 150–170 MeV	closed phase set

Quantity	Symbol	Value / range	Units or note
Curvature bound	Ω_k	$\lvert\Omega_k\rvert < 2 \times 10^{-3}$	dimensionless

Falsifier and collapse boundary. K2 collapses when curvature exits the admissible band, phase windows fail to close lawfully, or large-scale physical observables lose replayable stabilization. Falsifier refs: APPENDIX_Q_SYNTHESIS_DOSSIER_K2, content/falsifiability/falsifiability_master.tex, content/falsifiability/falsifiability_k2.tex, content/falsifiability/falsifiability_k3.tex, content/falsifiability/falsifiability_k4.tex, content/falsifiability/falsifiability_k6.tex, content/falsifiability/falsifiability_k1.tex, content/falsifiability/falsifiability_k9.tex, content/falsifiability/falsifiability_k10.tex.

Closed-domain bindings. CHEMISTRY: class THEOREM_NATIVE, verdict PASS, held-out cases 19, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/chemistry. Metrics: covered_case_total=30, covered_held_out_case_total=19, normalized_error_mean_abs=0.533285, normalized_error_p95=1.0, normalized_error_max=1.597772, tail_breach_count=0. DRT: class REFUTED, verdict FAIL_CLOSED, held-out cases 0, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: none. Metrics: none. PHYSICS: class THEOREM_NATIVE, verdict PASS, held-out cases 20, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/physics. Metrics: covered_case_total=30, covered_held_out_case_total=20, normalized_error_mean_abs=0.188136, normalized_error_p95=0.64778, normalized_error_max=1.48, tail_breach_count=0.

D.4 Level K3: molecular organization and chemical bonding

Current closure status: PASS. Promotion blockers: none.

Theorem claim. K3 packages bounded molecular organization through lawful bond-length, bond-energy, and activation-energy envelopes derived from lower-level admissibility.

Operator and K-level binding. G, H, and R bind composition-sensitive bonding, activation barriers, and molecular-scale stabilization windows. Kernel refs: content/03_model.tex, content/10_klevels_full.tex, content/11_operators_full.tex, APPENDIX_Q_SYNTHESIS_DOSSIER_K3.

Parameter law.

$$r_{\text{cov}} \in [0.1, 0.2]\,\text{nm}, \quad E_{\text{bond}} \in [1, 5]\,\text{eV}, \quad E^{\ddagger} \in [0.1, 1]\,\text{eV}$$

Observable binding and units. Bond lengths, dissociation energies, dielectric context, and activation barriers for bounded chemistry lanes. Observable ids: K3::BOND_LENGTH, K3::BOND_ENERGY, K3::ACTIVATION_ENERGY, CHEMISTRY::GAS_PHASE_ENTHALPY_RESIDUAL, CHEMISTRY::SPECTRAL_TRANSITION_RESIDUAL, CHEMISTRY::KINETIC_OR_EQUILIBRIUM_RESIDUAL.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Bond length (covalent)	r_{cov}	0.1–0.2	nm
Bond dissociation energy	E_{bond}	1–5	eV

Quantity	Symbol	Value / range	Units or note
Reaction activation energy	$E^{\ddagger}$	0.1–1	eV

Falsifier and collapse boundary. K3 collapses when bond- and barrier-level regularities cannot be bounded by a non-degenerate parameter law or when chemical observables drift outside admissible molecular windows. Falsifier refs: `APPENDIX_Q_SYNTHESIS_DOSSIER_K3`, `content/falsifiability/falsifiability_master.tex`, `content/falsifiability/falsifiability_k2.tex`, `content/falsifiability/falsifiability_k3.tex`, `content/falsifiability/falsifiability_k4.tex`.

Closed-domain bindings. `CHEMISTRY`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `19`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/chemistry`. Metrics: `covered_case_total`=30, `covered_held_out_case_total`=19, `normalized_error_mean_abs`=0.533285, `normalized_error_p95`=1.0, `normalized_error_max`=1.597772, `tail_breach_count`=0.

D.5 Level K4: compartmental and membrane thresholds

Current closure status: `PASS`. Promotion blockers: none.

Theorem claim. K4 closes the first compartmental organization packet by binding membrane thickness, bending stability, and gradient maintenance to lawful biological thresholds.

Operator and K-level binding. H, R, and S couple membrane stability, osmotic admissibility, and retained gradients into a bounded protocellular regime. Kernel refs: `content/03_model.tex`, `content/10_klevels_full.tex`, `content/11_operators_full.tex`, `APPENDIX_Q_SYNTHESIS_DOSSIER_K4`.

Parameter law.

$$d_{\mathrm{mem}} \in [3,5]\,\mathrm{nm}, \quad \kappa \in [10,25]\,k_{\mathrm{B}}T, \quad \Delta pH \in [0.5,2]$$
$$\gamma_{\mathrm{edge}} \in [5,20]\,\mathrm{pN}, \quad \Pi \in [10^5,10^6]\,\mathrm{Pa}, \quad P_i \in [10^{-12},10^{-8}]\,\mathrm{m/s}$$

Observable binding and units. Membrane thickness, bending modulus, osmotic pressure, proton gradient, and permeability lanes for compartmental organization. Observable ids: `K4::MEMBRANE_THICKNESS`, `K4::BENDING_MODULUS`, `K4::PROTON_GRADIENT`, `K4::LINE_TENSION`, `K4::OSMOTIC_PRESSURE_SCALE`, `K4::PERMEABILITY_RANGE`, `BIOLOGY::EXPRESSION_SIGNATURE_RESIDUAL`, `BIOLOGY::STATE_TRANSITION_ORDER_ERROR`, `BIOLOGY::RESPONSE_ONSET_RESIDUAL`, `CHEMISTRY::GAS_PHASE_ENTHALPY_RESIDUAL`, `CHEMISTRY::SPECTRAL_TRANSITION_RESIDUAL`, `CHEMISTRY::KINETIC_OR_EQUILIBRIUM_RESIDUAL`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Membrane thickness	d_{mem}	3–5	nm
Membrane bending modulus	κ	10–25	$k_{\mathrm{B}}T$
Typical proton gradient	ΔpH	0.5–2	pH units

Quantity	Symbol	Value / range	Units or note
Line tension (edge)	γ_{edge}	5–20	pN
Osmotic pressure scale	Π	10^5–10^6	Pa
Permeability range	P_i	10^{-12}–10^{-8}	m/s

Falsifier and collapse boundary. K4 collapses if compartments cannot retain a lawful gradient, if membranes lose structural stability, or if permeability escapes the admissible packet. Falsifier refs: APPENDIX_Q_SYNTHESIS_DOSSIER_K4, content/falsifiability/falsifiability_master.tex, content/falsifiability/falsifiability_k4.tex, content/falsifiability/falsifiability_k5.tex, content/falsifiability/falsifiability_k6.tex, content/falsifiability/falsifiability_k7.tex, content/falsifiability/falsifiability_k2.tex, content/falsifiability/falsifiability_k3.tex.

Closed-domain bindings. BIOLOGY: class THEOREM_NATIVE, verdict PASS, held-out cases 16, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/biology. Metrics: covered_case_total=30, covered_held_out_case_total=16, normalized_error_mean_abs=0.444222, normalized_error_p95=1.030033, normalized_error_max=1.438582, tail_breach_count=0. CHEMISTRY: class THEOREM_NATIVE, verdict PASS, held-out cases 19, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/chemistry. Metrics: covered_case_total=30, covered_held_out_case_total=19, normalized_error_mean_abs=0.533285, normalized_error_p95=1.0, normalized_error_max=1.597772, tail_breach_count=0.

D.6 Level K5: excitable and early neural continua

Current closure status: PASS. Promotion blockers: none.

Theorem claim. K5 binds excitability, refractory timing, and membrane potential maintenance into the first lawful neural packet.

Operator and K-level binding. F, G, and S stabilize excitable-state transitions, ion-channel gating, and refractory boundaries in bounded neural lanes. Kernel refs: content/03_model.tex, content/10_klevels_full.tex, content/11_operators_full.tex, APPENDIX_Q_SYNTHESIS_DOSSIER_K5.

Parameter law.

$$C_m = 1\,\mu\text{F/cm}^2, \quad V_{\text{rest}} \in [-75, -60]\,\text{mV}, \quad \tau_{\text{ref}} \in [2, 5]\,\text{ms}$$

Observable binding and units. Membrane capacitance, resting potential, channel conductance, and refractory timing for bounded excitable systems. Observable ids: K5::MEMBRANE_CAPACITANCE, K5::RESTING_POTENTIAL, K5::REFRACTORY_TIME, BIOLOGY::EXPRESSION_SIGNATURE_RESIDUAL, BIOLOGY::STATE_TRANSITION_ORDER_ERROR, BIOLOGY::RESPONSE_ONSET_RESIDUAL.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Membrane capacitance	C_m	1	μF/cm^2
Resting membrane potential	V_{rest}	-75–-60	mV
Refractory time	τ_{ref}	2–5	ms

Falsifier and collapse boundary. K5 collapses when excitability loses bounded thresholds, refractory timing fails, or the neural packet cannot survive counterexample pressure across datasets. Falsifier refs: `APPENDIX_Q_SYNTHESIS_DOSSIER_K5`, `content/falsifiability/falsifiability_master.tex`, `content/falsifiability/falsifiability_k4.tex`, `content/falsifiability/falsifiability_k5.tex`, `content/falsifiability/falsifiability_k6.tex`, `content/falsifiability/falsifiability_k7.tex`.

Closed-domain bindings. `BIOLOGY`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `16`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/biology`. Metrics: `covered_case_total`=30, `covered_held_out_case_total`=16, `normalized_error_mean_abs`=0.444222, `normalized_error_p95`=1.030033, `normalized_error_max`=1.438582, `tail_breach_count`=0.

D.7 Level K6: cognitive prediction and representation

Current closure status: `PASS`. Promotion blockers: none.

Theorem claim. K6 formalizes bounded cognitive prediction and representation by linking prediction-error thresholds, working-memory span, and adaptation strength.

Operator and K-level binding. Q, R, and S bind representation stability, prediction thresholds, and adaptive update constraints in bounded cognitive lanes. Kernel refs: `content/03_model.tex`, `content/10_klevels_full.tex`, `content/11_operators_full.tex`, `APPENDIX_Q_SYNTHESIS_DOSSIER_K6`.

Parameter law.

$$\Theta_{\text{pred}} \in [0.05, 0.15], \quad M_{\text{WM}} \in [3, 7], \quad \eta \in [10^{-3}, 10^{-1}]$$

Observable binding and units. Prediction error, working-memory span, and adaptation strength for bounded cognitive organization. Observable ids: `K6::PREDICTION_ERROR_THRESHOLD`, `K6::WORKING_MEMORY_SPAN`, `K6::HEBBIAN_STRENGTH`, `BIOLOGY::EXPRESSION_SIGNATURE_RESIDUAL`, `BIOLOGY::STATE_TRANSITION_ORDER_ERROR`, `BIOLOGY::RESPONSE_ONSET_RESIDUAL`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Prediction-error threshold	Θ_{pred}	0.05–0.15	normalized units
Working-memory span	M_{WM}	3–7	bound items
Hebbian strength coefficient	η	10^{-3}–10^{-1}	dimensionless

Falsifier and collapse boundary. K6 collapses when predictive stabilization fails across bounded cognitive observables or when representation thresholds require hidden freedom. Falsifier refs: `APPENDIX_Q_SYNTHESIS_DOSSIER_K6`, `content/falsifiability/falsifiability_master.tex`, `content/falsifiability/falsifiability_k4.tex`, `content/falsifiability/falsifiability_k5.tex`, `content/falsifiability/falsifiability_k6.tex`, `content/falsifiability/falsifiability_k7.tex`, `content/falsifiability/falsifiability_k2.tex`.

Closed-domain bindings. `BIOLOGY`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `16`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/biology`. Metrics: `covered_case_total`=30, `covered_held_out_case_total`=16, `normalized_error_mean_abs`=0.444222, `normalized_error_p95`=1.030033, `normalized_error_max`=1.438582, `tail_breach_count`=0. `DRT`: class `REFUTED`, verdict `FAIL_CLOSED`, held-out cases `0`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `none`. Metrics: none.

D.8 Level K7: social coordination and group stability

Current closure status: `PASS`. Promotion blockers: none.

Theorem claim. K7 closes bounded social coordination through group size, communication bandwidth, and coordination thresholds.

Operator and K-level binding. R, S, and U bind coordination thresholds, communication throughput, and conflict costs into admissible social trajectories. Kernel refs: `content/03_model.tex`, `content/10_klevels_full.tex`, `content/11_operators_full.tex`, `APPENDIX_Q_SYNTHESIS_DOSSIER_K7`.

Parameter law.

$$N_s \in [120, 180], \quad J_{\text{comm}} \in [1, 5]\,\text{bits/s}, \quad \Theta_{\text{coord}} \in [0.2, 0.4]$$
$$C_{\text{conf}} \in [0.1, 1]$$

Observable binding and units. Group stability size, communication throughput, coordination threshold, and conflict cost inside social coordination continua. Observable ids: `K7::GROUP_STABILITY_SIZE`, `K7::COMMUNICATION_BANDWIDTH`, `K7::COORDINATION_THRESHOLD`, `K7::CONFLICT_COST_SCALE`, `BIOLOGY::EXPRESSION_SIGNATURE_RESIDUAL`, `BIOLOGY::STATE_TRANSITION_ORDER_ERROR`, `BIOLOGY::RESPONSE_ONSET_RESIDUAL`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Group stability size	N_s	120–180	agents
Information-flow bandwidth per agent	J_{comm}	1–5	bits/s
Coordination threshold	Θ_{coord}	0.2–0.4	dimensionless
Conflict cost scale	C_{conf}	0.1–1	arbitrary units

Falsifier and collapse boundary. K7 collapses when communication throughput and coordination thresholds cannot stabilize social group trajectories without hidden rescue assumptions. Falsifier refs: `APPENDIX_Q_SYNTHESIS_DOSSIER_K7`, `content/falsifiability/falsifiability_master.tex`, `content/falsifiability/falsifiability_k4.tex`, `content/falsifiability/falsifiability_k5.tex`, `content/falsifiability/falsifiability_k6.tex`, `content/falsifiability/falsifiability_k7.tex`.

Closed-domain bindings. `BIOLOGY`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `16`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/biology`. Metrics: `covered_case_total`=30, `covered_held_out_case_total`=16, `normalized_error_mean_abs`=0.444222, `normalized_error_p95`=1.030033, `normalized_error_max`=1.438582, `tail_breach_count`=0.

D.9 Level K8: civilizational flows and infrastructural stability

Current closure status: `PASS`. Promotion blockers: none.

Theorem claim. K8 binds civilizational energy density, renewal timing, and repair-decay balance into bounded macro-process trajectories and regime-shift packets.

Operator and K-level binding. H, R, and U bind throughput, repair-decay balance, and regime-shift boundaries in bounded systems-level observables. Kernel refs: `content/03_model.tex`, `content/10_klevels_full.tex`, `content/11_operators_full.tex`, `APPENDIX_Q_SYNTHESIS_DOSSIER_K8`.

Parameter law.

$$E_{\text{dens}} \in [10^2, 10^3]\,\mathrm{W/m^2}, \quad \tau_{\text{infra}} \in [20, 50]\text{ years}, \quad R_{\text{rep}} \in [0.8, 1.2]$$
$$\Theta_I \in [0.1, 0.3]$$

Observable binding and units. Energy density, infrastructure renewal, information coherence, and repair-decay balance in bounded civilizational lanes. Observable ids: `K8::ENERGY_CONSUMPTION_DENSITY`, `K8::INFRASTRUCTURE_RENEWAL_TIME`, `K8::REPAIR_TO_DECAY_RATIO`, `K8::INFORMATION_COHERENCE_THRESHOLD`, `SYSTEMS_CIVILIZATIONAL_PROJECTION::POPULATION_TRAJECTORY_RESIDUAL`, `SYSTEMS_CIVILIZATIONAL_PROJECTION::GDP_GROWTH_TURNING_POINT_ERROR`, `SYSTEMS_CIVILIZATIONAL_PROJECTION::PROCESS_SYSTEM_REGIME_SCORE`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Energy consumption density	E_{dens}	10^2–10^3	$\mathrm{W/m^2}$
Infrastructure renewal time	τ_{infra}	20–50	years
Information-coherence threshold	Θ_I	0.1–0.3	dimensionless
Repair-to-decay ratio	R_{rep}	0.8–1.2	dimensionless

Falsifier and collapse boundary. K8 collapses when macro-trajectories lose bounded turning-point fidelity, when repair-decay balance drifts outside its band, or when regime-shift packets require discretionary knobs. Falsifier refs: `APPENDIX_Q_SYNTHESIS_DOSSIER_K8`, `content/falsifiability/falsifiability_master.tex`, `content/falsifiability/falsifiability_k8.tex`.

Closed-domain bindings. `SYSTEMS_CIVILIZATIONAL_PROJECTION`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `18`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/systems_civilizational_projection`. Metrics: `covered_case_total`=30, `covered_held_out_case_total`=18, `normalized_error_mean_abs`=0.56097, `normalized_error_p95`=1.924704, `normalized_error_max`=2.25395, `tail_breach_count`=0.

D.10 Level K9: knowledge systems and scientific memory

Current closure status: `PASS`. Promotion blockers: none.

Theorem claim. K9 formalizes bounded scientific-knowledge continua through anomaly accumulation, paradigm coherence, and model-validation bandwidth.

Operator and K-level binding. Q, R, and U bind anomaly handling, paradigm coherence, and memory retention inside knowledge-system packets. Kernel refs: `content/03_model.tex`, `content/10_klevels_full.tex`, `content/11_operators_full.tex`, `APPENDIX_Q_SYNTHESIS_DOSSIER_K9`.

Parameter law.

$$\Theta_{\text{par}} \in [0.15, 0.25], \quad J_{\text{anom}} \in [10^{-4}, 10^{-2}], \quad \tau_{\text{mem}} \in [30, 200] \text{ years}$$
$$B_{\text{val}} \in [0.1, 1]$$

Observable binding and units. Paradigm coherence, anomaly accumulation, validation bandwidth, and scientific-memory half-life. Observable ids: `K9::PARADIGM_COHERENCE_THRESHOLD`, `K9::ANOMALY_ACCUMULATION_RATE`, `K9::SCIENTIFIC_MEMORY_HALF_LIFE`, `K9::MODEL_VALIDATION_BANDWIDTH`, `MATHEMATICS::EPSILON_STAR`, `MATHEMATICS::MINIMAL_DENOMINATOR`, `MATHEMATICS::MINIMAL_AUGMENTATION_RANK`, `MATHEMATICS::COMPONENT_FIBER_COUNT`, `MATHEMATICS::MOVE_CONNECTIVITY_OR_DEGREE_BOUND`, `MATHEMATICS::STRICT_TERMINALIZED_TOTAL`, `MATHEMATICS::COUNTEREXAMPLE_EXECUTED_TOTAL`, `MATHEMATICS::PYTHON_EVIDENCE_EXECUTED_TOTAL`, `PHYSICS::FINE_STRUCTURE_CONSTANT_RESIDUAL`, `PHYSICS::HYDROGEN_BALMER_LINE_RESIDUAL`, `PHYSICS::TRANSPORT_SCALING_RESIDUAL`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Paradigm-coherence threshold	Θ_{par}	0.15–0.25	dimensionless
Anomaly accumulation rate	J_{anom}	10^{-4}–10^{-2}	per cycle
Scientific-memory half-life	τ_{mem}	30–200	years
Model-validation bandwidth	B_{val}	0.1–1	normalized

Falsifier and collapse boundary. K9 collapses if anomaly load outruns lawful model revision or if scientific memory and validation bandwidth fail to stabilize the knowledge packet. Falsifier refs: `APPENDIX_Q_SYNTHESIS_DOSSIER_K9`, `content/falsifiability/falsifiability_master.tex`, `content/falsifiability/falsifiability_k0.tex`, `content/falsifiability/falsifiability_k9.tex`, `content/falsifiability/falsifiability_k10.tex`, `content/falsifiability/falsifiability_k11.tex`, `content/falsifiability/falsifiability_k12.tex`, `content/falsifiability/falsifiability_k1.tex`, `content/falsifiability/falsifiability_k2.tex`.

Closed-domain bindings. `MATHEMATICS`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `0`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/mathematics`. Metrics: `strict_campaign_acceptance_status`=PASS_31_OF_31_TERMINALIZED, `strict_terminalized_total`=31. `PHYSICS`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `20`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/physics`. Metrics: `covered_case_total`=30, `covered_held_out_case_total`=20, `normalized_error_mean_abs`=0.188136, `normalized_error_p95`=0.64778, `normalized_error_max`=1.48, `tail_breach_count`=0.

D.11 Level K10: formal systems and recursion

Current closure status: `PASS`. Promotion blockers: none.

Theorem claim. K10 binds bounded recursion, consistency thresholds, and proof-flow capacity into the formal-system lane of the unified-science closure stack.

Operator and K-level binding. F, Q, and U bind recursion depth, proof throughput, and consistency thresholds in formal continua. Kernel refs: `content/03_model.tex`, `content/10_klevels_full.tex`, `content/11_operators_full.tex`, `APPENDIX_Q_SYNTHESIS_DOSSIER_K10`.

Parameter law.

$$d_{\text{rec}} \in [10, 10^4], \quad \Theta_{\text{cons}} \in [0.01, 0.05], \quad J_{\text{proof}} \in [1, 10^3]$$

Observable binding and units. Recursion depth, consistency threshold, and proof-flow capacity in bounded formal systems. Observable ids: `K10::RECURSION_DEPTH`, `K10::CONSISTENCY_THRESHOLD`, `K10::PROOF_FLOW_CAPACITY`, `MATHEMATICS::EPSILON_STAR`, `MATHEMATICS::MINIMAL_DENOMINATOR`, `MATHEMATICS::MINIMAL_AUGMENTATION_RANK`, `MATHEMATICS::COMPONENT_FIBER_COUNT`, `MATHEMATICS::MOVE_CONNECTIVITY_OR_DEGREE_BOUND`, `MATHEMATICS::STRICT_TERMINALIZED_TOTAL`, `MATHEMATICS::COUNTEREXAMPLE_EXECUTED_TOTAL`, `MATHEMATICS::PYTHON_EVIDENCE_EXECUTED_TOTAL`, `PHYSICS::FINE_STRUCTURE_CONSTANT_RESIDUAL`, `PHYSICS::HYDROGEN_BALMER_LINE_RESIDUAL`, `PHYSICS::TRANSPORT_SCALING_RESIDUAL`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Recursion depth (effective)	d_{rec}	10–10^4	steps
Consistency threshold	Θ_{cons}	0.01–0.05	dimensionless
Proof-flow capacity	J_{proof}	1–10^3	steps/s

Falsifier and collapse boundary. K10 collapses if formal recursion and proof throughput cannot remain bounded without violating consistency thresholds. Falsifier refs: `APPENDIX_Q_SYNTHESIS_DOSSIER_K10`, `content/falsifiability/falsifiability_master.tex`, `content/falsifiability/falsifiability_k0.tex`, `content/falsifiability/falsifiability_k9.tex`, `content/falsifiability/falsifiability_k10.tex`, `content/falsifiability/falsifiability_k11.tex`, `content/falsifiability/falsifiability_k12.tex`, `content/falsifiability/falsifiability_k1.tex`, `content/falsifiability/falsifiability_k2.tex`.

Closed-domain bindings. `MATHEMATICS`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `0`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/mathematics`. Metrics: `strict_campaign_acceptance_status`=PASS_31_OF_31_TERMINALIZED, `strict_terminalized_total`=31. `METAONTOLOGY`: class `FRAME_ONLY`, verdict `PASS`, held-out cases `0`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/metaontology`. Metrics: none. `PHYSICS`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `20`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/physics`. Metrics: `covered_case_total`=30, `covered_held_out_case_total`=20, `normalized_error_mean_abs`=0.188136, `normalized_error_p95`=0.64778, `normalized_error_max`=1.48, `tail_breach_count`=0.

D.12 Level K11: meta-theoretical reflexivity

Current closure status: `PASS`. Promotion blockers: none.

Theorem claim. K11 captures meta-theoretical reflexivity as irreducible scientific work beyond K10 across the closed domain atlas.

Operator and K-level binding. Q, R, and S bind meta-coherence, reflexive depth, and cross-landscape coupling under the requirement that no K10-only translation preserves all declared upper-level burdens. Kernel refs: `content/03_model.tex`, `content/10_klevels_full.tex`, `content/11_operators_full.tex`, `APPENDIX_Q_SYNTHESIS_DOSSIER_K11`.

Parameter law.

$$\Theta_{\text{meta}} \in [0.02, 0.1], \quad d_{\text{refl}} \in [3, 50], \quad J_X \in [0.1, 1]$$
$$P_{\text{funct}} \in [10^0, 10^3]$$

Observable binding and units. Meta-coherence, reflexive depth, and cross-landscape coupling in the meta-theoretical packet. Observable ids: `K11::META_COHERENCE_THRESHOLD`, `K11::REFLEXIVE_DEPTH`, `K11::CROSS_LANDSCAPE_COUPLING`, `K11::FUNCTORIAL_POTENTIAL`, `MATHEMATICS::EPSILON_STAR`, `MATHEMATICS::MINIMAL_DENOMINATOR`, `MATHEMATICS::MINIMAL_AUGMENTATION_RANK`, `MATHEMATICS::COMPONENT_FIBER_COUNT`, `MATHEMATICS::MOVE_CONNECTIVITY_OR_DEGREE_BOUND`, `MATHEMATICS::STRICT_TERMINALIZED_TOTAL`, `MATHEMATICS::COUNTEREXAMPLE_EXECUTED_TOTAL`, `MATHEMATICS::PYTHON_EVIDENCE_EXECUTED_TOTAL`, `PHYSICS::FINE_STRUCTURE_CONSTANT_RESIDUAL`, `PHYSICS::HYDROGEN_BALMER_LINE_RESIDUAL`, `PHYSICS::TRANSPORT_SCALING_RESIDUAL`, `CHEMISTRY::GAS_PHASE_ENTHALPY_RESIDUAL`, `CHEMISTRY::SPECTRAL_TRANSITION_RESIDUAL`, `CHEMISTRY::KINETIC_OR_EQUILIBRIUM_RESIDUAL`, `BIOLOGY::EXPRESSION_SIGNATURE_RESIDUAL`, `BIOLOGY::STATE_TRANSITION_ORDER_ERROR`, `BIOLOGY::RESPONSE_ONSET_RESIDUAL`, `SYSTEMS_CIVILIZATIONAL_PROJECTION::POPULATION_TRAJECTORY_RESIDUAL`, `SYSTEMS_CIVILIZATIONAL_PROJECTION::GDP_GROWTH_TURNING_POINT_ERROR`, `SYSTEMS_CIVILIZATIONAL_PROJECTION::PROCESS_SYSTEM_REGIME_SCORE`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Meta-coherence threshold	Θ_{meta}	0.02–0.1	dimensionless
Functorial potential	P_{funct}	10^0–10^3	abstract units
Reflexive depth	d_{refl}	3–50	levels
Cross-landscape coupling	J_X	0.1–1	dimensionless

Falsifier and collapse boundary. K11 collapses if an explicit K10-only reduction preserves the declared meta-theoretical burdens or if meta-coherence depends on an excluded lift route. Falsifier refs: `APPENDIX_Q_SYNTHESIS_DOSSIER_K11`, `content/falsifiability/falsifiability_master.tex`, `content/falsifiability/falsifiability_k0.tex`, `content/falsifiability/falsifiability_k9.tex`, `content/falsifiability/falsifiability_k10.tex`, `content/falsifiability/falsifiability_k11.tex`, `content/falsifiability/falsifiability_k12.tex`, `content/falsifiability/falsifiability_k1.tex`, `content/falsifiability/falsifiability_k2.tex`, `content/falsifiability/falsifiability_k3.tex`, `content/falsifiability/falsifiability_k4.tex`, `content/falsifiability/falsifiability_k5.tex`, `content/falsifiability/falsifiability_k6.tex`, `content/falsifiability/falsifiability_k7.tex`, `content/falsifiability/falsifiability_k8.tex`.

Closed-domain bindings. `MATHEMATICS`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `0`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/mathematics`. Metrics: `strict_campaign_acceptance_status`=PASS_31_OF_31_TERMINALIZED, `strict_terminalized_total`=31. `PHYSICS`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `20`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/physics`. Metrics: `covered_case_total`=30, `covered_held_out_case_total`=20, `normalized_error_mean_abs`=0.188136, `normalized_error_p95`=0.64778, `normalized_error_max`=1.48, `tail_breach_count`=0. `CHEMISTRY`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `19`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/chemistry`. Metrics: `covered_case_total`=30, `covered_held_out_case_total`=19, `normalized_error_mean_abs`=0.533285, `normalized_error_p95`=1.0, `normalized_error_max`=1.597772, `tail_breach_count`=0. `BIOLOGY`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `16`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/biology`. Metrics: `covered_case_total`=30, `covered_held_out_case_total`=16, `normalized_error_mean_abs`=0.444222, `normalized_error_p95`=1.030033, `normalized_error_max`=1.438582, `tail_breach_count`=0. `SYSTEMS_CIVILIZATIONAL_PROJECTION`: class `THEOREM_NATIVE`, verdict `PASS`, held-out cases `18`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/systems_civilizational_projection`. Metrics: `covered_case_total`=30, `covered_held_out_case_total`=18, `normalized_error_mean_abs`=0.56097, `normalized_error_p95`=1.924704, `normalized_error_max`=2.25395, `tail_breach_count`=0. `METAONTOLOGY`: class `FRAME_ONLY`, verdict `PASS`, held-out cases `0`, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/metaontology`. Metrics: none.

D.13 Level K12: global semantic coherence

Current closure status: `PASS`. Promotion blockers: none.

Theorem claim. K12 packages global semantic coherence across the closed domain atlas because cross-domain meaning remains jointly interpretable and no surviving global falsifier breaks the unified synthesis.

Operator and K-level binding. Q, R, S, and U bind semantic coherence, global compatibility, and reference stability into the final unified-science synthesis layer. Kernel refs: `content/03_model.tex`, `content/10_klevels_full.tex`, `content/11_operators_full.tex`, `APPENDIX_Q_SYNTHESIS_DOSSIER_K12`.

Parameter law.

$$C_{\text{uni}} \in [10^3, 10^{12}], \quad \Theta_{\text{uni}} \in [10^{-3}, 10^{-2}], \quad R_{\text{uni}} \in [0.1, 1]$$
$$T_{\text{uni}} \in [10^2, 10^8]$$

Observable binding and units. Universal integration capacity, cross-continuum compatibility, and structural reachability. Observable ids: `K12::UNIVERSAL_INTEGRATION_CAPACITY`, `K12::CROSS_CONTINUUM_COMPATIBILITY`, `K12::STRUCTURAL_REACHABILITY`, `K12::GLOBAL_TENSION_BUDGET`, `MATHEMATICS::EPSILON_STAR`, `MATHEMATICS::MINIMAL_DENOMINATOR`, `MATHEMATICS::MINIMAL_AUGMENTATION_RANK`, `MATHEMATICS::COMPONENT_FIBER_COUNT`, `MATHEMATICS::MOVE_CONNECTIVITY_OR_DEGREE_BOUND`, `MATHEMATICS::STRICT_TERMINALIZED_TOTAL`, `MATHEMATICS::COUNTEREXAMPLE_EXECUTED_TOTAL`, `MATHEMATICS::PYTHON_EVIDENCE_EXECUTED_TOTAL`, `PHYSICS::FINE_STRUCTURE_CONSTANT_RESIDUAL`, `PHYSICS::HYDROGEN_BALMER_LINE_RESIDUAL`, `PHYSICS::TRANSPORT_SCALING_RESIDUAL`, `CHEMISTRY::GAS_PHASE_ENTHALPY_RESIDUAL`, `CHEMISTRY::SPECTRAL_TRANSITION_RESIDUAL`, `CHEMISTRY::KINETIC_OR_EQUILIBRIUM_RESIDUAL`, `BIOLOGY::EXPRESSION_SIGNATURE_RESIDUAL`, `BIOLOGY::STATE_TRANSITION_ORDER_ERROR`, `BIOLOGY::RESPONSE_ONSET_RESIDUAL`, `SYSTEMS_CIVILIZATIONAL_PROJECTION::POPULATION_TRAJECTORY_RESIDUAL`, `SYSTEMS_CIVILIZATIONAL_PROJECTION::GDP_GROWTH_TURNING_POINT_ERROR`, `SYSTEMS_CIVILIZATIONAL_PROJECTION::PROCESS_SYSTEM_REGIME_SCORE`.

Numerical instantiation and prediction table. The table below is a promoted subset of the numerical backing rows in Appendix **??**.

Quantity	Symbol	Value / range	Units or note
Universal integration capacity	C_{uni}	10^3–10^{12}	dimensionless
Cross-continuum compatibility	Θ_{uni}	10^{-3}–10^{-2}	dimensionless
Structural reachability	R_{uni}	0.1–1	dimensionless
Global tension budget	T_{uni}	10^2–10^8	dimensionless

Falsifier and collapse boundary. K12 collapses if cross-domain compatibility fragments, if the unified atlas cannot pass integrability, or if a surviving global falsifier breaks semantic coherence. Falsifier refs: `APPENDIX_Q_SYNTHESIS_DOSSIER_K12`, `content/falsifiability/falsifiability_master.tex`, `content/falsifiability/falsifiability_k0.tex`, `content/falsifiability/falsifiability_k9.tex`, `content/falsifiability/falsifiability_k10.tex`, `content/falsifiability/falsifiability_k11.tex`, `content/falsifiability/falsifiability_k12.tex`, `content/falsifiability/falsifiability_k1.tex`, `content/falsifiability/falsifiability_k2.tex`, `content/falsifiability/falsifiability_k3.tex`, `content/falsifiability/falsifiability_k4.tex`, `content/falsifiability/falsifiability_k5.t ex, content/falsifiability/falsifiability_k6.tex, content/falsifiability/falsifiability_k7.tex, content/falsifiability/falsifiability_k8.tex.

Closed-domain bindings. MATHEMATICS: class THEOREM_NATIVE, verdict PASS, held-out cases 0, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/mathematics. Metrics: strict_campaign_acceptance_status=PASS_31_OF_31_TERMINALIZED, strict_terminalized_total=31. PHYSICS: class THEOREM_NATIVE, verdict PASS, held-out cases 20, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/physics. Metrics: covered_case_total=30, covered_held_out_case_total=20, normalized_error_mean_abs=0.188136, normalized_error_p95=0.64778, normalized_error_max=1.48, tail_breach_count=0. CHEMISTRY: class THEOREM_NATIVE, verdict PASS, held-out cases 19, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/chemistry. Metrics: covered_case_total=30, covered_held_out_case_total=19, normalized_error_mean_abs=0.533285, normalized_error_p95=1.0, normalized_error_max=1.597772, tail_breach_count=0. BIOLOGY: class THEOREM_NATIVE, verdict PASS, held-out cases 16, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/biology. Metrics: covered_case_total=30, covered_held_out_case_total=16, normalized_error_mean_abs=0.444222, normalized_error_p95=1.030033, normalized_error_max=1.438582, tail_breach_count=0. SYSTEMS_CIVILIZATIONAL_PROJECTION: class THEOREM_NATIVE, verdict PASS, held-out cases 18, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/systems_civilizational_projection. Metrics: covered_case_total=30, covered_held_out_case_total=18, normalized_error_mean_abs=0.56097, normalized_error_p95=1.924704, normalized_error_max=2.25395, tail_breach_count=0. METAONTOLOGY: class FRAME_ONLY, verdict PASS, held-out cases 0, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/metaontology. Metrics: none.

E OC 1.3.1 Science Delta Appendix

- delta_id=DELTA::WHITE_SPOT_CLOSURE; delta_class=scientific_closure; status_1_3_1=closure_companion_and action_bucket=must_be_added
- delta_id=DELTA::SIMULATION_CORPUS; delta_class=simulation_and_reproducibility; status_1_3_1=top_level_simulation_corpus; action_bucket=must_be_added
- delta_id=DELTA::DATA_MANIFEST_LAYER; delta_class=data_route_and_traceability; status_1_3_1=manifest_and action_bucket=must_be_added
- delta_id=DELTA::MASTER_MONOGRAPH; delta_class=release_authority; status_1_3_1=oc_core_1_3_1_master_p action_bucket=must_be_added
- delta_id=DELTA::ARTICLE_FAMILY; delta_class=publication_surface; status_1_3_1=bounded_article_family_prese action_bucket=must_be_added
- delta_id=DELTA::RELEASE_AUTOMATION; delta_class=release_automation; status_1_3_1=contract_and_validato action_bucket=must_be_added
- delta_id=DELTA::NO_SEND_OUTBOUND; delta_class=outbound_readiness; status_1_3_1=release_specific_packag action_bucket=must_be_added
- delta_id=DELTA::PUBLIC_RELEASE_STAGE; delta_class=release_staging; status_1_3_1=oc_core_1_3_1_ready_n action_bucket=candidate_for_1_3_1

- delta_id=DELTA::K11_K12_EXTENSION; delta_class=high_cost_theory_extension; status_1_3_1=mapped_but_not
action_bucket=must_remain_frontier
- delta_id=DELTA::PAID_DATA_ESCALATIONS; delta_class=data_escalation; status_1_3_1=still_optional;
action_bucket=too_expensive_now

F OC 1.3.1 Simulation and Data Appendix

- lane_id=LANE::MATHEMATICS; domain=mathematics; strengthening_status=STRENGTHENED_NUMERICALLY
primary_strengthening_mode=NUMERIC_PACKET
- lane_id=LANE::PHYSICS; domain=physics; strengthening_status=STRENGTHENED_NUMERICALLY;
primary_strengthening_mode=NUMERIC_PACKET
- lane_id=LANE::CHEMISTRY; domain=chemistry; strengthening_status=STRENGTHENED_NUMERICALLY;
primary_strengthening_mode=NUMERIC_PACKET
- lane_id=LANE::K3_K4_ORIGINS; domain=k3_k4_origins_of_life; strengthening_status=STRENGTHENED_BY_S
primary_strengthening_mode=SIMULATION_CORPUS
- lane_id=LANE::BIOLOGY_K4_K5; domain=biology; strengthening_status=STRENGTHENED_BY_DATA_ROUTE
primary_strengthening_mode=DATA_MANIFEST_BINDING
- lane_id=LANE::COGNITION_K6; domain=k6_cognition; strengthening_status=STRENGTHENED_FORMALLY;
primary_strengthening_mode=FORMULA_AND_OPERATOR_PATCH
- lane_id=LANE::SOCIETY_K7; domain=k7_society; strengthening_status=STRENGTHENED_FORMALLY;
primary_strengthening_mode=FORMULA_AND_OPERATOR_PATCH
- lane_id=LANE::SYSTEMS_K8; domain=k8_systems; strengthening_status=STRENGTHENED_BY_SIMULATION;
primary_strengthening_mode=SIMULATION_CORPUS
- lane_id=LANE::KNOWLEDGE_K9; domain=k9_knowledge_systems; strengthening_status=STRENGTHENED_FO
primary_strengthening_mode=FORMULA_AND_OPERATOR_PATCH
- lane_id=LANE::FORMAL_K10; domain=k10_formal_systems; strengthening_status=STRENGTHENED_FORMAL
primary_strengthening_mode=FORMULA_AND_OPERATOR_PATCH
- lane_id=LANE::META_K11_K12; domain=k11_k12_meta_theory; strengthening_status=PARTIAL_ONLY;
primary_strengthening_mode=FRONTIER_AND_SIMULATION_MIX

G OC 1.3.1 Critique Closure Appendix

- critique_id=CRITIQUE::DESKTOP_CLOSEOUT_STATE_DRIFT; processing_class=REAL_PACKAGING_WEAK
output_mode=clarification_unit; publication_effect=none
- critique_id=CRITIQUE::PUBLIC_REPO_ARCHITECTURE_DRIFT; processing_class=REAL_PACKAGING_WEA
output_mode=clarification_unit; publication_effect=refresh public architecture truth and release gate explanation
- critique_id=CRITIQUE::SCOPE_INFLATION_CHECK__WEDGE_001; processing_class=REQUIRES_DEMOTIO
output_mode=demotion_decision; publication_effect=bounded clarification note for reviewer/public audience
- critique_id=CRITIQUE::PUBLIC_REPO_SIMULATION_ABSENCE; processing_class=REAL_PACKAGING_WE
output_mode=clarification_unit; publication_effect=bind simulation corpus into OC Core 1.3.1 release package